

ENVIRONMENTAL SCIENCE

(For First Year B.Tech. Students)

(Designed as per the revised syllabus of
Pondicherry University effective from 2013)

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Syllabus

ENVIRONMENTAL SCIENCE

UNIT I - ENVIRONMENT AND ENERGY RESOURCES

Environmental segments – atmosphere, hydrosphere, lithosphere and biosphere. Atmospheric layers. Pollution definition and classification. Pollutants classification. Forest resources - use and over exploitation, deforestation, forest management. Water resources - use and conflicts over water, dams - benefits and problems. Mineral resources - mineral wealth of India, environmental effects of extracting & using mineral resources. Food resources - world food problems, environmental impact of modern Agriculture - fertilizer and pesticides. Energy resources – growing needs, renewable and non-renewable energy resources and use of alternate energy sources. From unsustainable to sustainable development.

UNIT II - ECOSYSTEM AND BIODIVERSITY

Concept of an ecosystem - structure and function of an ecosystem. Producers, consumers, and decomposers. Energy flow in the ecosystem. Food chains, food webs and ecological pyramids. Introduction, types, characteristic features, structure and function of forest, grassland, desert and aquatic (fresh water, estuarine and marine) ecosystems. Biodiversity – definition, genetic species and ecosystem diversity. Value of biodiversity - consumptive use, productive use, social, ethical, aesthetic and option values. Hot spots of biodiversity. Threats to biodiversity, habitat loss, poaching of wildlife, human wildlife conflicts. Endangered and endemic species. Conservation of biodiversity - in-situ and ex-situ conservation of biodiversity.

UNIT III - AIR POLLUTION

Definition and classification. Chemical and photochemical reaction in different layers of atmosphere. Causes, sources, effects and control measures of air pollutants - oxides of Nitrogen, oxides of Carbon, oxides of Sulfur, hydrocarbons, chloro-fluoro carbons and particulates. Mechanism and effects of air pollution phenomenon - Global Warming, Ozone Depletion, Acid Rain, Sulfurous Smog and Photochemical Smog.

UNIT IV – WATER AND LAND POLLUTION

Water pollution - causes and effects of organic water pollutants - pesticides, insecticides, detergents and surfactants. Causes and effects of inorganic water pollutants - heavy metal pollution due to Hg, Pb, Cr & Cu. Water pollution control and monitoring - DO, COD, BOD & TOC. Land Pollution - Solid waste management - causes, effect and control measures of urban and industrial wastes. Thermal and radioactive pollution.

UNIT V - POLLUTION CONTROL AND MONITORING

Basic concepts and instrumentation of IR, UV-VIS, atomic absorption spectrometry, Gas Chromatography and Conductometry. Analysis of air pollutants - NO_x, CO_x, SO_x, H₂S, Hydrocarbons and particulates.

Text Books:

1. K. Raghavan Nambiar, "Text Book of Environmental Studies" 2nd Ed, Scitech Publications (India) Pvt Ltd, India 2010 (For Units I & II)
2. A. K. De, "Environmental chemistry" 7th Ed; New age international (P) Ltd, New Delhi, 2010. (For Units III, IV & V)

Reference Books:

1. B.K. Sharma, "Environmental Chemistry" 11th Ed, KRISHNA Prakashan Media (P) Ltd, Meerut, 2007.
2. S.S.Dara, and D.D. Mishra "A text book of Environmental chemistry and pollution control, 5th Ed, S. Chand and Company Ltd, New Delhi, 2012.
3. Richard T. Wright, Environmental Science: Toward a Sustainable Future, 10th edition, Prentice Hall, 2008.
4. G. S. Sodhi, Fundamental concept of environmental chemistry, I Ed, Alpha Science International Ltd, India 2000.

UNIT I ENVIRONMENT AND ENERGY RESOURCES

ENVIRONMENTAL SEGMENTS

The environment consists of four segments. These are,

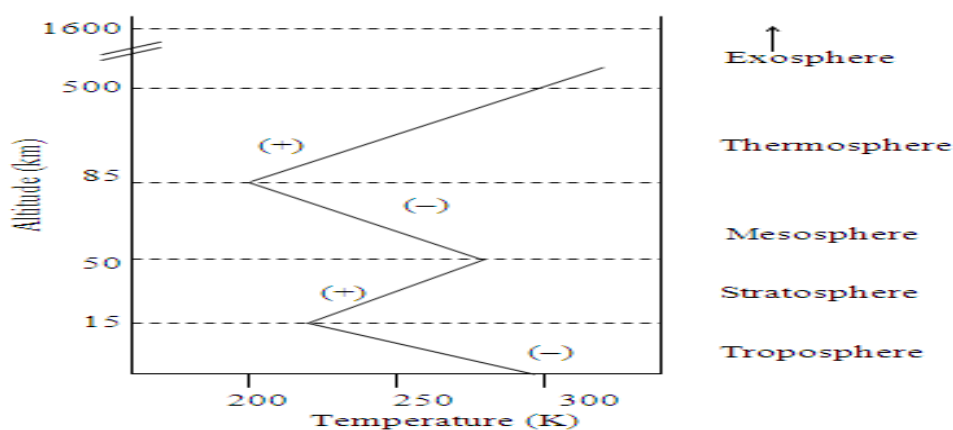
Atmosphere (ii) Hydrosphere (iii) Lithosphere and (iv) Biosphere

ATMOSPHERE

Atmosphere is the protective thick gaseous surrounding the earth, which sustains life on earth and saves it from unfriendly environment of outer space. It extends to a height of about 1600 kms from the earth's surface.

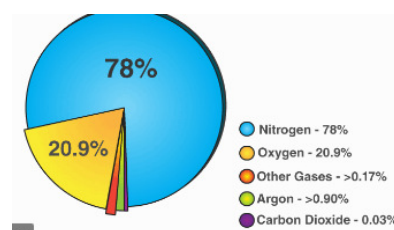
Atmosphere Structure

Depending upon the temperature, the atmosphere is further divided into five regions. These include Troposphere, Stratosphere, Mesosphere, Thermosphere and Exosphere.



Atmosphere components

The components of atmosphere are Nitrogen (78%), Oxygen (21%) and water vapour (0.15). The minor components are argon and carbon dioxide. Some trace elements present in the atmosphere are He, Ne, Kr, Xe, H₂, CH₄, O₃, NH₃, CO, NO_x, I, SO₂.



Region	Altitude (km)	T (°C)	Chemical species	Functions
Troposphere	0 – 11	15 to – 56 – ve laps	N ₂ , O ₂ , CO ₂ , water vapour, trace elements	Lowest layer in which organisms live. It is turbulent, dusty zone, containing much water vapours and clouds and thus the region of strong air movements and cloud formation.
Stratosphere	11 – 50	– 56 to – 2 + ve laps	Ozone, N ₂ , O ₂ and water vapour	Rich in ozone, which absorbs harmful UV radiations from the sun and acts as protective shield for life on earth.
Mesosphere	50 – 85	– 2 to – 92	O ₂ ⁺ , NO ⁺ , N ₂ , O ₂	Due to very low absorption of UV rays of ozone, the temperature is

		– ve laps		low.
Thermosphere	85 – 500	– 92 to 1200 + laps	O_2^+ , O^+ , NO^+	Gaseous components are ionized and form electrically charged particles. These particles reflect radio waves back to earth surface and enable to have wireless communication.
Exosphere	500 – 1600	above 1200 + ve laps	H_2 , He	Uppermost layer which has high temperature due to solar radiation. Earth's magnetic field becomes more important than gravity in the distribution of atomic particles.

HYDROSPHERE

At first life is originated in water. Hydrosphere covers more than 75% of earth surface. In the earth surface water availability,

97% - Oceans 2% - Glaciers and Ice caps

1% - Fresh water for Human consumption

Properties

- universal solvent
- high surface tension and high latent heat
- poor thermal conductor and has high viscosity, and buoyant medium

Uses

- 30% - Irrigation 12% - Industries 7% - Domestic purposes

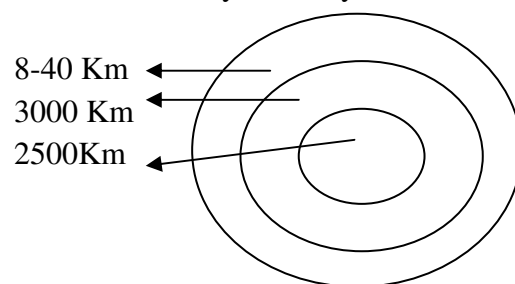
LITHOSPHERE

The solid component of the earth is called lithosphere. It contains mainly three layers.

Crust (outer solid zone covered with soil)

Mantle (molten state)

Core (central fluid or vapour of Ni and Fe)

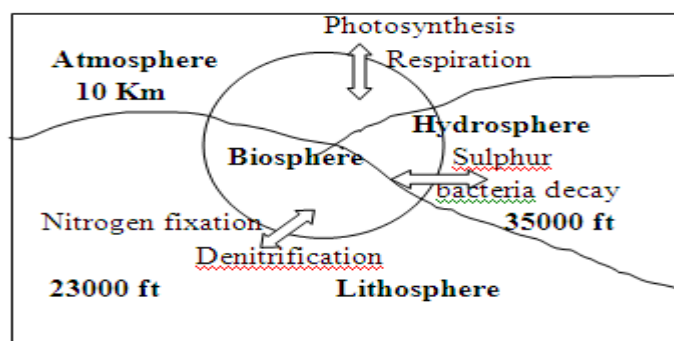


Properties and uses

- Soil useful for plant growth
- Store house of minerals reservoir of water
- conserver of soil fertility, producer of vegetative crops
- house of wild life
- Soil microbes bond in soil particles decompose the nutrients released in food chain
- Support for all terrestrial organisms

BIOSPHERE

Biosphere is the interaction of living organism with environment.



POLLUTION

Definition

(i) Pollution may be defined as “an undesirable change in the physical, chemical or biological characteristics of our air, water and land that may or will harmfully, affect human life, the lives of desirable species, living condition and cultural assets.

(ii) Pollution may be defined as addition or excessive addition of certain materials to physical environment (water, air, land) making it less fit or unfit for life.

Classification of Pollution

According to origin	According to environment	According to pollutant
(i) Natural Ex: Forest fire, Earth Quake, Volcanic eruptions (ii) Artificial Ex: Automobile exhaust, Industrial water	(i) Air pollution Ex: toxic effluents from power plants (ii) Water pollution Ex: waste from industries, sewages (iii) Soil or land pollution Ex: fertilizers, acid rain, Pesticides	Radioactive, Sewage, Pesticide, Thermal, Plastic, Smoke, Smog, Acid rain, Soap, detergent, Oil, Noise, Marine, Slit, Chemical, Oceanic, etc.

POLLUTANT

Definition

Pollutants are undesirable substances which are present in the wrong place at the wrong time in the wrong quantity.

Pollutant Classification

According to Form	According to Existence	According to Eco system
(i) Primary (emitted from source) Ex: SO ₂ , NO ₂ (ii) Secondary (derived from primary pollutants) Ex: PAN, PBN, PPN	(i) Quantitative (concentration reaches beyond threshold value) Ex: CO ₂ , NO (ii) Qualitative (do not occur in environment but are passed into it by human beings) Ex: DDT, Herbicides	(i) Bio-degradable (decomposed easily) Ex: domestic sewage, animal, plant debris (ii) Non- Bio degradable (decompose slowly or do not decompose) Ex: DDT, Plastics, Hg and Pb salts

FOREST RESOURCES

A resource is anything we get from the environment to meet our needs and desires. A forest is a biotic community, predominantly of trees, shrubs or any other woody vegetation usually with a closed canopy. Forest varies a great deal in composition and density and is distinct from meadows and pastures contribute substantially to the economic development of a country.

Types of Forest in INDIA

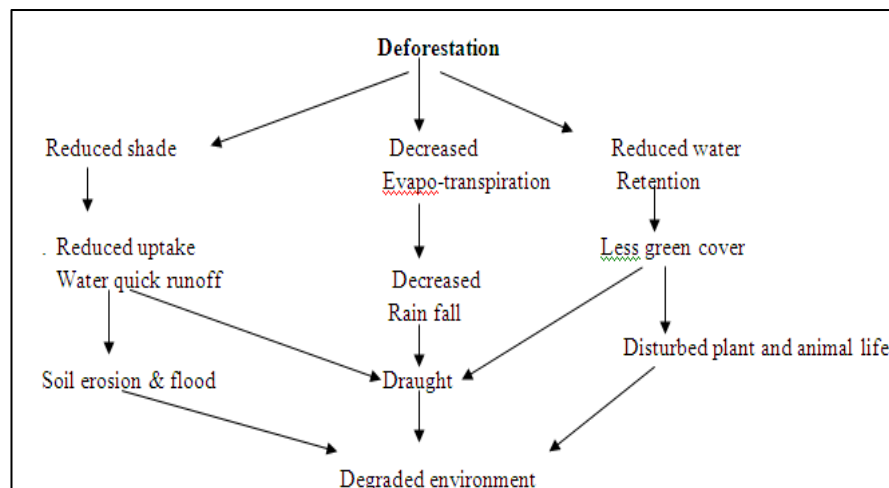
- (i) Moist tropical forest
- (ii) Dry tropical forest
- (iii) Montage sub-tropical forest
- (iv) Montage temperate forests
- (v) Sub-alpine forests
- (vi) Alpine scrubs

Importance of forests

- Forests - source of timbers, wood fuel, bamboos & other valuable products.
- Source of medical plants.
- Guard against erosion of land, damage of water sheds, foods and sedimentation.
- Shelters for variety of wildlife.
- Provide grazing for livestock.
- Sources of raw material for timber, charcoal, paper, resin, gum etc.
- Drugs, essential oils and herbs are extracted from trees.
- Maintains the temperature balance and helps in reducing global warming.
- Trees absorb noise and thereby reducing noise pollution.
- Forests reduce pollution by absorbing CO₂, particulate and other harmful gases.

DEFORESTATION

Removing the various wealth of forests is termed as deforestation. Forest in developing countries are undergoing deforestation and is a serious threat to economy, quality of life and future environment.



Causes of deforestation

- Rapid explosion of human and livestock population.
- Increased requirement of timber and fuel wood because of increase in population.
- Expansion of agricultural land.
- Enhancing grazing by cattle.
- Construction of dams, power stations, highways hydroelectric projects, opencast and limestone mines and other projects.
- Development of forest resource based industries.
- Fires and extreme weather conditions.

Effects of deforestation

- Soil erosion and flood formation
- Deserts formation
- Rain fall declines (or) less rainfall
- Loss of fertile soil
- less vegetation growth

- Climate of a region getting affected
- Loss of flora and fauna
- Loss of biodiversity
- Disturbances in ecological balance
- Increase in CO₂ concentration that leads to global warming
- Economic loss
 - Lowering of water table.

Measures to reduce deforestation

- Tree-felling should be matched with tree-planting programmes
- Use of fire wood should be discouraged.
- Use of other sources of energy like biogas, solar energy to be used.
- Awareness and understanding should be created among the people.
- Forest dwellers should have access to subsidized source of fuel, fodder, building material, A ban or freeze (**i.e. Moratorium**) of 15 to 20 years should be imposed.
- During moratorium, an extensive afforestation with people's participation should be followed.
- Environmentally sound action plans based on scientific research should be adopted.
- Community forest should be developed around villages.
- Protection of standing forests should be done.
- The national forest policy 1952 stated that 1/3 of geographic area of country should be under forest.
- Forest conservation act 1980 was enacted with a view to check indiscriminate deservation and diversion of forest land to non- forest purpose.

Forest conservation

- (i) Forest conservation and management program are based on two basic principles:
 - (a) Sustainable supply of tree products and services to people and industry
- (b) Maintenance of long time ecological balance through protection, restoration and conservation
 - (ii) Afforestation should be done in areas unfit for agriculture, along highways and rivers, around play ground and parks.
 - (iii) A special programme of trees plantation called **Vanamahotsava** is held in the month of **August & February** in our country. It should be made more popular & effective.
 - (iv) Forest should be protected from fire and pests by modern methods like modern fire fighting equipments and fumigation and aerial spray of fungicides
 - (v) Modern methods of forest management like irrigation, fertilizers, bacterial and mycorrhizal inoculation disease and pest management control of weeds, breeding of elite trees and tissue ultra techniques.

Afforestation programmes

Protection and conservation forestry

- (i) Protection of degraded forests
- (ii) Managed scientifically for producing forest without causing any negative environmental impact
- (iii) Forest areas are designated as National parks and Sanctuaries.

Production of commercial forestry

- (i) Intensive plantation of trees on available lands and land not under agriculture
- (ii) Fast growing trees are raised using modern techniques
- (iii) The production forestry programme includes:
 - (a) **Social forestry:** Social forestry programme involves rising of tree for fire wood, fodder, agricultural implement on public and common lands, roads, rail sides etc. A number of indigenous and/or exotic tree species which can meet the requirement of rural community are used for plantation in social forestry programme.
 - (b) **Agro forestry:** This programme aims at surviving an ancient practice of using the same land for farming and forestry. Two well known systems of agro forestry programmes **taunya system and jhum** cultivation.
 - **Taunya** system involves growing crops between rows of planted trees
 - **Jhaum or shifting cultivation** involves felling and burning of forest followed by cultivation of crops for few years, and abandoning cultivation to allow forest's regrowth.
 - (c) **Urban forestry programme:**

It aimed at growing ornamental and fruit trees in urban areas along roads, in private compounds, vacant lands and common parks. It has great aesthetic value for the citizens.

CHIPKO MOVEMENT

Chipko is a Hindi word which means embrace or to hug. This is the first and foremost movement against indiscriminate felling of trees in India was fought by bisnoi woman **Amrita Devi** of Khejarili village in Tehri Garhwal district in UP. It came into limelight in 1973 in Gopeshwar in Chamoli district when the local people hugged trees and prevented this cutting.

This movement continued under the leadership of **Sh.Sunder lal Bahuguna** (an environmentalist & jouranalist) and **Chandi prasad Bhatt**. They with village people under took a peace march about 3000 km, and the movement got success with the following out comes.

- (i) U.P government abolished the contractor system.
 - (ii) FDC (forest Development Corporation) was formed.
- Chipko movement flew to various parts of the country. In Karnataka this movement is known as **Appiko movement**. It has three objectives, **ulisu (to conserve)**, **belesu (to grow)**, **balasu (rational use)**.

ROLE OF INDIVIDUAL IN FOREST AND WILDLIFE CONSERVATION

Forest Conservation

- Don't cut trees
- Plant more trees
- Minimize the use of paper
- Minimize the use of wood as fuel
- Save plants which have medicinal value

Wild life Conservation

- Hunting should be banned
- Do not use skin of animals
- Minimize use of meat
- Rare animals and birds should be protected
- Don't encourage magic programmes with animals

OVER GRAZING

Over grazing is the process of eating away the forest vegetation without giving chance to regenerate. Grassing animals (cattle, horse, etc) play a significant role in rural life and food to urban areas of our country.

Effects:

- Reduction in the diversity of plant species
- Due to loss vegetation, the soil is eroded by water
- Not suitable for cultivation
- Regeneration capacity also gets affected
- Desertification takes place.
- Soil moisture is removed
- Loss of soil structure and soil fertility
- Affects the regeneration of the grass
- Decrease in the recycling of organic compounds

DESERTIFICATION

Conversion of fertile land into a desert by natural or man-made activities is called desertification. It is also a form of land degradation. This is characterized by devegetation, depletion of ground water, salination and soil erosion.

Causes

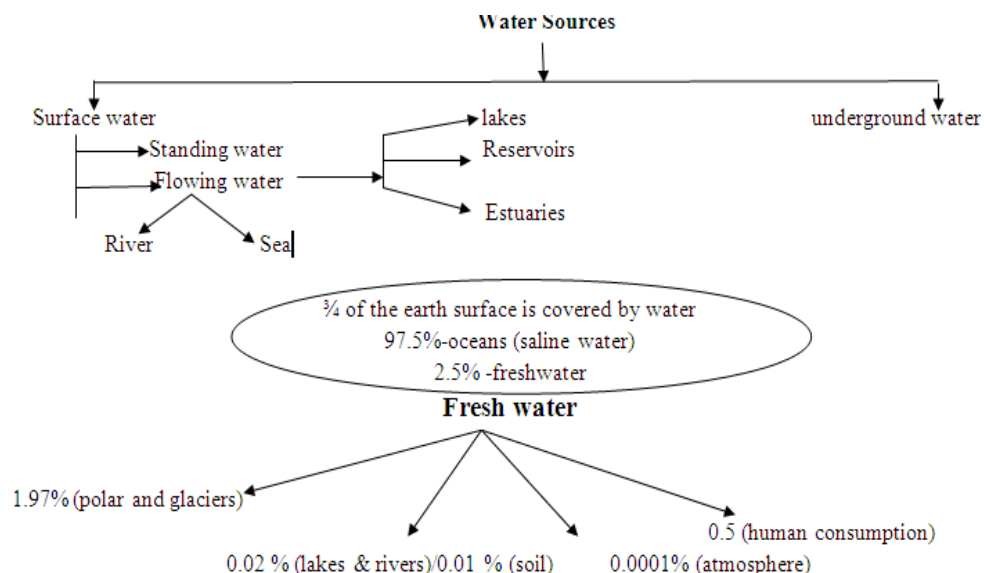
- Deforestation and Overgrazing
- Mining
- Climatic change
- Overutilization of ground water
- Disposal of toxic wastes
- Excess use of fertilizers and pesticides

Control measures

- Restoring forests and gross covers
- Mulching (use of artificial protective covering)
- Growing plant species
- Rising trees as wind breaks

WATER RESOURCES

Water is an indispensable natural resource on this earth on which all life depends. It is main constituents of hydrosphere and it is a renewable resource.



The water we use keeps on cycling endlessly through the environment which is called as **hydrological cycle**. The chief sources of water supply are ground water, surface water, sea water and rain water. **Over exploitation of water**

Human beings depend on water for almost every developmental activity. Water is used for

- Drinking
- Transportation
- Disposal of industrial wastes
- Irrigation
- Washing
- As coolant

A rapid rise in population and expansion in industry and agriculture have increased the demand for water manifold and use of water has increased 4-8% per year.

Problems related with water resources:

- (i) About 40% of the people lives in arid and semi arid regions. These people have to spend substantial amount of time, effort & energy in obtaining water.
- (ii) To meet the requirement of population, surface waters are over drawn and drying up of wetlands.
- (iii) Ground water also dries out because of over removal.
- (iv) Excessive irrigation in semi arid and arid region cause salt accumulation in soil.
- (iv) Depletion of ground water along coastal areas lead to the movement of saline water into freshwater, Estuaries also became more saline.
- (v) Heavy rainfall on soils results in rapid runoff causing soil erosion.

FLOODS

- Inundation of large land area with water for several days in continuation is called flood.
- Floods have been regular feature of some parts of **India and Bangladesh** causing huge loss of agriculture, life & property.
- In India, **Assam, West Bengal, Bihar, Orissa and UP** are worst affected states.

Causes of floods

- Heavy rainfall
- Melting ice or snow
- Exceeds the carrying capacity of receiving river system

Control measures

- Flood banks are built along the rivers
- Building dams upstream
- The management of rainfall and resultant runoff is very important (based on natural unit called watershed). **Watershed** is an area bounded by the divide line of water flow
- Networking of rivers is being proposed

DROUGHT

The conditions of dryness for prolonged period is called drought. It results when the average rain fall for an area below normal. Drought causes famine and starvation of human and animal population

Causes

- a. Deforestation b. Overgrazing c. Mining

Control measures

- Afforestation programmes
- Dry farming techniques and water conservation schemes

CONFLICT OVER WATER

Indispensability of water and its unequal distribution has often lead to interstate or international disputes. It is foreseen that water could be the source of the world's next big conflict if state or nation do not start co-operating to share their resources.

International Conflicts

- India & Pakistan fight to share water from **Indus**
- Mexico & USA – **Colorado** river
- India & Bangladesh – **Brahmaputra** river
- Iran and Iraq- **Shatt-al-Arab** water.

National Conflicts

- Sharing of **Cauvery** water by Karnataka & Tamilnadu
- **Krishna** water by Karnataka & Andhra Pradesh
- The **Bhavani** water by Tamilnadu and Kerala
- **Sutlej – Yamuna** water by Punjab and Haryana

The water crises of India have been solved **by inter linking of rivers**. The linking of **Ganga and Cauvery** has been an agenda for many years. The inter linking aimed to resolve problems of drought and water shortage. Recently on the direction of Supreme Court of India, set up a task force to work out the process of linking **Himalayan river with Peninsular rivers**. This mega project estimated to cost 560,000 cores.

Conservation and Management of Water

- (i) Traditional method of irrigation (sprinkle irrigation, drip irrigation)
- (ii) Recycling of used water in industries
- (iii) Reduction of domestic water wastage by waste water treatment plants
- (iv) Rain water harvesting
- (v) Production of water sheds and afforestation

Water management approaches

- Construction of dams & water reservoirs
- Desalinization of sea water & saline ground water
- Diversion of water bodies through canals
- Regular dredging and desiltation of rivers, stream and other water bodies.

DAM – Benefits and Problems

River valley projects with big dams play a key role in the development process. India has the distinction of having the largest number of rivers valley projects.

Benefits:

- Hydro power generation
- Agricultural development/promote navigation, fishery
- Recharge ground water
- Wild life population increases in dams surroundings
- Potential for economic upliftment and growth
- Help in checking floods and famines
- Generate electricity and reduce water and power shortage

Problems of dam construction:

- Submerge forests
- Displace local people
- Cause water logging and siltation and may result in earthquake
- Stagnation and water logging near reservoir
- Breeding of vectors and spread of vector-borne diseases
- Micro climatic changes / Flash floods
- Loss of land fertility along river since sediments carrying nutrients get deposited in reservoir.

We need water throughout the year, but the main source of water, rain falls only during a period of 3-4 months in a year and 80-85% of water lost as runaway water. Thus to meet country's requirements this is the only possible way through dam building at appropriate sites.

ROLE OF INDIVIDUAL IN WATER CONSERVATION

- Don't keep water taps running while brushing, shaving, washing or lathing
- In washing machines, fill the machine only to the level required for your clothes
- Install water-saving toilets that use not more than 6 liters per flush
- Check for water leaks in pipes and toilets and repair them properly. A small pin-hole sized leak will lead to the wastage of 640 liters of water in a month
- Reuse soapy water of washing for washing off courtyards, drive ways
- Water the plants in your kitchen garden and the lawns in the evening when evaporation losses are minimum. Never water the plants in mid-day
- Use drip irrigation and sprinkling irrigation to improve irrigation efficiency and reduce evaporation
- Install a small system to capture rain water and collect normally wasted used water from sinks, cloth-washes, bath tubs which can be used for watering plants
- Build rain water harvesting system in your houses. Even the president of India is doing this.

MINERAL RESOURCE

Minerals are exhaustible, non-renewable resources found in the earth crust. **Mining** is the act of extracting ores, coal, etc from the earth.

Minerals:

- (i) Metallic (bauxite, Hematite)
- (ii) Non metallic (Quartz, Diamond)

Uses

- Development of industrial plants
- Generation of energy (e.g.) Coal, Lignite,
- Construction, housing, settlements
- Defense equipments-weaponry, ornaments
- Transportation means
- Communication –telephone wires, cables, electronic devices
- Medicinal system- Ayurvedic
- Agriculture – as fertilizers, seed dressing & Jewelry

Major Minerals available in INDIA

Mineral (Metallic)	Place
Coal & Lignite	West Bengal, Orissa, Jharkhand, MP, AP
Uranium (Pitch blend/Uranite ore)	Jharkhand, AP (Nellore, Nagaonda), Meghalaya, Rajasthan
Aluminum (Bauxite)	Jharkhand, West Bengal, Maharashtra P, Tamilnadu
Iron (Hematite & Magnetite)	Jharkhand, Orissa, MP, AP. Tamilnadu, Karnataka, Goa
Copper (Copper Pyrites)	Rajasthan Bihar, Jharkhand, Karnataka, MP, West Bengal, AP,
Minerals (Non-Metallic)	Place
Diamond	MP
China Clay	Rajasthan & West Bengal
Calcite	Rajasthan & Gujarat
Asbestos	Rajasthan & Bihar
Sulphur	Tamilnadu
Mica	Bihar & AP
Gypsum	Rajasthan & Tamilnadu
Feldspar	Rajasthan & Tamilnadu

ENVIRONMENTAL IMPACTS AND EFFECTS OF MINING

The mining, processing and disposal of minerals have negative effects on the environment. The main harms are,

(i) Devegetation and defacing of Landscape

The top soil is removed from the mining area. It disturbs and damages the land.

(ii) Ground water contamination and surface water pollution

Mining disturbs the natural hydrological process and also pollute ground water & surface water.

(iii) Subsidence of land

Mining results in tilting of buildings, cracks in houses, buckling of roads, bending of rail tracks and leaking of gas from cracked pipelines leading to serious disasters.

(iv) Air pollution

Extraction and processing of ores emits enormous quantities of air pollutants. Ex: SPM, soot, metal particles etc.

(v) Occupational health hazards

Miners often suffer from serious respiratory, skin diseases, and lung diseases.

Remedial measures

- Eco-Friendly mining technology.
- Low – Grade ores can be better utilized by using microbial leaching technique (Thiobacilleous ferrooxidanes used for extracting gold embedded in FeS ore). The ore are inoculated with bacteria to remove impurities.
- Restoration of mined areas by re-vegetation.
- Prevention of drainage and conform the standards of air emission.
- Recycling process.
- New technologies to recycle them.
- In order to reduce the demand for pure metals, alloys should be developed.
- Alternate Non-Conservational energy sources, such as Wind, Nuclear, Hydro energies are developed

SOIL EROSION

Soil erosion is removal of top fertile soil by water, wind, oceanic waves and glaciers.

Causes

- (i) Continuous flow of water
- (ii) Wind action
- (iii) Deforestation
- (iv) Landslides

Effects of Soil erosion

- Protection of soil from impact of rain drops
- To slow down the water from concentrating and moving down the slope in narrow path
- Slow down the water movement, when it flows along the slope
- To encourage more water to enter the soil
- Increase the size of the soil particles
- Reduction in the wind velocity near the ground by growing vegetation cover and ridging the land
- To grow the strips of stubble or other vegetation cover, this might catch and hold the moving
- particles of the soil. Keeping in view the above principle ecologists have derived several methods.

Role of Individual in Soil conservation

- While constructing your houses, don't uproot the trees as far as possible. Plant the distributed areas with a fast growing native ground cover
- Grow different types of ornamental plants, herbs and trees in your garden .Grow grass in the open areas which will bind the soil and prevent its erosion
- Make compost from your kitchen waste and use it for your kitchen garden or flower pots
- Do not irrigate plants using a strong flow of water, as it would wash off the soil
- Better use sprinkling irrigation
- Use green manure and much in the garden and kitchen garden which will protect the soil
- If you own agricultural fields, do not over irrigate your fields without proper drainage to prevent water logging and salination
- Use mixed cropping so that some specific soil nutrients not get depleted.

FOOD RESOURCES

The main sources of human food are plants and animals. About 76% of food comes from the crops mostly from cereal and the remaining from grazing livestock and fishes. **The food and agriculture Organization (FAO) of UN** estimated that on an average the minimum caloric intake on a global scale **2,500 cal/day**. People receiving less than 90% of these minimum dietary calories are **undernourished** and if it is less than 80% means **seriously undernourished**.

Malnutrition

Besides the minimum caloric intake we also need proteins, minerals etc. Deficiency or lack of nutrition often leads to **malnutrition** result several diseases.

Mal nourishment may lead to major problems, such as,

Disease	Deficiency	Effect
Marasmus	lack of protein	progressive emaciation
Kwashiorkor	lack of protein	failure of neural development in child
Anemia	lack of iron	inability to absorb Fe
Goiter & Hypothyroidism	lack of Iodine	cause of developmental abnormalities
Pellagra	Lack of Vitamins.	deficiency of tryptophan & Lysine
Chronic hunger		people have enough food to stay alive but not enough to lead satisfactory and productive lives

World food problems

- World population is growing every year and so the demand of food is also increasing constantly. Although world food production has increased almost three times, rapid population growth especially in LDC (less developed countries) has out stripped the food production.
- The world food summit 1996 has set the target to reduce the number of under nourished to half by 2015.
 - The FAO estimate 840 million people are remain chronically hungry. In India 300 million people are food insecure and poverty stricken.

Reasons for world food problems

- Only less percentage of land is fit for cultivation
- The food grains like rice, wheat, corn and vegetables are staple foods of people all over the world
- Soil erosion, water logging, water pollution, affect agriculture
- Urbanization – deterioration of agricultural lands
- Imbalance among supply of food
- Demand for food in developing countries
- Poverty
- Environmental effects
- Economic affairs
- Political systems
- Environmental ethics.

Food problem in India

(i) Quantitative aspect of food problem

16% of world population & only 2% agricultural area

(ii) Qualitative aspect of food problem

Nutritious food per day per person -3000 calories.

In India, 2000 cal, Canada 3060 cal, America 3090 cal, UK 3290 cal, Japan 2280 cal.

Reasons for unbalanced diet

- Less production of protective food
- Lack of money for the purchase of nutritious articles
- Meat, fish and egg are not used due to religious causes
- No knowledge about balanced diet

Distributive aspect of food problem:

The policies adopted by the government for food procurement, price control & sales are not effective due to completion in government officials. Black marketing by trades is next problem.

Economic aspect of food problem:

The price of food grains are soaring high while income has no proportionate increase. Hence poor people are unable to avail the meals for two times a day.

Environmental impacts of modern agriculture:

- It makes use of **hybrid seeds** of selected and single crop variety.
- Lots of energy subsidies in the high tech equipments and lots of energy subsidies in the form of fertilizers, pesticides, and irrigation water.
- The food production has increased tremendously, evidenced by “**Green revolution**”, however it also gave rise to several problematic off-shoots.

Impacts related to high yielding varieties: (HYV)

- (i) Encourages **monoculture** (i.e.) the same genotype is grown over vast areas.
- (ii) In case of an attack by some pathogen, then total devastation of crop due to uniform conditions which helps in rapid spread of the disease.

Fertilizers:

In early days farmers used to supplement the soil with organic manure in order to increase the crop yield now farmers are using more fertilizers than the crop requires, as a result great amount of them is leached out of soil and contaminate the sub-soil water resources. These are non-biodegradable and create environmental problems.

Effects:**(i) Micronutrient imbalance**

Farmers use the macronutrients (N, P, K) indiscriminately to boost up crop growth. Excessive use of this cause micronutrient imbalance. For example in Punjab & Haryana has caused deficiency of the micronutrient Zn in the soils, which affects soil fertility.

(ii) Nitrate pollution (blue baby syndrome)

Nitrogenous fertilizers applied in the fields often leach deep into the soil and ultimately contaminate the ground water. When the concentration exceeds 25 mg/l, they cause the serious health hazard **Blue baby syndrome (or) methemoglobinemia**. This affects infants to the maximum extent. In Denmark, England France, Germany, Netherland & India the problem exists.

Impact of phosphates (Eutrophication)

Excessive use of N and P in fertilizers particularly rich in Phosphate affects water bodies like river, lakes, ponds, etc. The nutrients N & P get washed from the fields and along with run of water reach the water bodies causing them to become over nourished. This over nourishment gives way to growth of undesirable aquatic plants. As a result water bodies get affected by harmful algae. This is known as **algal bloom or eutrophication**.

The algal species quickly complete their life cycle. So the oxygen is consumed in the process of decomposition and very soon the water gets depleted of dissolved oxygen.

Effects of using pesticides

Pesticides are substances used to kill pest in order to improve the crop yield. In yearly days farmer used to control pest by using salt, smoke and insect repelling plants. But now a day first generation pesticide S, Ar, Pb, Hg and second generation pesticide like DDT are used. These pesticides protect our crops at the same times, they have serious side effects.

Creating resistance in pest and producing new pests

Some individuals of pest species usually survive even after pesticide spray. The survivors give rise to highly resistant generations. 20 species of pests are now known which are immune to all types of pesticides and known as “**super pests**”

Death of non–target organisms

Many insecticides are broad spectrum poisons which not only kills the pests but also non–target species which are useful to us.

Biological magnification

Many of the pesticides are non bio-degradable and keep on accumulating in the food chain. This process is called biological magnification.

WATER LOGGING

Over irrigation of croplands leads to water logging. This gradually forms continuous column with water table. Under water logged conditions pores–spaces in the soil air gets fully drenched with water and the soil air gets depleted. So the roots of the plants do not get air for respiration. In Punjab & Haryana this problem exists.

Reasons for water logging

- Heavy rain
- Poor drainage
- Poor irrigation management
- Over-watering- Excess water supply to the land

Effects of water logging

- Poor soil quality
- Water pollution
- Growth of plants and trees affected
- Vegetation can turn to yellow and the growth is stunted and thin
- Bare patches of soil appear

Remedy for water logging

- Perverting excessive irrigation
- Sub surface drainage technology
- Bio-drainage with trees like eucalyptus
- Only apply water when necessary
- Increase deep rooting vegetation
- Proper drainage and management

SALINITY PROBLEMS

Saline soils are accumulation of soluble salt like NaCl, Na₂S, CaCl₂, MgCl₂ in the soil profile. This problem exists in India and all so all over the world. EX: Punjab & Haryana

Causes:

- Excessive irrigation
- Canal water /ground water irrigation
- Poor drainage / Use of excess fertilizers
- Flood
- Evaporation / Transpiration

Effect:

- Stunted plant growth / Lowers crop yield
- Some time plants are destroyed
- Land become impotent & no crop can be cultivated
- Loss of fertility
- Loss of productivity & Loss of economy
- Pollution of land and water

Remedy:

- Flush them out by applying good quality water
- Underground network of perfected drainage pipes for flushing out salts

The control soil salinity research institute (CSSR) at Karnal, Haryana has the success story of converters Zarifa Viram village to Zarifa Abad (i.e.) from the barren land to productive land.

Role of Individual in Food Conservation

- Take minimum amount of food
- Don't cook food
- Don't store large amount of food grains
- Protect food items from pests and insects
- Avoid over eating

ENERGY RESOURCES

Energy is defined as the “capacity to do work”. Energy is available on the earth in number of forms.

Development of energy

Wood → coal → oil → alternate energy

Growing energy needs

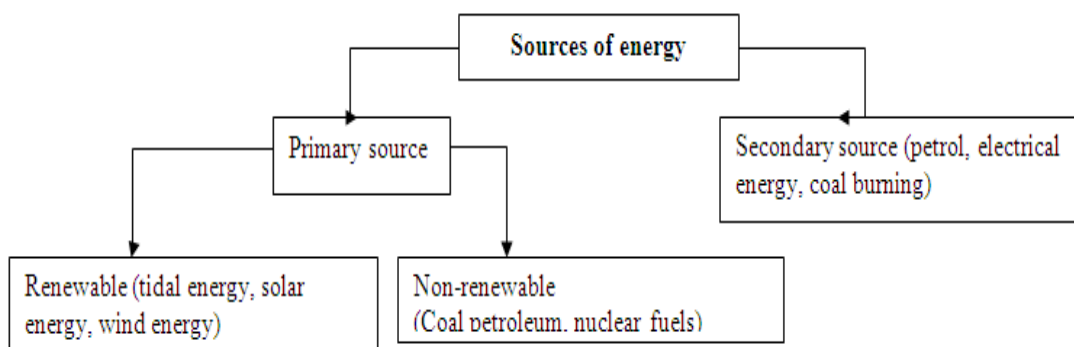
Development in different sectors relies largely upon energy. With the demands of growing population the world is facing further energy deficit.

Energy distribution –world scenario

Category	Countries	Popul ation	Energy used
Developed	USA, Canada	5%	25%
Developing	Bhutan, India, Nepal, Ethiopia	High	Less

T

the fossil fuels supply 95% of commercial energy of the world resources and are not going to last for many more years. Our life style is changing very fast and from a simple way of life to a luxurious life style.



RENEWABLE AND NON-RENEWABLE ENERGY RESOURCES

(a) Non-renewable energy resources

Non-renewable energy resources are natural resources which cannot be regenerated once they are exhausted. There are some non-renewable energy resources, They are,

- Coal
- Petroleum
- Natural gas
- Nuclear energy

(i) Coal

Coal is a solid fuel which is found inside the earth crust where they have been formed coal through heat and pressure over millions of years. It contains carbon, sulphur and nitrogen. India has 5% of world coal. It is not good because of poor heat capacity. China is the leading producer of coal and USA second place. Coal is used for industries, cooking heating thermal power plants, etc.

Coal Types:

- (i) Anthracite (90% of carbon)
- (ii) Bituminous or soft coal (80% carbon)
- (iii) Lignite or Brown Coal (70% carbon)

Disadvantages:

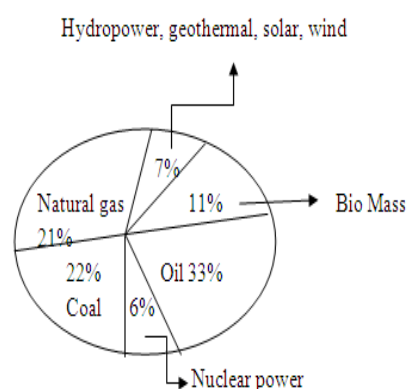
- Produces CO₂ when burnt and causes global warming
- Coal contains impurities like S and N produces toxic gases during burning

(ii) Petroleum

Petroleum or crude oil is a thick liquid. It is a complex mixture of alkaline hydrocarbon and small amount of S, O and N as impurities.

Occurrences:

Petroleum formed by the decomposition of dead plants and animals that were buried under lake and at high temp and pressure for millions of years. Hence it has to be



purified and refined by fractional distillation. During distillation at various temperatures, we get various products like petroleum gas, petrol, diesel, fuel oil, paraffine oil, etc.

(iii) Natural gas

It is a mixture of 50-90% of CH_4 and small amount of the other hydro carbon. Its calorific value range from 12,000-14,000 kcal/m³. It is used as domestic and industrial fuel

(iv) Nuclear energy

It is obtained through nuclear fission or fusion reaction of selected radioactive materials which yield large amount of heat and energy.

(b) Renewable energy resources / non –conventional energy sources / alternate energy resources

- These sources can be regenerated continuously in nature.
- They are available in great amount in nature and regenerate in a relatively short period of time.

Wind energy resources:

High speed winds have lot of energy. Energy recovered from high speed of the wind is called wind energy.

Wind mills:

The blades of the wind mill keep on rotating continuously due to high speed of the wind. The rotational motion of the blades drives number of machines like water pump, electric generators etc.



Wind form:

Large number of wind mills are installed and join together to form wind farm. It produces electricity.

Advantages:

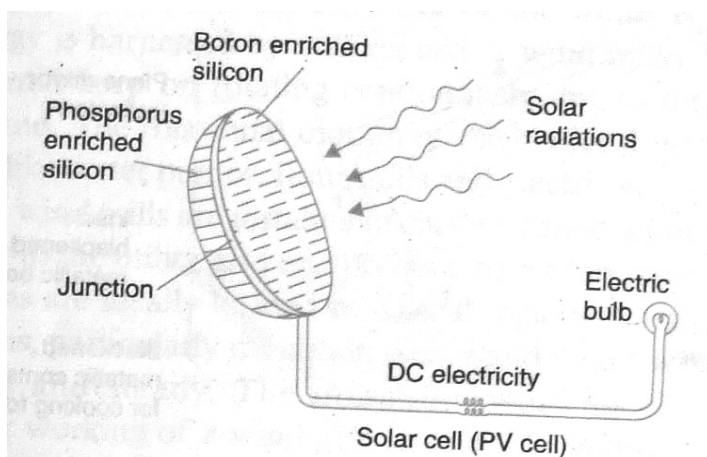
- Very cheap
- Doesn't cause any pollution

Solar energy resources:

This is the most important and easily available energy. The energy we get directly from the sun is called solar energy. Some of the solar energy harvesting devices are,

Solar cells (or) Photovoltaic cells (PV cells) Solar cell

It contain p-type semiconductor (Si doped with B) and n-type semi conductor (Si doped with P) they are in close contact with each other when solar rays fall on the top layers of p-type semi conductor, the electrons from the valance band get promoted, to conduction band the potential difference causes flow of electrons.



Uses: Used in street light, calculator, run TV, radio and water pumps.

Solar battery:

A group of solar cells joined together in a definite pattern which form a solar battery. It produces more electricity. It is used to solve conventional electricity problems in remote areas.

Some other devices using solar energy are,

- Solar heat collectors
- solar water heater
- solar furnace
- solar power plant
- solar cookers

Ocean energy resources:

Use the sea's energy (either wave action or tidal flow) to generate electricity.

Ocean thermal energy:

Energy available due to the difference in temperature of surface of tropical oceans and at deeper level called ocean thermal energy.

Uses:

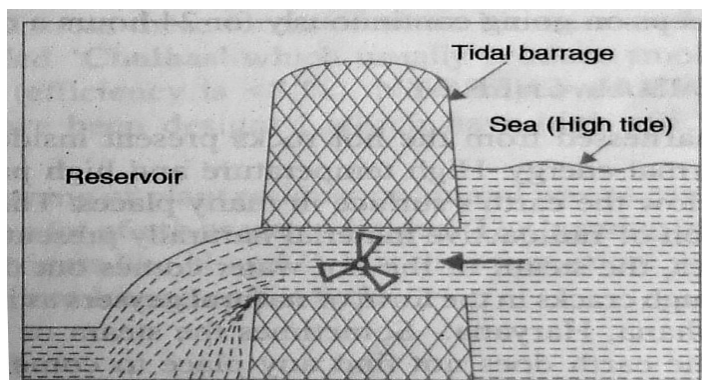
The warm surface water used to boil NH_3 and cool deep water used to condense the vapour into liquid. This is used to rotate the turbine.

Tidal energy:

Lot of energy is inherent in the process of rise and fall of the tides. Scientists have tried to utilize this tidal energy which is external and pollution free.

Uses:

Rotate turbine & generate electricity.



Geo-thermal energy resources:

In volcanic regions of earth, heat from the molten inlet offers a huge and sustainable energy source. This is known as geo-thermal energy. In some places, the heated water comes to the earth surface as hot springs. This natural heat comes from the fission of radio-active material present in the rocks in the interior the earth.

Uses:

- (i) Produces electricity
- (ii) For space heating/boiling water
- (iii) Turbo generators

Hydropower energy resources:

Hydro electric energy is produced from kinetic energy of water falling from a height. Hilly and high land areas are suitable for this purpose.

Hydropower energy

Water falling from a height turns turbines at the bottom of dams to generate electricity.

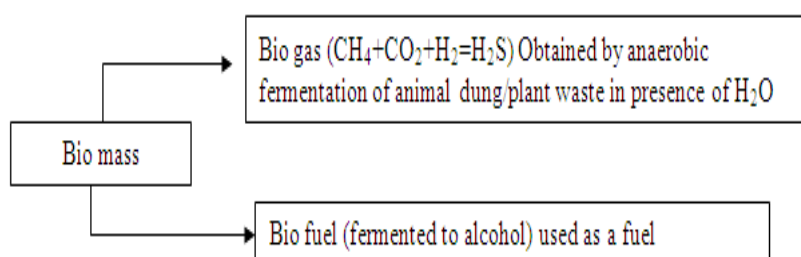
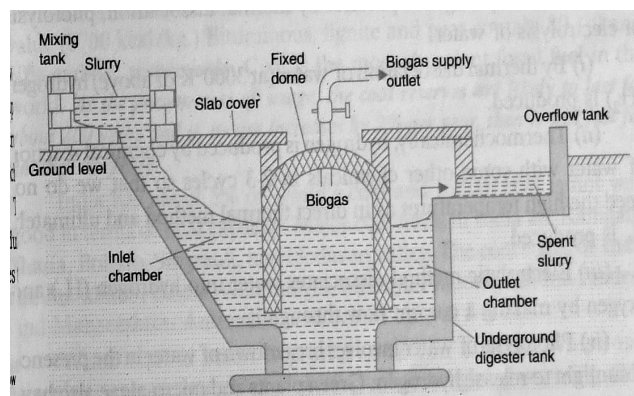
Advantages: Economically cheap.

Bio mass energy

All forms of biological matter living and non living that stores biochemical energy is termed as biomass. It includes plant and animal waste, wood, crop residues cattle dung, manure, swage, agricultural waste, etc.

Conversion of biomass:

Thermo chemical process- combustion, pyrolysis, Liquefiers. Biological process – enzymatic, anaerobic/aerobic process



Advantages:

Low cost, Supplied directly no storage problems

ROLE OF INDIVIDUAL IN ENERGY CONSERVATION

- Turn off lights, fans and other appliances when not in use
- Obtain as much heat as possible from natural sources. Dry the clothes in sun instead of drier if it is a sunny day.
- Use solar cooker for cooking your food on sunny days which will be more nutritious and will cut down on your LPG expenses
- Build your homes with provisions for sunspace which will keep your house warmer and will provide more light
- Grow deciduous trees and climbers at proper places outside your home to cut off intense heat of summers and get at cool breeze and shade. This will cut off your electricity charges on coolers and air-conditioner. A big tree is estimated to have a cooling effect equivalent to 5 air conditioners. The deciduous trees shed their leaves in winter therefore they do not put hindrance to sunlight and heat.
- Drive less make fewer trips and use public transportations whenever possible. You can share by joining a car-pool if you regularly have to go to same place.
- Add more insulation to your house during winter close the windows at night. During summer close windows during days if using an A.C otherwise loss of heat would be more, consuming more electricity.
- Instead of using the heat convertor more often wear adequate woollens.
- Recycle and reuse glass, metals and paper.
- Try riding bicycle or just walk down small distances instead of using your car or scooter.
- Lower the cooling load on air conditioner by increasing the thermostat setting as 3-5% Electricity is saved for every one degree rise in temperature setting.

FROM UNSUSTAINABLE TO SUSTAINABLE DEVELOPMENT

- ❖ The practices of the past, if continued further, will not allow sustenance of life on earth, more particularly human beings. We have already taken too much from nature, and caused serious damage to the environment.
- ❖ Nature's regenerative capacities have been progressively weakened, bringing deterioration day by day. In order to safeguard the sustenance of life and future of the humanity, we have to change our approach from unsustainable to sustainable development.
- ❖ Economic policies and developmental activities should be planned in such a way that they shall not hamper the environment any more. Similarly, over-consciousness about the environment should not affect the economic and other developmental activities. **Developmental activities towards betterment of humanity and environmental safety should go hand in hand.** A judicious balance between developmental activities and environmental protection should be ensured. It is possible only through sustainable development.

What do we mean by sustainable development?

- **World Commission on Environment and Development (WCED)** (Our common Future, 1987, also called **Brundt land report**) identified and named the **“sustainable development, which is the development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”**
- This concept of sustainable development has linked environmental care with development. The idea underlying this concept is to improve the quality of human life while living within the carrying capacity of the supporting ecosystem. Man has to practice sustainable utilization of natural resources.
- The deviation from unsustainable to sustainable development marks the dawn of awakening and wisdom of human beings to the problems caused by their own past reckless actions.
- Regulations, management, surveillance and accountability should become part of our life in dealing with environment.
- Eco development, that is maintenance of ecological balance, should happen quickly. People should cultivate 'environment thinking' while embarking on development.

What should be done to achieve sustainable development without affecting the environment?

Monitoring the following is required

- Climatic changes
- Biodiversity
- Disposal of pollution
- Generation of industrial effluents

According to **Dr. M. S. Swaminathan**, the following nine elements characteristic should be taken care of to promote ecologically sound agriculture:

Land	Water
Energy	Nutrient Supply
Genetic diversity	Pest management
Post harvest system	Systems approach
Location specific research and development	

- A production efficiency of minimum environment damage through energy transition should be commenced
- For a stable world population .A demographic transition should be attempted.
- There should be a clear shift in resource transition from non-renewable to renewable.
- There should also be political transition for global objectives among developed and Developing Nations.

Issues concerning Food

Food problem (related to growing population) has to be tackled by introduction of high-yielding varieties, development of aquaculture, fish farming, etc.

Issues concerning Energy

To tackle energy crisis, the shift has to be towards non-conventional sources like solar, wind, tidal, etc.

Issues concerning pollution

Steps to minimize pollution are of urgent need.

QUESTIONS

PART -A

1. Define environment. Mention the components of an environment.
2. Mention the different atmospheric layers.
3. What are the functions of lithosphere?
4. What is atmosphere? What are its functions?
5. What is hydrosphere?
6. Define Biosphere?
7. Define pollution. How it is classified?
8. Classify Pollutants.
9. What are primary and secondary pollutants? Give example.
10. What are the environmental effects of mining?
11. Define Overgrazing.
12. What is meant by Soil erosion?
13. What is desertification? Give two reasons for it.
14. Give any two benefits and problems caused by construction of dams.
15. What is meant by resources?
16. Differentiate renewable and non-renewable energy resources.
17. What are the effects of deforestation?
18. What are the environmental impacts of ground water usage?
19. What is Sustainable development?
20. What are the advantages of rain water harvesting?
21. What is meant by energy plantation?
22. Define food security.
23. List out the urban problem related to energy.
24. Give an account of water crises and conflicts over water.
25. What is eco-friendly farming and organic farming?
26. Discuss the measures to conserve water.
27. What are the impacts of modern agriculture?
28. What is Bio-accumulation Or Bio-magnification?

29. What is meant by Eutrophication?

PART-B

1. Write note on chemical and photo chemical reactions in atmosphere. **(Jan 11)**
2. Describe the various structures of atmosphere and their functions. **(Jan 11)**
3. List out the environmental effects of extracting mineral resources. **(Jan 11)**
4. What is deforestation? What are the effects of deforestation? **(Jan 15, May 14)**
5. Explain the adverse environmental impacts of modern agriculture. **(Jan 15)**
6. Discuss how the big dams have affected forests and tribal people. **(Jan 15)**
7. Explain how the alternate energy sources play an important role in environmental impact? **(Jan 15)**
8. Explain types of renewable and non-renewable energy resources. **(Jan 14)**
9. What are the effects of over utilization ground water? **(Jan 14, May 14)**
10. Explain the following: (a) water logging (b) Salination (c) Floods (d) Over grazing (e) Eutrophication (f) drought (g) Blue baby syndrome. **(May 14)**
11. Explain environmental causes of land degradation. Give its effects. **(May 13)**
12. Give an account of non-renewable energy resources & its significance. **(May 14)**
13. What are the effects of over utilization of fertilizers? **(Jan 13)**
14. What is solar energy? Describe methods of harvesting solar energy. **(May 11)**
15. Explain about world food problems and changes caused by overgrazing and agriculture.
16. What is the difference in resource utilization between developed and developing countries?
17. How Bhopal tragedy connected with agriculture?
18. Enumerate the desired qualities of an ideal pesticide.
19. What are the advantages of organic farming over the conventional one?
20. As an individual, in what way you can contribute in conservation of nature in the following: (a) Saving energy (b) Reducing waste (c) Lowering your forest impacts (d) Reducing toxic substances. **(Jan 11)**
21. Discuss equitable use of resource for sustainable life styles.
22. What do mean by sustainable development? What are the major measures to attain sustainability?
23. Differentiate between (i) deforestation and desertification (ii) forest degradation and deforestation (iii) Afforestation and agro forestry
24. Discuss the problems of exploitation of natural resources
25. What is the importance of forest resources?
26. In what way forests are immense value to us? Justify answer with 10 points.
27. Discuss the forests statistics of world and India.
28. How population pressure has affected forest wealth of India.
29. Mention some major projects which has degraded the forests in India
30. What are steps taken by the government of India to tackle deforestation problem
31. Write note on Sardar Sarovar Narmada project, Andolan movement, Chipko movement.
32. What are the methods of conserving water resource.
33. What are the environmental impacts of ground water usage?

34. Should we go for big dams or small ones? Give argument in favor of your answer.
35. How water is important to human body. Give any five points.
36. Mention few world's largest rivers
37. What is wet land? What are its uses? Give example in India
38. Why dams necessary? Justify with 10 points
39. Define meteorological droughts
40. "Water could be the source of the world's next big conflict" Explain.
41. What are the impacts of river valley projects?
42. Explain the values of minerals as natural resources.
43. How can mineral resources can be conserved?
44. What is jhum cultivation and taungya system of agro forestry?
45. Discuss the food distribution around the world and their problems.
46. List any five pesticides used in modern agriculture. What are their impacts on the environment?
47. Write note on (a) world food problem (b) malnutrition (c) undernourishment (d) over nutrition (e) nutritional deficiency diseases. (f) balanced diet (g) Hunger hot spots of the world
48. Write note on (a) superpest (b) shifting cultivation (c) Blue baby syndrome
49. Write short note on (a) biomass energy (b) nuclear energy (c) solar energy (d) wind farms (e) wind energy (f) geothermal energy (g) solar cell (h) Biogas plant (i) hydro energy (j) tidal energy
50. Compare and contrast nuclear power with solar power from the environmental point of view.
51. Compare various types of energy with respect to its suitability for Indian conditions.
52. Comment upon the types of energy harnessed from oceans
53. Give the energy scenario in Indian context.
54. Explain the solar energy trapping and application.
55. How solar energy can be converted into other forms of energy?
56. Explain the global nuclear energy scenario and environmental impacts.
57. Why is urban requirement of energy more than rural requirement?

UNIT II - ECOSYSTEM AND BIODIVERSITY

Ecosystem - Definition

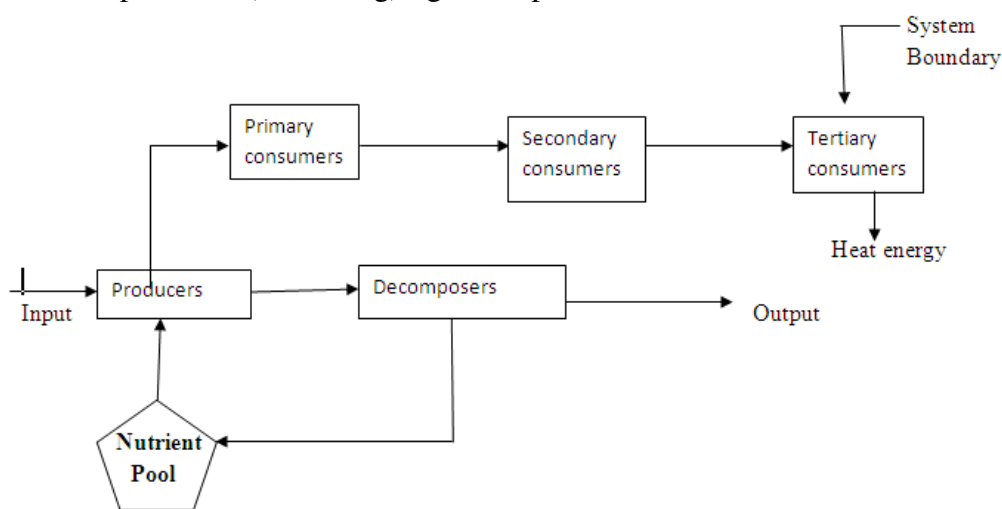
The term eco system was defined by A.Q.D.Tansley in 1935. According to him, an ecosystem is one which is resulting from the interaction of all the living and nonliving factors of any particular environment.

STRUCTURE OF ECO SYSTEM

The structure of an eco-system is characterized by the physical organization of biotic and abiotic component. The two structural features of any ecosystem are,

Biotic components - (Living) plants, animals and other organisms.

Abiotic components - (non-living) light, temperature, water, nutrients.

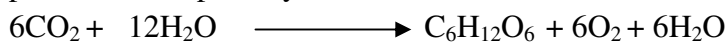


Biotic components

Biotic compounds are basically divided into following categories:

- (a) Producers of autotrophs (e.g. plants)
 - (b) Consumers or heterotrophs (e.g. animals)
 - (i) Primary consumers-herbivores (plant eaters)
 - (ii) Secondary consumers-carnivores (meat eaters)
 - (iii) Tertiary or top consumers-omnivores (plant and animal eaters)
 - (c) Detrivores (eat dead bodies)
 - (d) Decomposers (e.g. Micro organisms)
- (a) Producers (autotrophs) (auto-self, trophs-food)**

Producers are green plant or autotrophs which can synthesize their own food by the process called photosynthesis.



(b) Consumers (Heterotrophs) (Hetero-other, trophs-food)

Consumers are heterotrophic organism which directly or indirectly depends on producers for energy which is produced in the form of food.

Plant eating species (e.g. insects, rabbit, goat, deer, cow, etc.)

Animal eating species (e.g. fish, lion, tiger, etc.)

Consumers are further classified as,

- (i) Primary consumer (Herbivores) (plant eating):** They feed on producers. These organisms which depend upon green plants for their food.
 - e.g. rabbit, insects, goat, deer, cow, etc.

(ii) **Secondary consumers - 1° carnivores (meat eaters):** They feed on primary consumer. They directly depend on herbivores for their food.

➤ e.g. snake, frog, cat, etc.

(iii) **Tertiary or top consumers - 2° carnivores (eat plant and animal matter):** They feed on secondary consumers. They kill and eat the secondary or primary consumers.

➤ e.g. lion, tiger, etc.



(c) **Detrivores (eat dead bodies)**

They are small animals, which feed on dead bodies of other organisms, such as termites, ants, parasites and earthworm.

(d) **Decomposers (e.g. microorganisms)**

Decomposers are generally saprophytic microbes such as bacteria and fungi which decomposes the dead organisms. They decompose organic complexes into simple organic substances, and ultimately into inorganic nutrients. These inorganic nutrients together with other organic substances are then utilized by the producers for the synthesis of their own food.

Abiotic component (non-living organism)

Abiotic components are non-living chemical and physical factors in the environment.

➤ e.g. climate, soil, water, air, energy, nutrients, etc.

- **Physical factors**

Sunlight, temperature, rainfall, wind, soil type, water availability are some of the important physical features which have strong influence on ecosystem.

- **Chemical factors**

They are the source of essential nutrients.

e.g. carbon, nitrogen, oxygen, hydrogen, phosphorus, sulphur, various organic substances (proteins, lipid and carbohydrates) present in the soil and water largely influence the functioning of ecosystem.

FUNCTION OF ECOSYSTEM

The function of an eco-system broadly includes processes such as energy flow, nutrient cycling, productivity etc. The major functional attributes of an ecosystem are,

(a) Energy flow

(b) Food chain

(c) Food web

(d) Ecological pyramids

(a) Energy flow in an eco-system

- Energy is ability to do work.
- The ultimate source of energy is the sun.
- In ecosystem, solar energy is stored in producers (photosynthesis) which is transformed by consumer of different tropic levels.
- The transformation of solar energy from one tropic level to another by consumer is called **energy flow**.
- The behavior of energy in the ecosystem follows **the laws of thermodynamics**.

❖ First law of thermodynamics

- This law states that “energy can neither be created nor be destroyed but it can be transformed from one form to another”.
- **e.g.** The solar energy captured by the green plants (producers) gets converted into chemical energy and later it flows to the consumers.

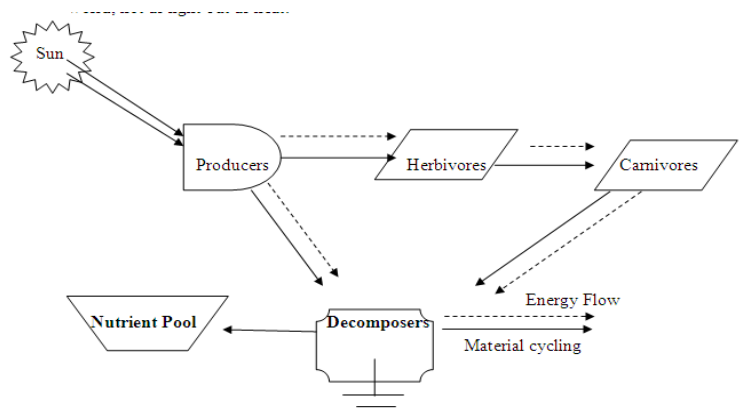
❖ Second law of thermodynamics

- This law states that “whenever energy is transferred there is a loss of energy in the form of heat”.
- **e.g.** This take place when energy is transferred between tropic levels.
- There will be loss of energy (about 80-90%) in the form of heat as it move from one level to another level. The amount of energy transferred to next level gradually decreases.

Energy Flow through an Ecosystem

Flow of energy in an eco-system

- The solar energy enters in the form of light rays.
- About 30% of sun light reaching earth atmosphere is reflected back in to space.
- About 20% of solar energy is used in heating atmosphere
- About 50% is used up in heating land, vegetation and water.
- Only 1% or less sunlight is used in photosynthesis by producers.
- Energy flows from the sun to the producers and then to the consumers and at each transfer a large proportion of energy is lost as heat.
- So less energy is left for the consumers at the upper most tropic level.
- Thus solar energy that entered the living system through the producers goes back into the non-living world, not as light but as heat.



(b) Food chain

- The sequence of eating and being eaten in an ecosystem is known as food chain.
- All organisms living or dead are potential food for some other organisms
- Thus there is essentially no waste of a natural ecosystem.
- In an ecosystem, food is the primary source of energy.
- Green plants alone are able to trap solar energy and convert it into chemical energy which is stored in the form of food.
- All other organisms are consumers and they depend upon producers (plants).
- **e.g.** A caterpillar eats plant leaf. A sparrow eats caterpillar. A cat eats sparrow.
- When they all die, they are all consumed by microorganisms like bacteria, and fungi (decomposed) which breaks down the organic matter and convert it into simple inorganic substance that can be again used by the plants.

Types of food chain

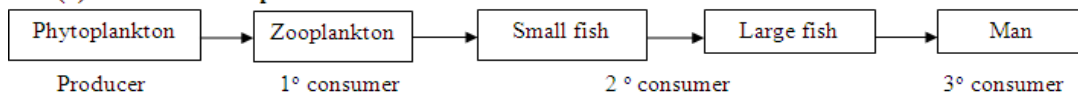
Mainly food chains are of two types. They are Grazing food chain and Detritus food chain.

Examples

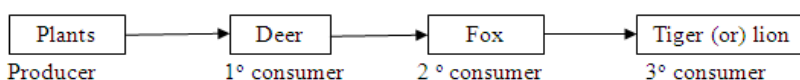
(i) Food chain in grassland



(ii) Food chain in a pond



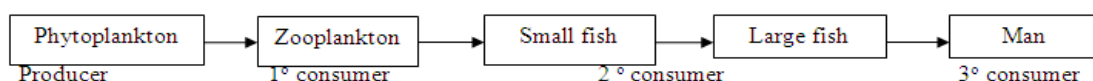
(iii) Food chain in forest



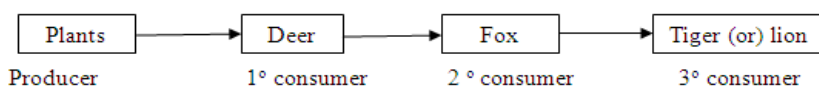
(a) Grazing food chain

It starts from plants (producers) to primary carnivores and so on.

e.g. (i) Pond ecosystem



(ii) Forest ecosystem

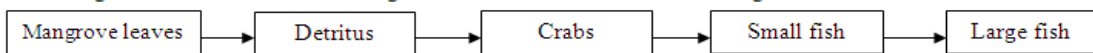


(iii) Marine ecosystem



(b) Detritus food chain

It starts from dead organic matter. The dead organic matter is decomposed into detritus by microorganisms like bacteria and fungi. Detritus food chain based on mangrove leaves as follows:



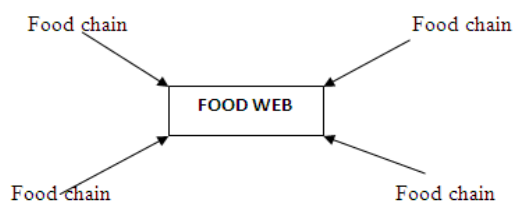
(c) Food web

- The **interlocking pattern** of various food chains in an ecosystem is known as food web.

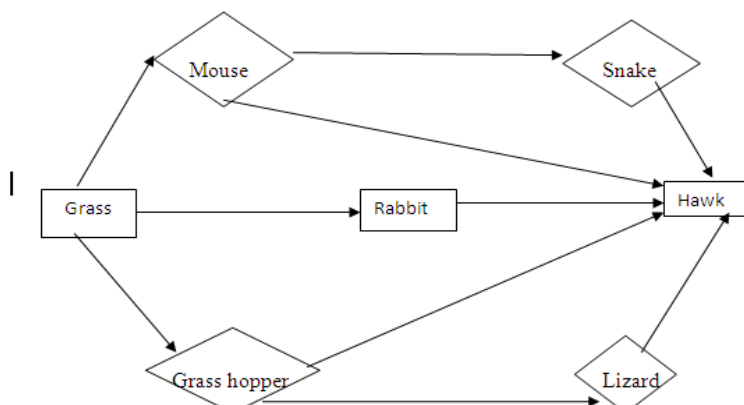
- Food relationship cannot be explained only in terms of a single food chain, the same organism may derive food from more than one trophic level and even the same organisms may be eaten by several organisms at higher trophic level.

- e.g. Rat is eaten by both the hawk and the snake.

- Thus in nature, a network of food chain is found and these



e.g. for food web



interconnected foodhains from a structure called **food web**.

- The food web becomes more complicated because of variation in taste,availability and compulsion. E.g. Tigers normally do not eat fish or crabs but in sundarbans they are forced to feed on them.

Difference between food chain and food web

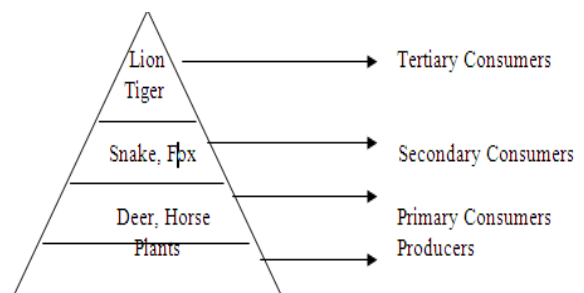
- In food chain, if one species gets affected then species in subsequent tropic levels are also affected
- But in food web if one species gets affected it does not affect the other tropic levels so seriously. There are number of options available at each tropic level.

(d) Ecological pyramids

- Ecosystem comprises various tropic levels.
- They are called producers, primary consumers, secondary consumers and Energy passes from one level to another level.
- Each transfer some energy is lost as heat.
- Each level receives less energy as compared to the previous level.
- The energy level gradually decreased in the food chain, **forming a pyramid like structure**. This relationship is called **ecological pyramid**.

Types of Pyramids

- (i) Pyramids of number
- (ii) Pyramids of energy
- (iii)Pyramids of biomass

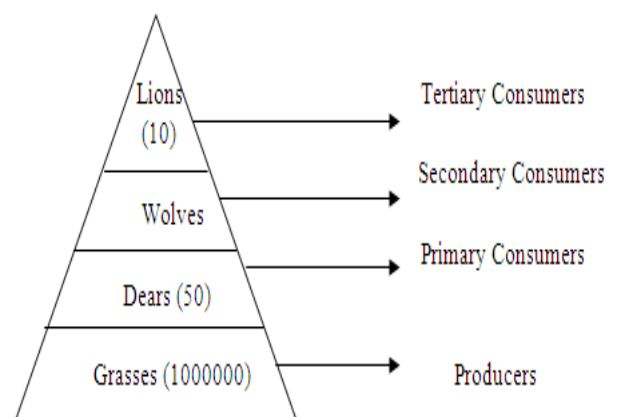


(i) Pyramids of number

- In this there is decreases in the number of individual from the lower level to the higher levels.
- The producers have highest number, whereas the number decreases in each successive level and become the lowest at the apex of the pyramid, i.e., number of individuals are highest at producer level and lowest at top consumer level.
- The number of individuals decreases from the base to the apex but the size of the organism increases. This can be found in grassland and pond ecosystem.

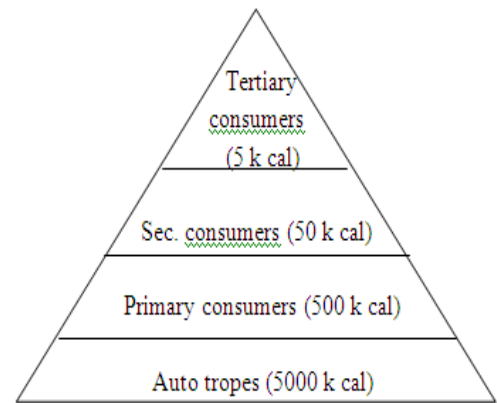
For example, in the grass land ecosystems,

- The lower level is occupied with grasses and they are abundantly present which is called producer.
- The second level is occupied by the deers. It is the primary consumers and its number is lower when compared to the grasses.
- The secondary consumers are wolves which occupy the third level, whose numbers are lower when compared to deers.
- The tertiary consumer is lion which occupies the next tropic levels. The number of last tropic level is the lowest.



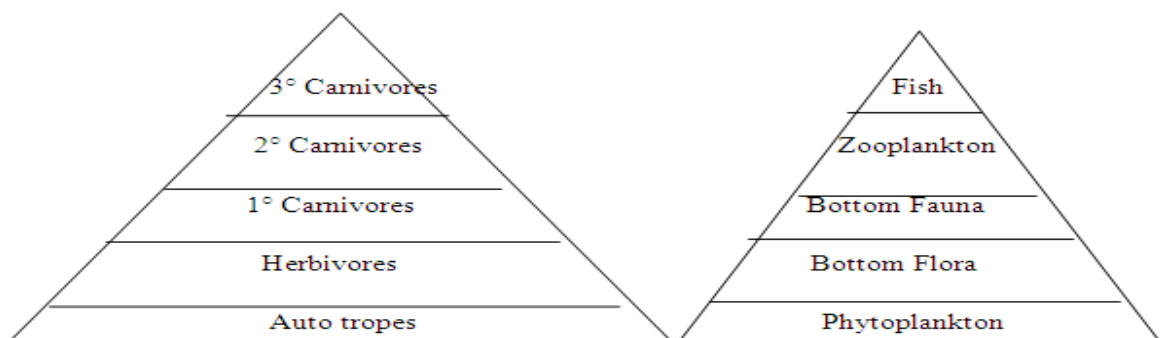
(ii) Pyramids of energy

- It represents the amount of energy present in each trophic level.
- There will be gradual decrease in availability of energy from autotrophs to higher levels.
- Greater amount of energy is available at the producer level than at the primary consumer level.
- During this transfer there is a considerable loss of energy in the form heat and about 90% of energy is lost.
- Thus the energy transfer from one level to another level is only 10%.
- Hence there is sharp decrease in energy at each level as move from lower to higher level.



(iii) Pyramids of biomass

- Biomass may be defined as the total weight of dry matter (dry weight) present in the ecosystem.
- In short, census of population multiplied by weight of an average individual in it.
- It gives an estimate of the weight of population called biomass.



In **terrestrial environment** the pyramid of biomass consists of

Autotrophs → Herbivores → 1° Carnivores → 2° Carnivores → 3° Carnivores

In case of **marine environment** the pyramid of biomass starts from phytoplankton to fish at the top.

Phytoplankton → Flora → Bottom Fauna → Zooplankton → Fish

Biome

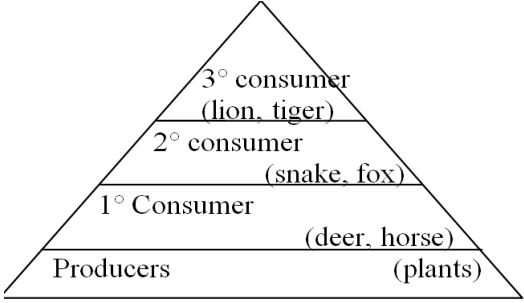
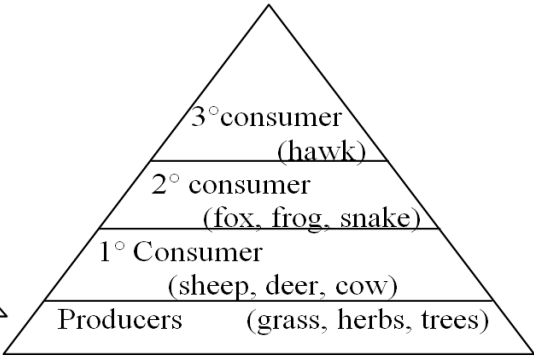
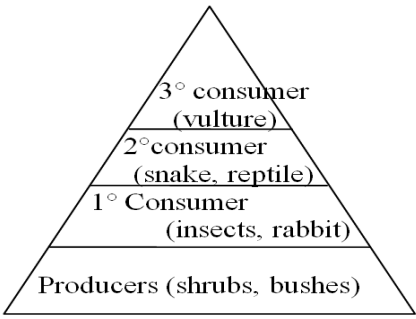
The earth has many different combinations of environmental conditions that support the amazing range of organism around us, which results in “Biodiversity”.

Biome can be considered life zones, environments with similar climatic, topographic, and soil conditions, and roughly comparable biological communities. Some of these are,

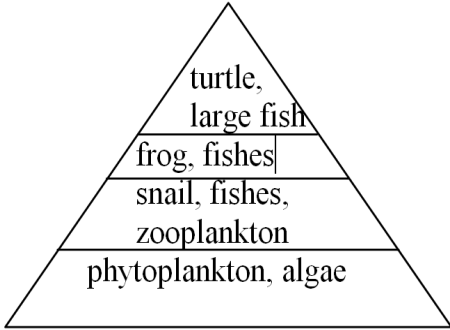
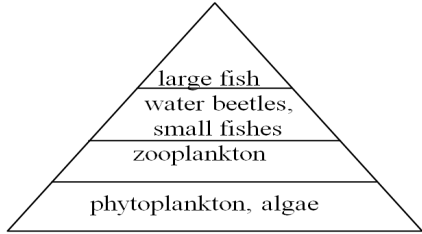
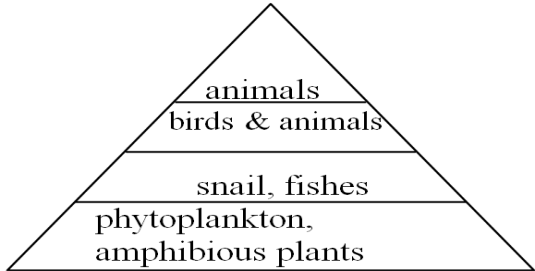
- Forest ecosystem
- Grasslands
- Deserts
- Aquatic systems

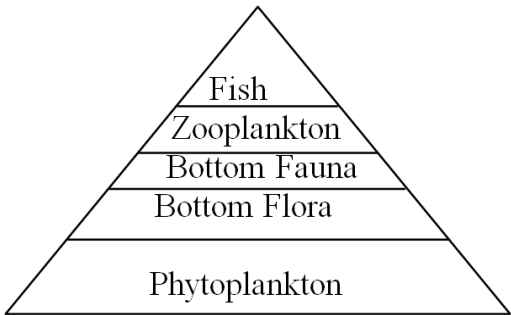
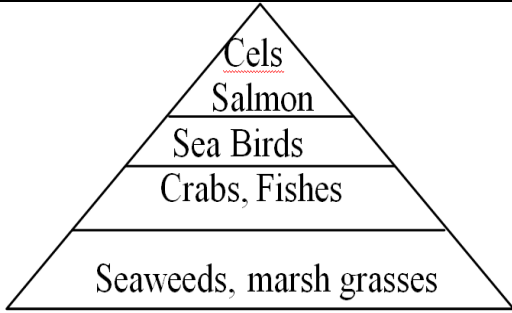
Recognizing biome gives us general idea on the plants and animals in the particular environment and their adaptation. The above ecosystems are discussed in detail in the following tables.

Introduction, types, characteristic features, structure and function of eco system			
	FOREST ECOSYSTEM	GRASSLAND ECOSYSTEM	DESERT ECOSYSTEM
INTRODUCTION	These are the ecosystem having large number of trees that support many animals and birds The forest occupies nearly 40% of the world's land area In India, it occupies only 19% of its total land area	Grasslands are dominated by grass species but sometimes allow the growth of a few trees, shrubs Rainfall is average and it is uneven. Overgrazing leads to degradation of these grasslands resulting in desertification	Desert occupied about 35% of our world's land area It characterized by less than 25 cm rainfall Atmosphere is dry
TYPES	Tropical rain forest Tropical deciduous forest, tropical scrub forest, Temperate rain forest Temperate deciduous forest	Tropical grassland Temperate grassland Polar grasslands	Tropical dessert Temperate desert Cold desert
CHARACTERISTICS	Forests are characterized by warm temperature and adequate rainfall Forest maintains climate and rainfall Forests support many wild animals and protect biodiversity. Soil is rich in organic matter and nutrients	Grassland ecosystem is a plain land occupied by grasses It is ideal place for grazing animals It is characterized by low or uneven rainfall Soil is very rich in nutrients and organic matter	The desert air is dry and the climate is hot Annual rainfall less than 25 cm Vegetation is poor The soil is very poor in nutrients and organic matter
STRUCTURE	Abiotic: Climate, minerals. Biotic Producers: trees, shrubs Consumer: 1° consumer: ants, flies, insects. 2° consumer: snake, bird, fox 3° consumer: lion, tiger. Decomposer: bacteria and fungi	Abiotic: Nutrients C, H, O, N, P, S etc. Biotic Producer: grass, shrubs Consumer: 1° consumer: cow, deer, buffaloes 2° consumers: snake, lizard, birds 3° consumer: eagle Decomposer: bacteria and fungi	Abiotic: Temperature, sunlight Biotic Producers: trees, shrubs, grass Consumer: Reptiles, insects, camels Decomposer: bacteria, fungi

FUNCTIONS	Producers: produce food through Photosynthesis Consumer: 1° consumer: depend on producer food 2° consumer: depend on 1° consumer 3° consumer: depend on 2° consumer Decomposer: decompose dead plants and animals	Producers: produce food through Photosynthesis Consumer: 1° consumer: depend on producer for food 2° consumer: depend on 1° consumer for food 3° consumer: depend on 2° consumer for food Decomposer: decompose dead plants and animals	Producers: produce food through Photosynthesis Consumer: Producers are very less, animals are also very low, depend on producer for food Decomposer: Desert has poor vegetation with low amount of dead organisms, decomposed by few fungi and bacteria
FOOD CHAIN	Plants → Herbivore → Carnivores elephant, deer lion, tiger ↓ ↓ Death → Decomposers	Grass → grasshopper, mouse → hawk rabbit ↓ (lizard, snake)	Shrubs → Insects → Snake Vulture
ECOLOGICAL PYRAMID			

AQUATIC ECOSYSTEMS (FRESHWATER)			
	POND	LAKE	RIVER
INTRODUCTION	Ponds are standing water bodies and are usually shallow which play a very important role in the villages. It is a small fresh water aquatic ecosystem It contains several types of algae, aquatic plants, insect, fishes, and birds. They are used for washing, bathing, drinking etc and get polluted easily. It receives enough water during rainy season.	Lakes are standing water bodies and are usually large. It is a medium fresh water aquatic ecosystem. It contains several types of rooted and floating plants, phytoplankton and zooplankton. Used for various purposes.	River is a flowing water bodies. Rivers are fresh water aquatic ecosystem. The river water are rich in biotic diversity. Flowing water transport nutrients and carries waste product from many aquatic organism. Used for various purposes.
TYPES		Oligotrophic, Geotrophic, Dystrophic, Volcanic, Mesomictic, Artificial lakes.	
CHARACTERISTICS	Pond is temporary It is a standing water body It gets polluted easily due to limited amount of water Used for domestic purpose	Lake is permanent It is a standing water body Used for irrigation and drinking purpose	River is permanent It is a fresh water body and free flowing water system Due to mixing of water dissolved oxygen content is more River deposits brings number of nutrients
	Abiotic: Temperature, water, light, organic and inorganic compounds. Biotic: Producers: Phytoplankton, microphytes (algae, large floating plants) Consumer: 1° Consumer: Zooplankton (fish)	Abiotic: Temperature, light, water, organic and inorganic compounds. Biotic: Producers: Phytoplankton Consumer: 1° Consumer: Zooplankton	Abiotic: Temperature, light, nutrients, organic and inorganic compounds. Biotic: Producers: Phytoplankton and other amphibious plants Consumer : 1° consumer: insects, snails, fishes

STRUCTURE	2° Consumer: Insects (beetles, fish) 3° consumer: Large fish Decomposer: Bacteria & fungi	2° Consumer : water beetles, small fish 3° Consumer : Large fish Decomposer: bacteria & fungi	2° Consumer: Birds and animals Decomposer: Bacteria & fungi
FUNCTIONS	Producers: produce food on their own. They are green plants. (photosynthesis) Consumer: 1° consumer: depend on producer 2° consumer: depend on 1° consumer 3° consumer: depend on 2° consumer Decomposer: decompose dead plants and animals	Producers: produce food on their own. They are the green plants.(photosynthesis) Consumer: 1° consumer: depend on producer for food 2° consumer: depend on 1° consumer 3° consumer: depend on 2° consumer Decomposer: decompose dead plants and animals	Producers: produce food on their own. They are green plants.(photosynthesis) Consumer: 1° consumer: depend on producer 2° consumer: depend on 1° consumer 3° consumer: depend on 2° consumer Decomposer: decompose dead plants and animals
FOOD CHAIN	Algae → small fish → Frog ↓ turtle	phytoplankton → zooplankton ↓ waterbeetles ← small fish ↓ large fish	phytoplankton → snail, fishes ↓ birds and animals ↓ animals
ECOLOGICAL PYRAMID			

	MARINE ECOSYSTEM	ESTUARINE ECOSYSTEM
INDRODUCTION	It comprises about 70% of the earth surface. The ocean environment is characterized by its high concentration of salts and minerals. It supplies huge variety of sea product and drugs.	An estuary is a partially enclosed coastal area at the mouth of a river where sea water mixes with fresh water. It is strongly affected by the tidal action. Estuaries are generally abundant of nutrients
CHARACTERISTICS	It occupies a large surface area with saline water. It is rich in biodiversity. Temperature varies from 0° C to about 28° C It provides iron, phosphorus, oil, natural gas.	Estuaries have a rich biodiversity and many of the species are endemic (widespread) Half of the life is spent in freshwater and the remaining half in salty water. The living organisms in estuarine ecosystem have ability to tolerate.
STRUCTURE	Abiotic: Temperature, light, Mg, K salts. Biotic: Produces: Phytoplankton (algae, large floating plants) Consumer: Fishes, sea snake, whale, seals Decomposer: Bacteria and fungi	Abiotic: Temperature, Na and K salts and various nutrients Biotic: Produces: Phytoplankton, grasses Consumer: Crabs, sea birds, small fishes. Decomposer: Bacteria and fungi
FUNCTIONS	Producers: produce food on their own. Consumer: depend on producer for their food Decomposer: decompose the dead organic matter	Producers: produce food on their own. Consumer: depend on producer for their food Decomposer: decompose the dead organic
FOOD CHAIN	Phytoplankton → Zooplankton → Fish → Man	Phytoplankton, Seaweeds, march grasses } Crabs, Fishes } Sea Birds } Cels, Salmon } Bacteria, fungi
ECOLOGICAL PYRAMID		

Ecological succession

Ecological succession is defined as the process in which communities of plant and animal species in a particular area are replaced over time by a series of different and often more complex communities. The types of succession are,

- **Primary succession:** In areas that have not already been occupied by any community.
- **Secondary succession:** Previously existing community is removed and new community is developed.
- **Allogenic succession:** Interaction of external physical force.
- **Autogenic succession:** Developmental process generated in the system.

BIO-DIVERSITY

- Bio means life and diversity means variety, hence biodiversity refers wide variety of life on the earth.
- Bio diversity is defined as “richness of species (micro-organisms, plants and animals) occurring in a given habitat”.
- Biodiversity protects the fresh air, clean water and productive land.

Biodiversity Classification

Biodiversity is generally classified into three types.

(a) Generic diversity

- Basic source of varieties of living organisms on earth.
- Raw material from which new species arises.
- Refers to the variation of genes within species.
- Variation may be in the entire gene or in chromosomal structure.
- Responsible for both the similarities and the differences between organisms.

Example

- All varieties of banana belong to only one species. Although it has variation in colour, shape, size and mineral content.
- All varieties of rice belong to only species, but there are thousands of wild and cultivated varieties of rice which show variation at the genetic level with difference in their color, size, shape and nutrient content.
- To conserve genetic diversity, different populations of a species must be conserved.

(b) Species diversity

- Species diversity refers to the variation in living organism.
- Surface of earth have many species.
- E.g.: Plants, animals, microorganisms.
- The total number of species present on the global could not be determined correctly.
- Plants, animals and microorganisms interact with each other and with the abiotic component of the environment.

Example

- Apple, grapes, mango, rice, comes under the category of plant.
- Lion, tiger, deer comes under animal species.
- Fungus, bacteria comes under microorganism species.

(c) Ecosystem diversity

- Ecosystem diversity is the variation in biological communities.

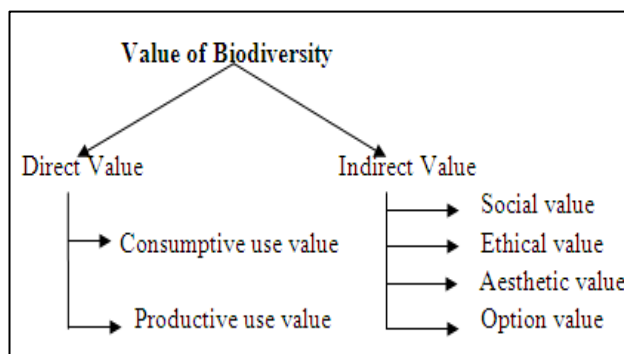
- In an ecosystem, biotic component (plant, animal, microorganism) interacting with one another and then with non-living elements (soil, air, water, etc) i.e. abiotic components.

Example

- River ecosystem which includes the fish, aquatic insects, variety of plants.
- Agricultural ecosystem depends upon human activity for the maintenance.

VALUE OF BIODIVERSITY

- Biosphere is a life supporting system.
- Each species in this system has its own role and importance.
- We know only about few species out of numerous that exist.
- Every year many species, out of known and unknown, are lost.
- The rate of their loss far exceeds than their exploration by us.
- Due to lack of our understanding, it is difficult to determine the role of the specific species.
- So preserving a diversity of life on the earth has come to be an accepted goal.
 - It is important to think about the value of biodiversity. The value of biodiversity can broadly be divided into two categories.



(a) Direct use value

- ❖ Bio resource makes direct contribution to human welfare without purchasing.
- ❖ Two types of direct values are,

(i) Consumptive use value

- Biodiversity products are harvested and consumed directly.
- Many medicinal plant and herbs are consumed for manufacture of drugs directly.
- In German more than 2,500 species of plants used for medicinal purpose in homeopathy.
- India uses 3000 species of plant in Ayurveda, Homeopathy and Unanisisystem of medicines.
- More than 30 medicines have been prepared from Neem tree for eye irritation and diabetics.
- Large number of wild plants and animals are consumed by human as food.
- Firewood's are directly consumed by village and tribal people.
- Fossil fuels like coal, petroleum and natural gas are also the product of biodiversity

(ii) Productive use value

Products are derived from plant and animals. They are sold and marketed.

Animal	Animal products	Plant and Animals	Industries
Silkworm	Silk	Wood	Paper, Pulp, Plywood
Sheep	Wool	Cotton	Textile
Elephants	Tusk	Fruits & vegetables	Food
Animals	Leather	Leather	Leather
Fish & animals	Food	Pearl	Pearls

(b) Indirect use value

The Indirect values involve the functions performed by biodiversity which are not of any direct use. It has many types.

(i) Social use value

These are the values associated with the social life, religious aspects of the people. Some of the Social values of the biodiversity are,

- Elephant, horse, cow, camel and other used for transport.
- Domestic animals are used for production of individual from the unanticipated threats or danger.
- Many plants are considered as holy plant in our country. E.g. tulsi and lotus.
- The leaves and fruits of these plants are used to worship.
- Many animals are also considered as holy animals. e.g. cow, snake, peacock, etc.

(ii) Ethical value (existence value)

- Existence value of biodiversity.
- “All life must be preserved”.
- The concept of “live and let others live”.
- We have to think that all species have right to exist.
- Every species, which exists in, nature, gives us benefit directly or indirectly.
- We are not deriving anything directly or indirectly from kangaroo, zebra, but we all strongly feel that these species should exist in nature.
- This is the ethical value or existence value attached to each species.
- Man holds great responsibility towards preserving and conserving other species.

(iii) Aesthetic value (beauty)

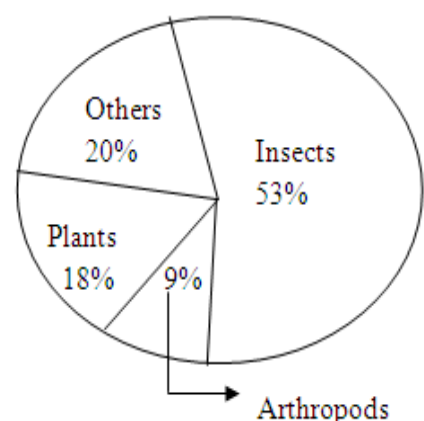
- The beautiful nature of plant and animals insists us to protect the biodiversity.
- The most important value of biodiversity is **eco-tourism**.
- People from far places spend a lot of time and money to visit the beautiful areas to enjoy the aesthetic value of biodiversity.
- This type of tourism is known as eco-tourism.
- Eco-tourism is coming up as new industry and a source of revenue for countries.
- Elephants, horse, camels and others are used for ceremonial purpose.
- Neem and mango leaves are used during festival time.
- Many ornamental plants are used for house decoration.

(iv) Optional value

- These values include the potentials of biodiversity that are presently unknown and need to be explored.
- i.e. any species may be proved to be a valuable species after someday.
- Thus optional value is the value of knowing that there are biological resources existing on this biosphere which may be proved to be important in future.
- There is a possibility that we may have some potential cure for AIDS or cancer existing within the depths of a marine ecosystem, or in a tropical rainforest.

HOT SPOTS OF BIODIVERSITY

- Distribution of biodiversity is not uniform on the earth.
- Areas which exhibit high species of richness as well as high species of endemism are termed as **hot spots of biodiversity**.
- British ecologist **Norman Myers** formulates the biodiversity hotspots concept in 1988.



- He identified only 12 such hot spot areas.
- Later **Myers** (2000) identified 25 hot spots of biodiversity on a global level, out of which two are present in India namely Eastern Himalayas and Western Ghats.
- At present he identified 34 hot spots of biodiversity.
- The hotspots are considered as outstanding example of evolution of varieties of species and at the same time extinction of species.
- According to World Resource Institution, the approximate number of known living species is about 14 million.
- Out of these the different percentage of various species is shown below.

Bio diversity at National level

- Among the plant rich countries of the world, India is the 10th place
- Among the Asian countries it lies at the 4th place.
- In considering number of endemic species of the world, India is at 11th place.
- It remains at the 10th place in diversity of mammals.

Hot spots of India

Among 34 hot spots, two are found in India namely **Western Ghats and Eastern Himalayas**. The Indian hot spots are not only rich in flora and endemic species of plants but also rich in reptiles, amphibians and mammals.

Eastern Himalayas

- ❖ Eastern Himalayan hot spot comes under Indo-Burma hot spot which covers about 2 million sq. km. tropic Asia, East of Indian subcontinent.
- ❖ The area comprises Nepal, Bhutan and neighboring states of northern India.
- ❖ Eastern Himalaya consists of about 9000 plant species, in which 39% are endemic.
- ❖ Among them, 55 flowering plants are endemic to this area but these are rare in other places.
- ❖ Eastern Himalayas are also rich in wild plant.
- ❖ e.g. Rice, banana, chili, sugarcane, ginger, citrus, etc.
- ❖ Tea is reported to be cultivated in this region for the last 40,000 years.
- ❖ 63% mammals are from Eastern Himalayas
- ❖ 60% Indian birds are from North East.
- ❖ Huge wealth of fungi, insects, birds, mammals have been found in this region.

Western Ghats

- ❖ Western Ghats are parallel to western coast of Indian peninsula in Maharashtra, Karnataka, Tamil Nadu and Kerala
- ❖ 40% of total endemic species of plants
- ❖ 62% of amphibians
- ❖ 50% of lizards.
- ❖ contain several trees species of great commercial value such as Indian redwood But they have now been cleared for many areas.
- ❖ Many animal species such as Asian elephant, Indian tiger are found in this region.
- ❖ The western slopes of the mountains receive heavy rain fall whereas the eastern slopes are drier.
- ❖ Nearly 1500 endemic plant species are found from Western Ghats.
- ❖ Out of 5 locations identified by IUCN as threatened, 3 of them are found in this region.

INDIA – THE MEGA DIVERSITY NATION

- ❖ India ranks sixth among twelve mega diversity countries in the world.
- ❖ The total number of species in the world is estimated to be around 5 to 30 million

❖ Out of which 14 million species have been described.

(a) The total number of plant species is estimated = 45000.

These are classified as under:

- Flowering plants = 15000
- Gymnosperms = 64
- Bryophytes = 2843
- Pteridophytes = 1012
- Lichens = 1940
- Fungi = 23000
- Out of the above number of endemic to India = 4900
- Highly threatened = 1500

(b) Total number of fauna estimated = 81000 species

These are classified as

- Mollusks = 5000
- Arthropods = 57000
- Fishes = 2546
- Amphibians = 204
- Reptiles = 428
- Birds = 1228
- Mammals = 372

Out of above, 62% species are endemic to India.

❖ Besides above estimate, a host of microorganisms are to be described.

❖ The marine habitat of India covering an area of over 7500 km

❖ Cost line extending about 200 nautical miles is yet to be fully explored.

❖ Wild strains of coffee, tobacco and sugarcane are cultivated.

Cultivated species of India

- Cereals and Millets = 51
- Fruit species = 104
- Spices = 27
- Fiber plant species = 24
- Oil seeds = 12

Also several wild strains of coffee, tobacco and sugarcane are cultivated.

Taxa	Number of species		Percentage India: world
	India	world	
Protochordata	116	2173	5.34
Pisces	2546	21723	11.72
Amphibian	206	5145	4.00
Reptile	485	5680	8.54
Aves	1228	9672	12.60
Mammalia	375	4629	8.03
Total	126656	1719183	7.36

Biosphere Reserves of India

Biosphere reserve are terrestrial and ecosystems, which are internationally recognized within the framework of UNESCO's Man and Biosphere (MAB) Programme.

India has 13 biosphere reserves. They are,

Nilgiri	Nanda Devi
Nokrek	Manas
Sunderbans	Gulf of Mannar
Great Nicobar	Similipal
Dibru – Saikhowa	Dihang Dibang
Pachmarhi	Kanchenjunga
Agasthyamalai	

Also, it has the following **world heritage sites**

- (i) Kaziranga National park
- (ii) Keloladeo National park
- (iii) Manas National park
- (iv) Nanada devi National park

India has six **Ramasar wetlands**

- (i) Chilka Lake
- (ii) Keoladeo National park
- (iii) Walar lake
- (iv) Harike lake
- (v) Loktak lake
- (vi) Sambhar lake

All these views say that India is a mega diversity nation.

Biodiversity at local level

Tamilnadu has rich biological diversity which consists of 5 national parks, 20 wild life sanctuaries and two biosphere reserves.

Some of the important locations in Tamil Nadu are,

- Anamalai Wild life Sanctuary •Mudumalai Wild life Sanctuary
- Mundanthurai Wild life Sanctuary •Vedanthangal Birds Sanctuary
- Guindy National park •Chennai Crocodile Bank
- Kanthankulam Birds Bank •Srivilliputhur Grizzled squirrel sanctuary
- Gulf of mannar •Marina National park
- Parambikulam sanctuary

Biodiversity at local level is classified into four classes based on species richness and distribution.

Point richness:

It refers to richness of species that can be found at a single point in a given space.

Alpha richness:

It refers to the number of species found in small homogeneous area.

Beta richness:

It refers to the rate of change in species composition across different habitats.

Gamma richness:

It refers to the rate of change of species in a large landscape area.

THREATS TO BIODIVERSITY

- If natural ecosystem gets distributed it leads to reduction in biodiversity.
- This disturbance is mainly due to human population and industrializations.
- The rate at which species are disappearing has increased over the last 150 years.
- Human activities have removed two or three species per decade, in last 150 years.
- If the present trend continues, millions of plant, animals and microorganism may be destroyed in the middle of the twenty-first century.

VARIOUS THREATS TO INDIAN BIODIVERSITY

- Human population raises, an ever increasing demand for land, house, raw materials, fuels, food etc, also equally rises.
- If the present trend continues millions of animals, plants and micro organisms may be destroyed in the next few decades.
- Deforestation and the subsequent soil erosion result in loss of biodiversity.

Threat to biodiversity is mainly due to:

- Habitat loss
- Poaching of wild life
- Man- wildlife conflicts
- Extension of agriculture
- Filling of wet lands
- Destruction of coastal areas
- Uncontrolled commercial exploitation

(a) Loss of Habitat

Habitat is the sum of the environmental factors such as food, water and shelter, which is need for given species to live comfortably and reproduce in a given area. Habitat loss threatens to a wide range of plants and animals. The loss of habitat is mainly due to the following reasons.

- Agricultural activities
- Extraction (including mining, fishing, logging, harvesting)
- Developmental activities (human settlements, industrial and associated infrastructure, dams)
- Human activities (grazing, hunting)
- Production of drugs (pharmaceutical companies collects wild plants)
- Forest encroachment, construction of dams
- Discharge of industrial effluents and chemicals kills the aquatic organism
- Continuous flooding on wetlands leads to loss plants and animals
- Introduction of transmission lines, roads

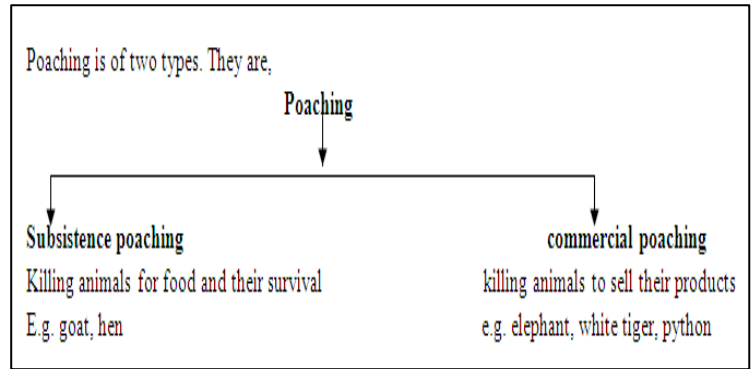
(b) Poaching of wild life

Poaching means killing or hunting of animals. Hunting is highly responsible for the depletion of many species. It leads to loss of animal diversity.

- **e.g.** American passenger pigeon. 200 years ago this was the world's most abundant bird with a population of three to five billion. In spite of this abundance, these species are hunted and the entire population gets reduced within 20 years.
- The current global fish catcher is about 160 million tones.
- Marine animal such as dolphins of about 4,00,000 killed annually.

Factors influencing poaching

- Due to increase in human population, animals are killed illegally for their meat, skin, internal organs
- Wild life is sold for medicines, skin and other products such as ivory, horns
- Illegal wild life smuggling is high in Asia, North America and Europe because of the huge profit
- In 1950 an estimated 1,00,000 tigers existed in the world. But today less than 7,750 tigers exist in the world. Mostly because of poaching for their furs and bones
- In Rwanda and Zaire, male gorilla is hunted for its body parts, head and hands
- In Brazil the blue morpho butterfly is poached for making attractive trays and other objects
- In USA, the snowy large egrets are poached for its white plumes so as to keep it in ladies hats
- Blubber was used to prepare lamp oil and lubricating oil
- Baleen was used to prepare combs and other similar articles
- Elephant feet are used to make ash trays



Remedy measure

- Illegal hunting should be stopped immediately
- Biodiversity laws should be strengthened
- We should not purchase fur coat, purses or bag items made of crocodile skin or python skin
- Complete and comprehensive ban on illicit trade in endangered species
- Networking and exchange of views with national and international agencies involved in eradication of such illicit trade
- Prepare a database of wildlife related crimes as well as criminals and using it for apprehending and prosecuting offenders

(c) Man – Wildlife – conflict

Man is continuously interacting with many ecosystems through several activities like hunting, food and fuel and infrastructure development. Each activities is responsible directly or indirectly to create sufficient disturbance to many wild life species.

Factors influencing man-wild life conflict

Rapidly growing human populations interferes with wildlife, creating man-wildlife conflicts and reasons are,

- ❖ Due to reduction of forest areas, wildlife move outside and attack the agriculture field and humans
- ❖ Usually the ill, weak and injured animals have a tendency to attack man
- ❖ The village put electric wiring around their croup fields in order to protect the crops from the wild life. This can create injury to the elephant and they become violent. So man–elephant conflict takes place in this region
- ❖ Expansion of human population causes shrinkage of habitats
- ❖ E.g. Elephant require large range of land for their survival. But with the extension of agricultural land and industries elephants are forced to come out.

- ❖ As a result conflict starts between man and elephant. Several elephants were killed because of the damage done by the elephants to the farmer
- ❖ In Sambalpur, Orissa, 200 people were killed by elephant and 98 elephant have been killed by man and 30 elephants are badly injured
- ❖ Human encroachment into the forest area raises a conflict between man and the wildlife, perhaps because it is an issue of both.
- ❖ e.g. Dwindling habitats of tigers, elephants, rhinos and bears due to shrinking forest cover
- ❖ The cash compensation paid by the government in lieu of the damage caused to the farmer's crop is not enough
- ❖ In Mysore, a farmer gets compensation of Rs. 400/- per quintal of expected yield while the market price is 2400/- per quintal. The agonized farmer therefore gets revengeful and kills the wild animals

Remedy measure

- ❖ Adequate fruits and water should be made available for the wild life with in forest zones
- ❖ Solar powered fencing should be provided along with electric current proof to prevent the animals to enter into fields
- ❖ Tiger Conservation Project (TCP) has made provisions for making available vehicles, tranquillizer guns, binoculars and radio sets to tactfully deal with any imminent danger
- ❖ Adequate crop compensation and cattle compensation scheme must be started
- ❖ Wildlife corridors should be provided for mass migration of big animals during unfavorable periods

ENDANGERED SPECIES

It is otherwise known as “**Threatened species**”. Endangered animals are those animals **in danger of extinction**.

- ❖ An endangered species is a population of organisms which is at risk of becoming extinct because it is very few in numbers;
- ❖ It is threatened by changing environment, deforestation, human activities, etc.
- ❖ Such organisms are very important from the point of view of conservation.
- ❖ The International Union for Conservation of Nature and resources (IUCN) publishes the **Red Data Book** which includes the list of endangered species of plants and animals.
- ❖ The Red Data book gives the warning signals for threatened species of animals, which if not protected, are likely to become extinct.

In India nearly

- 450 plant species
- 53 species of mammals
- 69 species of birds
- 23 species of reptiles
- 3 species of amphibians
- 3 species of fish
- 2 species of mollusks have been identified as endangered species

India is the 2nd rank for the number of threatened mammals and it is 6th rank for the number of threatened birds.

A few species of endangered reptile, birds, mammals and plant are,

- **Reptiles** - tortoises, python, green sea turtle
- **Birds** - peacock, pelican, Indian bustard
- **Mammals** - Indian wolf, red fox, sloth bear, tiger, Indian lion, golden cat, desert cat
- **Plants** - A large number of medicinal plants, sandal wood tree

Factors affecting Endangered species

- Human disposes their waste product on land and river.
- Land pollutants travels through the food chain and accumulate in the tissue of the living organism, finally it leads to death.
- Extinction of wild animals takes place due to over exploitation of the natural resources.
- Climate change
- Unnecessary killing of wild animals.

ENDEMIC SPECIES

Plants and animals are distributed unevenly on the global level. Some plants and animals are widespread while other has some restricted distributions. The species, which are found only in a particular region, are known as endemic species.

An endemic species is one that is **only found in a particular region and nowhere else in the world**. Since these species are not widespread and may be confined to only one or two protected areas, they are of great conservation concern.

India possesses a large number of endemic species (plants and vertebrate species)

- 33% of flowering plants are endemic to our country
- 53% are fresh water fishes
- 60% amphibians
- 36% reptiles
- 10% mammals

Area rich in endemism are,

- ❖ North - East India
- ❖ The western Ghats
- ❖ Eastern Himalayas
- ❖ Andaman and Nicobar islands

It can be observed that Western Ghats area is about 4% and East India is about 5.2% of the total geographical area of our country, which have a rich biodiversity. These zones have innumerable habitats, biotic communities and ecosystems. These zones are inhabited by a large number of endemic species.

Four endemic species of conservation significance occur in Western Ghats are,

- (i) Lion – tailed macaque (*Macaca Silenus*)
- (ii) Nilgiri langur (*Presbytis johnii*)
- (iii) Brown palm civet (*Paradoxurus jerdoni*)
- (iv) Nilgiri tahr (*Hemitragus hylocris*)

CONSERVATION OF BIODIVERSITY

Conservation of biodiversity is concerned with the protection of species and their population.

Importance of conservation

- (i) Biodiversity provides food, raw material for clothing, for shelter, fuel and medicines
- (ii) Medicinal drugs, food and other important raw materials can be derived from plants and animals
- (iii) It provides immediate benefits to the society such as recreation and tourism
- (iv) Conservation of biodiversity is very useful to the human kind to meet the challenge of global warming and ozone layer depletion. Hence it is essential to conserve the diversity.

Types of biodiversity

There are two types of biodiversity conservation. They are,

- (a) In – situ conservation (within habitat)
- (b) Ex – situ conservation (outside habitat)

(a) IN - SITU CONSERVATION

- In situ conservation involves protection of plant and animal species within its natural habitat.
- Natural habitats maintained under in - situ conservation are called “protected areas”.
- Protected areas (national park, wildlife sanctuaries, etc) are the most important in - situ conservation method.

At present we have,

In situ conservation	Numbers available
Biosphere reserves	7
National parks	80
Wildlife sanctuaries	420

These conservations covering about 4% of the geographic area.

Biosphere reserves

Biosphere reserves are special category of protected areas.

- It is used to protect species for long term.
- It protects maximum number of species.
- It is also useful for educational and research purpose.
- There are 14 biosphere reserves in India.
- Out of 14 biosphere reserves, 3 have been included in the world network of biosphere reserves.

Name of biosphere	India (state)
Gulf of Mannar	Tamil Nadu
Nilgiri	Tamil Nadu, Karnataka, Kerala
Sundarbans	West Bengal

- In India, the first biosphere reserve is Nilgiri which spread over Kerala, Tamil Nadu and Karnataka.
- Biosphere reserves may have one or more national parks.
- For example Nilgiri biosphere reserve has two national parks namely Bandipur National park and Nagarhole National park.
- In a biosphere reserve wild life populations as well as traditional life styles of tribal and various domesticated plants and animals genetic resources are also protected.

There are three district zones in a biosphere reserve. They are,

1. Core zone
2. Buffer zone
3. Transition zone

Core zone: The core zone is also known as natural zone, which comprises an undistributed and legally protected eco system.

Buffer zone: The buffer zone, the outer part of the core zone, is managed to accommodate a greater variety of resource, research and educational activities.

Transition zone: The transition zone, the outermost part of the biosphere reserve, comprises reserve management and the local people, where in activities such as settlements, cropping, forestry, recreation and other economic uses allowed with conservation goals.

National parks

- Earlier, the national parks, such as Yellowstone park in USA and the royal park near Sydney, Australia, were chosen because of their beauty and recreational value but now the national parks have been given a status of selected areas dedicated to conservation of wildlife along with its beauty.
- It is also used for enjoyment through tourism without affecting the environment.
- Grazing of domestic animals, all other forests activities are prohibited with in a national park.
- According to the national wildlife database, there are 92 existing national parks in India covering an area of 37,921 sq. km.
- In India, first national park named Hailey national park, which is now known as Corbett national park, Uttarakhand in the year 1936.

State-wise national parks in India

State/UT	National park	Area covered (km)
Assam	5	1968
Jammu & Kashmir	4	4680
Madhya Pradesh	9	3656
Tamil Nadu	5	307
Uttarakhand	6	4077
India	2	37,921

Some important national parks in India, their location and important flora fauna are given below:

Name of park	State	Flora	Fauna
Corbett	Uttarakhand	Moist and dry forest	Tiger, leopard
Guindy	Tamil Nadu	Evergreen forest and semi-evergreen forest and grasslands	Elephant, leopard and gaur
Periyar	Kerala	Dry mixed deciduous forest, bamboo	Tiger, leopard, elephant, wild dog, jackal, wild bear
Great Himalayan	Himachal Pradesh	Oak forest	Leopard, civet cat, Brown and black bear
Gir	Gujarat	Dry teak forest, desert thorn forest	Lion, panther
Similipal	Orissa	Semi – evergreen forest and grassland	42 species of Mammals, 242 species of birds, 29 species reptiles

Wild – life sanctuaries

- Like national parks, wild – life sanctuaries are also protected areas dedicated to protect wildlife, but they are aimed to conserve animal species only.
- In these protected areas, killing, shooting, hunting or capturing of wildlife is prohibited.
- The chief wildlife warden of the state is authorized to grant permission to enter and stay in a sanctuary for study, research, photography, tourism etc.
- According to national database resource, there are 492 existing wildlife sanctuaries in India, covering an area of the 1,16,992 sq. km.

State – wise national park in India

State /UTs	No. of WLS	Area Covered (km)
Assam	15	943
Jammu & Kashmir	15	10312
Madhya Pradesh	25	1758
Tamil Nadu	19	2539
Uttarakhand	6	2413
India	492	1,16,992

Select Wildlife Sanctuaries of India

Name of the sanctuary	State/UT	Important fauna
Anamalai sanctuary	Tamil Nadu	Elephant, tiger, Panther, leopard, wild dog
Mudumalai wildlife sanctuary	Tamil Nadu	Elephant, sambhar, barking deer , Mouse deer, wild dog, wild cat
Keolapara Ghana bird sanctuary	Rajasthan	Birds such as Siberian crane ,egrets, spoon bill etc
Periyar sanctuary	Kerala	Elephant, leopard, sambhar, egrets, gaur
Vedanthangal bird sanctuary	Tamil Nadu	Birds

(b) EX-SITU CONSERVATION

Ex-situ conservation involves protection of plant and animals in suitable place outside their natural habitats. Ex-situ conservation of biodiversity implies conservation outside the habits by ethics perpetuating sample populations in genetic resource centre, zoos, botanical gardens, culture collection, etc., or in the form of gene pools and gamete storage for fish.

Ex-situ conservation also includes setting on germplasm banks for seeds, pollen, semen, ova, cells, etc. Ex-situ conservation includes the following methods.

(a) Zoos

- ❖ Zoos or Zoological gardens are the places where wild animals are kept for public showing. Some of the zoos contain rare animals. They have recorded success with captive breeding of animals (e.g. birth of a calf to rhino in a zoo of Bihar in September, 2000). The Pere David's deer and the European Bison, for example, have lost their natural habitats many decades ago, but they have been kept alive solely through zoos.
- ❖ In India, there are 15 large zoos (total area 2317 ha), 17 medium zoos (total area 983 ha), 32 small zoos (total area 3148 ha) and many mini zoos (total area 2827 ha). The Central Zoo Authority (CZA) of India provides financial assistance to zoos and also approves exchanges and donations of animals both domestic and foreign for breeding programmes.

(b) Botanical Garden

- ❖ They play an important role in the conservation of plant species. There are several instances where plants believed to be extinct were found living only in a botanical garden, e.g., *sophora toroniro*. The Botanical Survey of India (BSI) with its main botanic garden at sibpur and the regional experimental gardens established at different places to suit the adaptation of plants occurring in different ecological conditions in the country have in their holding rare, endemic and threatened plants.
- ❖ Records of threatened plants that are in cultivation have been kept in Green Books. The Indian Green book prepared by BSI lists 10 such species which are rare, endangered or endemic but all of which are growing in living state in some botanical gardens of BSI in the country. Currently, botanical gardens have assumed a special role as centers where active research and action oriented plan for conservation can be developed.

(c) Seed Banks

- ❖ The seed bank is an excellent method of conserving diversity. The seeds have to store under minimal life so that they can be kept for several years without accumulations of mutations and with a minimum loss of viability. Such sort of conservation is practiced by the help of cold storage in seed banks where seeds are stored for long durations.
- ❖ The preservation of unmanageable seeds, i.e., those which cannot be stored at low moisture content, present problems for certain species, especially many fruit and forest species. Efforts to develop improved methods for storing recalcitrant seeds involve a three-fold strategy; dry storage, cryogenic storage and storage of fully hydrated seeds at ambient or sub ambient temperatures.
- ❖ Quite recently, a seed bank has been set up at the National Bureau of Plant Genetic Resources (NBPGR), New Delhi for the preservation of land races and cultivators have replaced banks is associated with a world network of gene resource centers coordinated by the International Bureau of Plant Genetic Resources (IBPGR) of FAO. Seed bank of NBPGR is also involved in the preservation of plants of wild origin of potential economic value, or of scientific, aesthetic or ecological importance.

(d) Pollen Storage

- ❖ Pollen storage is of great importance in the conservation of genetic diversity. However, the life of pollen is much shorter than seed and a maximum of 3 years in some species has been reported.

(e) Tissue Culture

- ❖ It is best method of preservation of genes through establishment of tissue banks. Tissue culture technique can be extended to endangered species or for those which may otherwise require very varied climatic conditions and can be maintained at one place in aseptic cultures.
- ❖ The advantage of this technique are economy of space, relatively low maintenance cost, protection of genotypes from natural calamities and diseases if growing in open and easy international exchange of healthy stock.

Advantages of Ex-Situ conservation

- Organism is assured of food, shelter and security, and hence can have longer life span and breeding.
- Chances of survival are enhanced under human care and secure conditions.
- This method also provides possibility of using genetic techniques to improve the concerned species.

Disadvantages of Ex-Situ conservation

- This method can be employed only for a few selected species, because of limitations of space, finances and facilities to the concerned institution.
- This conservation under a set of favorable environmental conditions deprives the organism, the opportunity to adopt the ever changing natural environment. As a result new life-forms cannot evolve and gene-pool gets stagnant.

Benefits of Biodiversity conservation

- It is possible to increase and improve our food supplies by developing new crop plants, fish supplies, and animal sources.
- Increased use of biological control agents using natural enemies to control pests.
- Potential sources of genes to improve quality, resistance, and vigor of conventional crops through hybridization and genetic engineering.
- Increased use of natural products for medicine, drugs and poisons.
- Recognition of environmental services performed by wild organisms, including pollination, biodegradation, soil aeration, fertilization, gas exchanges, water storage.
- Sources of warning signs to offset the lack of health screening tests for many chemicals and pollutants.
- Aesthetic value of interesting wildlife and plant life.

QUESTIONS

PART - A

1. Define ecology and ecosystems.
2. What are the types of an ecosystem?
3. Name any four types ecosystems.
4. What are the basic components of an ecosystem?
5. What is a biotic community?
6. What are autotrophs and heterotrophs?
7. Enumerate the characteristics of an ecosystem.
8. What are difference between (a) Ecosystem & Biome (b) Biosphere and biomes.
9. What are 'producers', 'Consumers' and 'Decomposers', give Example.
10. Give examples of animals that are herbivores, carnivores, omnivores
11. List the biotic & abiotic components of an ecosystem.
12. What are the four types of consumers?
13. What is meant by energy flow in an ecosystem?
14. What are food chains and food webs? Discuss their significance with example.
15. Differentiate between Grazing food chain and Detritus food chain.
16. What are the types of grass land eco-system?
17. Distinguish between food chain and food web.
18. Define biodiversity. Explain the different types of bio diversity
19. What are the major threats to biodiversity?
20. Name two hotspots of biodiversity in India.
21. Define endangered species.
22. What are the criteria for recognizing hot spots?
23. What are the characteristics of estuaries?
24. What is Red Data Book? Name some endangered species of our country.
25. Give an example for food chain in forest.

26. What is meant by IUCN Red list, UNCED, WB, WWF?
27. What are the different zones in a lake ecosystem?
28. Give a simplified food web in terrestrial ecosystem.
29. Define ecological succession. Write different types of ecological succession.
30. Differentiate natural and artificial ecosystem.
31. What do NBPGR and NBAGR stand for? Where are they located?
32. Mention few biosphere reserve centers of Tamil nadu
33. What is difference between national park, wild life sanctuary and reserve forest?

PART - B

1. Discuss the components of ecosystem and explain. **(Jan 15)**
2. Explain energy flow of ecosystem with help of thermodynamics law. **(Jan 14)**
3. Explain different types of ecological pyramids giving examples. **(Jan 15)**
4. What are ecological pyramids? Explain why some of these pyramids are upright while others are inverted different eco system. **(May 14)**
5. Show the structure and functions of forest ecosystem. **(May 14)**
6. Explain any one of the strategy adopted to conserve biodiversity. **(May 14)**
7. Discuss the major threats to Indian biodiversity. **(May 14)**
8. Describe the productive use and social values of biodiversity. **(Jan 13)**
9. Name and discuss the values that can be assigned to biodiversity. **(Jan 15)**
10. Describe the structure and functions of grass land ecosystem. **(May 13)**
11. Explain the in-situ conservation of biodiversity. **(Jan 13)**
12. "India is a mega diversity nation"- Explain. **(Jan 13)**
13. Explain in-situ and ex-situ conservation of biodiversity with examples.
14. Comment on Indian biodiversity with special reference as mega diverse nation.
15. What do you meant by values of bio-diversity? Explain its types with examples.
16. What is meant by alpha, beta and gamma richness? Discuss, giving examples.
17. What are hotspots of biodiversity? Which are the hotspots found in India?
Discuss their salient features.
18. In the biosphere energy follows a liner path way, while matter follows a circular path way. Explain with illustration.
19. Write short notes on (a) tropical rain forests (b) Savannas (c) Arctic Tundra.
20. Discuss the major features of a stream (river) that differ from that of a lake.
21. Discuss the salient features of an estuarine ecosystem.
22. In what way estuarine ecosystem differs from Lake Ecosystem.
23. What are the major causes of man wildlife conflicts? Discuss the remedial steps for the conflict.
24. Enumerate five important biosphere reserves, national parks (with important wild life) and wild life sanctuaries (with major wild life) of India. Also mention the state where they are located.
25. Name animals for whose protection and conservations specific projects have been launched in India. Explain the reasons for loss of our biodiversity
26. . How the study of biodiversity is beneficial to us?
31. Name types of plants for which gene sanctuaries in India exist.
Explain the efforts taken towards conservation of Biodiversity in India.
27. Is biotechnology threat to biodiversity? Substantiate your answer with example.

UNIT III

AIR POLLUTION

Air pollution - Definition

“Any change in physical, chemical and biological properties of air caused by human activities and natural resources which may cause harmful effects” is known as Air pollution.

The presence of one or more contaminant such as dust, gas, mist, odour, smoke, smog or vapour in the outdoor atmosphere, in quantities of characteristics and of duration so as to be injurious to human, plant or animal life or to property or which interferes with the life and property is known as Air Pollution.

SOURCES OF AIR POLLUTION

(i) Natural sources

- volcanic eruptions
- Forest fires
- Natural organic and inorganic decays
- Vegetative decay
- Marsh gases
- Deflation of sand and dust
- Extra terrestrial bodies
- Soil debris and comets
- Green plants through evaporation, transpiration release huge amount of CO₂
- In the lower atmosphere, the natural chemical reactions lead to the conversion of gases or vapours into solids and liquid products by oxidation, reduction, combination, combustion, condensation or polymerization.

(ii) Man made sources

(a) Over Population

The rapid explosion of population is one of the most important factors of air pollution. This creates several serious problems including worsening of the conditions of the environment.

(b) Deforestation

Plants maintain the balance of CO₂ and O₂ in the nature. By deforestation the balance of CO₂ and O₂ in the nature is getting affected.

(c) Burning of fossil fuels and fires

The conventional sources of energy are wood, and fossil fuels like coal, oil and natural gas. The industries need fossil fuels to drive machines and for many other purposes.

The imperfect mode of burning of this fuel will emit pollutants such as,

- Carbon particles, metallic dust particles, resins, tars, aerosols, solid oxides, sulphates and nitrates
- Nitrogen compounds such as NO, NO₂, N₂O
- Sulphur compounds such as SO₂, SO₃, H₂S
- Halogens like F, Cl, Br and I
- Radioactive substances such as γ particles, C-14, Sr-90, Cs-137

(d) Emission from vehicles

The automobile exhausts are responsible for more than 75% of total air pollution. The imperfect mode of burning of this fuel will emit pollutants such as,

- Fine particles such as carbon particle, metallic dust particles, resins, tars, aerosols, solid oxides, sulphates and nitrates etc, which are less than 100μ in diameter
- Coarse particles which includes carbon particles and heavy dust which are over 200μ diameter
- Nitrogen compounds such as NO, NO₂, N₂O
- Sulphur compounds such as SO₂, SO₃, H₂S
- Halogens like F, Cl, Br and I
- Radioactive substances such as γ particles, C-14, Sr-90, Cs-137

(e) Rapid Industrialization

- A large number of industries such as chemical industries, paper and pulp mills, cotton mills, metallurgical plants and smelters, petroleum refineries, mining and synthetic rubber industries are responsible for about 20% of air pollution.
- The most common pollutants are CO₂, CO, SO₂, H₂S, NO, NO₂
- In addition, smoke coming out from chimneys also contains small particles of dust, carbon, metals, other solids, liquids and even radioactive materials which get mixed in the smoke and pollute the air.
- Accidental release of poisonous gases due to carelessness of authorities and other faults, from the industries also pollute the air badly.

(f) Agricultural activities

- Different types of biocides such as pesticides, insecticides, herbicides cause air pollution.
- These poisonous substances carried away by wind to different places, during their spray over crops and thus make the air foul for human health and other animals.

(g) Wars

- The air pollution is caused by various types of explosives and nuclear explosions used in wars.
- Nuclear reactions, nuclear weapon testing, chemical processing plants, hospitals, research laboratories and modern technical institutions contribute numerous dangerous radionuclides to air.
- Nuclear effluents released into the air are potentially more hazardous to man, animals and plants.
- Several nucleides like C-14, Sr-90, I-131, Kr-85, Cd-137, H-3 pose a serious threat to atmosphere.

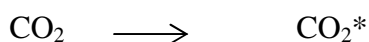
REACTIONS IN ATMOSPHERE

The entire atmosphere extends to about 200 km and 95% of the air molecules lie within 20 km on the earth's surface. Chemical reactions occurring in troposphere, stratosphere, mesosphere and thermosphere affect living organisms on the planet earth.

(a) Reactions in Troposphere

Troposphere is the most active zone of the atmosphere, in which the visible events of weather like rain, lighting and hurricane occur. CO₂ and water vapour of tropospheric environment exert a strong influence on climatic condition. They absorb considerable portion of the outgoing radiation and CO₂ molecules get changed into a vibrationally excited molecule.

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The excited molecule may collide with another molecule converting the excess vibrational energy into heat. This effect of CO_2 in increasing the earth's temperature is commonly known as Green House Effect.

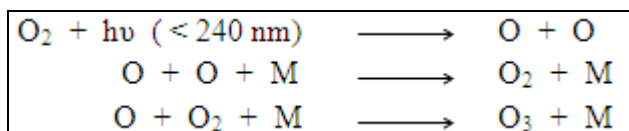
Water too, like CO_2 , absorbs infrared radiation, again resulting in rise of atmospheric temperature. But increasing amounts of particulates, dust, smoke and smog in the atmosphere can scatter and reflect solar radiation back into space, which will tend to lower the atmospheric temperature.

The chemistry of lower atmosphere consists of gases released by natural as well as man made sources at the earth's surface.

- The major emissions are those of SO_2 , H_2S , CO , CO_2 , N_2 , NO , NO_2 , NH_3 and hydrocarbons.
- CO_2 and N_2 are not reactive.
- H_2S , NH_3 and CH_4 are the products of anaerobic biological decay.
- H_2S is readily oxidized to SO_2 which is then oxidized to H_2SO_4 and comes down as acid rain.
- NH_3 is produced naturally, but it is oxidized to NO_2 and finally to HNO_3 which is also washed out by rain.
- Some NH_3 also reacts directly with acidic aerosols containing H_2SO_4 or HNO_3 and forms $(\text{NH}_4)_2\text{SO}_4$ or NH_4NO_3 .
- Atmospheric pollutants are present largely in the troposphere and lower stratosphere.
- A major burden of air pollution is carried by troposphere.

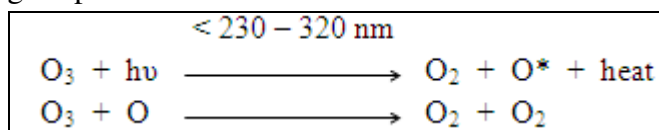
(b) Reactions in Stratosphere

Oxygen plays an important role in the troposphere, whereas ozone plays a significant role in the stratosphere. Ozone which is not very stable, is formed in the stratosphere by photochemical reaction of oxygen as,



where M is a third body like nitrogen or any other inert molecule, which cause decomposition of ozone.

Ozone so formed undergoes photo dissociation as follows:



The third body absorbs excess heat generated by the above reaction and stabilizes the ozone molecule. Thus there exists dynamic equilibrium between formation and decomposition of ozone molecule. Ozone acts as a protective cover against severe UV radiation. Without ozone, life on the planet would be rapidly destroyed.

The species of oxygen in the upper atmosphere include O_2 , O , O^* , O^+ and O_2^+ along with O_3 are formed by photochemical dissociation, ionization etc. For example,

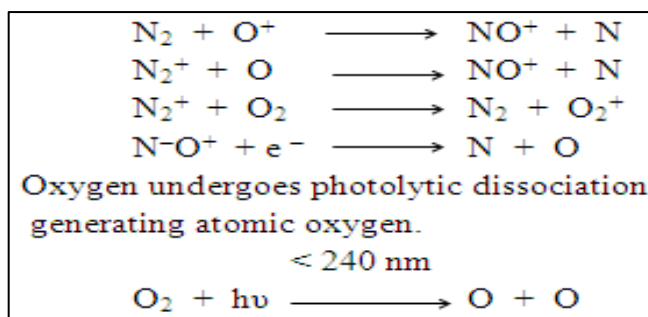


Now ozone depletion is taking place at tremendous rate due to NO_x , chlorofluorocarbons, SO_2 , CH_4 , etc.

(c) Reactions in Mesosphere and Ionosphere

Mesosphere lies above the stratosphere at an altitude range of 45-85 km. The ionosphere which overlaps the mesosphere extends from 50 km to thousands of km.

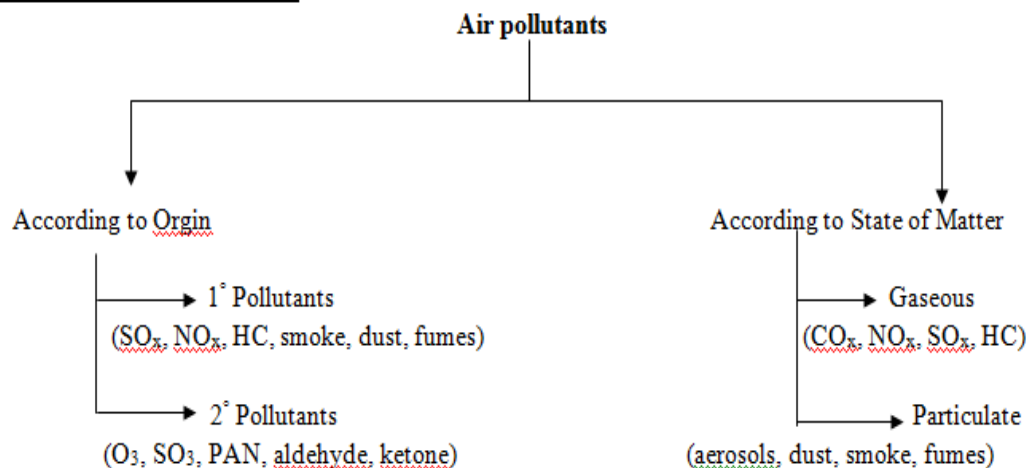
Ionic species, such as O_2^+ , O^+ and NO^+ are observed at a height of 85 km or above, which are produced by sun's UV radiation by photo ionization. The species so produced undergo ion-molecule reactions as follows.



Some of these species exist in electronically excited state and their decay to the ground state causes 'auroras' as observed in the northern hemisphere. Fast moving electrons and photons coming mainly from sun are responsible for excitation.

SOURCES, EFFECTS AND CONTROL MEASURES OF AIR POLLUTANTS

Classification of air pollutants



Among various Air pollutants, the major primary pollutants and most significant are,

- Oxides of Nitrogen (NO_x),
- Oxides of Sulphur (SO_x),
- Oxides of Carbon (CO & CO_2),
- Hydrocarbons (C_xH_y),
- Particulates,
- Chloro fluoro carbons

Title	Oxides of Nitrogen	Oxides of Sulphur	Oxides of carbon	Hydrocarbon
	NO, NO ₂ , N ₂ O	SO ₂ , SO ₃	CO, CO ₂	C _x H _y
<u>Sources</u>	<u>Natural Sources:</u> • cosmic rays emission • natural bacterial action <u>Man Made Sources:</u> • fuel combustion in automobiles and industries • byproducts of some chemical industries • rising fiber balls associated with atmospheric nuclear explosion.	<u>Natural Sources:</u> • volcanic eruption <u>Man Made Sources:</u> • fossil fuel combustion • industries • transportation • thermal power plants • manufacture of sulphuric acid • fertilizers • smelting industries • power plants, refineries.	<u>CO</u> <u>Natural Sources:</u> • forest fires, natural gas • marsh gas, volcanoes • decomposition of chlorophyll • hemoglobin breakdown • methane by decomposition of organic matter <u>Man Made Sources:</u> • auto mobile Exhaust • incomplete combustion of fossil fuels • burning of agricultural products • industrial processes • high temperature furnace <u>CO₂</u> <u>Natural Sources:</u> <u>Man Made Sources:</u> • volcanic eruptions • burning of solid fuels for domestic purposes • thermal power plants • Industries	<u>Natural sources</u> • trees emit huge quantities of hydrocarbons • eucalyptus, cotton wood, oak, sweet gum and spruce trees emit isoprene compounds • methane is produced by bacteria during anaerobic decomposition of organic matter in soil, water and sediments bacteria $[\text{CH}_2\text{O}] \longrightarrow \text{CH}_4 + \text{CO}_2$ <u>Man made sources</u> • automobiles, industrial processes, processing, storage and transfer of products • evaporation of organic solvents in air during industrial operation such as in paints, varnishes, lacquers, undercoating • gasoline hydrocarbons evaporate and get mixed with air, increasing their concentration in atmosphere

<u>Effects on plants</u>	Higher concentrations of NO₂ damages • leaves of plants • retard photosynthetic activity • cause chlorosis • leaf spotting • breakdown of plant tissues • affects metabolism in plant tissues of bean and tomato • NO_x leads to the production of secondary pollutants such as O₃, PAN and smog which damage the vegetation	Higher concentrations of SO₂ damages vegetable crops • affects plant growth • affect nutrient quality of plant products • acute exposure kills leaf tissues causing leaf necrosis • chronic exposure causes bleaching of leaf pigments • cotton, wheat, barley and apple are most sensitive • species like cucumber, oats and spinach are damaged even in low concentration • H₂SO₄ mists and aerosols are extensively toxic to plants and soil fertility • high levels of SO₂ reduce pH of leaf tissues of some trees	Higher concentrations of CO damages • leaf curling • reduced leaf size • leaf drop • premature and interference with cellular respiration • when exposed for long duration, it inhibits nitrogen fixation ability of bacteria • at 10, 000 ppm CO causes reduction in chlorophyll	• Hydrocarbons, photochemical oxidants are injurious to plants and it may cause chlorosis • acetylene, ethylene and propylene show extreme toxicity towards plants and damaging growth of vegetation. • ethylene hydrocarbons inhibit plant growth and damage leaf tissues and death of flowering plants
<u>Effects on Human beings</u>	• health effects of NO₂ vary with degree of exposure • 50-100 ppm - inflammation of lung tissues • 500-600 - death of victim • higher doses bronchitis and respiratory problems • NO bonds with hemoglobin and reduces oxygen	London or Sulphurous smog called killer smog had been associated with increase in SO₂ content in air It causes • eye irritation • headache • cough • breathing problem	effects of CO • it causes symptoms of CO poisoning at 200 ppm for about 8 hours, because the gas readily combines with hemoglobin for which it has about 200 times more affinity than oxygen and forms carboxy hemoglobin. • this results in hypoxia (oxygen	• hydrocarbons at high concentration (500-1000 ppm) have carcinogenic effects on lungs and cause swelling • aromatic hydrocarbons inhalation causes much irritation to mucus membrane • secondary pollutants like PAN produced by hydrocarbon results

	carrying capacity • cigarette smokes develop lung disease as it contains 350 to 1500 ppm of NO _x • nitric and nitrous acid cause respiratory, digestive and nervous ailments • higher NO _x cause inflammation, bleeding, pneumonia, oxygen deficiency and lung cancer • NO _x are extremely dangerous to human health and their acute effects are sometimes more severe than that of carbon monoxide	• chronic asthma • cancer • heart diseases • diabetes • allergy in skin	deficiency) and it causes reduced mental activity, headache, weak eye sight and irregularity in psychomotor activity • it leads to fertility problems, premature births, spontaneous abortions • cigarette smokers inhale CO • persons who are most likely to be exposed to CO are garbage workers, traffic policemen, vehicle drivers, urban shopkeepers	in formation of smog which causes irritation of eyes, nose, throat and respiratory distress • excess of hydrocarbon increases mucus secretion as a result of which respiratory tracts are blocked • group of hydrocarbons such as benzopyrene are carcinogenic and cause cancer, affecting DNA and cell growth • methane is a severe gas pollutant create narcotic effects on human • cyclic hydrocarbons affect nervous system, cause loss of coordination
<u>Effects on material</u>	• NO _x fades away a number of textile dyes • NO ₂ and hydrocarbons produces peroxides which combine with ozone and cause crack in rubber • NO ₂ produces aerosols which damage nylon fibers • oxides of nitrogen are involved in generation	SO ₂ attacks rapidly • the marble • limestone • roofing • slate • electrical contacts • textiles • paper and buildings acid rain produced by oxidation of SO ₂ corrodes metals & attacks fibers and washes out basic materials like lime from soil	<u>Effects of CO₂</u> CO ₂ is non-toxic; therefore it is not harmful to human health, unlike CO • CO ₂ is utilized by green plants to prepare starch during photosynthesis. It produces adverse physiological effects only at very high levels • CO ₂ molecules absorb heat energy and tend to prevent long wave infrared heat radiation from earth	• photochemical reactions increase number of hydrocarbons in atmosphere and ozone level increases • hydrocarbon combines with NO _x and forms photochemical smog, it affects textiles, paper, rubber, polymers • damages long chains of carbon atoms losing tensile strength of polymers

	secondary pollutants such PAN and peroxy organic compounds and responsible for photochemical smog which is toxic to materials • NO_x is responsible for production of nitric acid which is washed down as acid rain which reduces fertility of soil	• SO₂ causes acid rain which rapidly attack on marble and is known as Stone Leprosy $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{CO}_2 + \text{H}_2\text{O}$ • SO₂ polluted air accelerates corrosion rates of metals such as Fe, steel, Zn and Cu	from escaping into space and deflect back to earth • increased amount of CO₂ in air is mainly responsible for global warming • increased temperature due to global warming may also lead to melting of polar ice caps and hence flooding of low lying areas all over the globe may occur	
<u>Control measures</u>	• flame temperature is decreased by injecting re-circulated flue gases or steam • by two stage combustion processes: 1st fuel is fired at a high temperature and mixed with 90 to 95% of stoichiometric air, 2nd partial cooling and excess of air is injected with greater than air-fuel ratio - catalytic decomposition - chemical sorption - catalytic reduction - spark retard is an effective measure to control NO_x - catalytic exhaust reaction	• removal of sulphur from fuel before burning • use of low content sulphur fuels and removal of SO_x from flue gases • using other energy sources such as natural gas and use of nuclear power to generate electricity • SO₂ from flue gases removed by introducing limestone into combustion zone of furnace $\text{CaCO}_3 + \text{SO}_2 \xrightarrow{\text{O}_2} \text{CaSO}_4 + \text{CO}_2$ • smelting industries adopting some catalytic oxidation methods to convert SO₂ into H₂SO₄ which can use internally	• by engine modification to have fuel efficient (lean) mixtures to reduce CO emissions • using mass transport system • using non-conventional sources of energy	• control of hydrocarbon depends on control of primary precursors i.e. hydrocarbons and NO_x • hydrocarbons from auto exhaust emissions can be controlled by applying techniques like incineration, absorption, adsorption and condensation $\begin{array}{lcl} \text{Hydrocarbon} & \xrightarrow{\quad} & \text{CO} + \text{H}_2\text{O} \\ \text{CO} & \xrightarrow{\quad} & \text{CO}_2 \end{array}$ • adopting these methods, hydrocarbon pollutants can be converted to less harmful end products

PARTICULATE POLLUTANTS

Air borne small, solid particles and liquid droplets are commonly known as particulates. They vary in size ranging from a diameter of 0.0002μ to 500μ . Particulates possess large surface areas and hence appear to form attractive sites for sorption of various organic and inorganic matters.

Types of particulates

Particulate pollutants may be classified according to their optical properties, nature and size as follows:

(i) Smoke

Smoke particulates consist of solid and liquid particles ranging from 0.05 to 1.0 micron which are formed during incomplete combustion of carbonaceous materials.

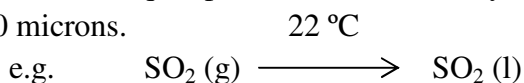
e.g. It includes smoke of gaseous pollutants like oxides of sulphur, oxides of nitrogen, carbon monoxide and hydrocarbons.

(ii) Dust

Dust is composed of fine solid particles and their size range from 1 micron to 100 microns. These particles are formed by (a) entrained materials in mechanical processing like sand from sand blasting (b) the particles directly from materials when it undergoes a mechanical operation like saw dust from wood works and (c) directly from the material being processed like coal.

(iii) Mist

Mist or liquid particles are formed by condensation of a vapour, having a size less than 10 microns.



(iv) Spray

It constitutes liquid particles obtained from the parent liquid by the process of mechanical disintegration like atomization.

(v) Fumes

Fumes are generally obtained by the condensation of vapours by sublimation, distillation, boiling, calcination and by several other chemical reactions.

e.g. Generally organic solvents, metals and metallic oxides form fume particles having less than 1 micron.

Sources of Particulates

(a) Natural sources

- volcanic eruption
- blowing of dust and soil by the wind
- spraying of salt and various other solid particles by the seas and oceans
- natural process injects nearly 800-2000 million tones of particulate matter each year into atmosphere

(b) Man made sources

- combustion of fossil fuels
- industrial operation like mining and polishing
- industries related with the production of pesticides, fertilizers, textiles and chemicals
- industrial fugitive processes
- non industrial fugitive processes such as construction of buildings bridges, dams
- agricultural activities, roadways dust, fire dust

- transportation sources
- fly ash from thermal power plants
- man made activities emit nearly 450 million tones of particulates every year into the atmosphere

According to the composition, particulate is divided into two types:

- (i) Organic particulate (ii) Inorganic particulate

(i) Organic Particulate matter (OPM)

It includes polycyclic aromatic hydrocarbons (PAH) like chrysene, benzo fluoranthene, benzo α -pyrene, benzidine etc. The size of OPM is $20 \mu\text{g}/\text{m}^3$.

Sources: OPM originates mainly from combustion of fuels, automobiles and vegetation.

Effect: (i) OPM is mainly carcinogenic in nature.

(ii) PAH compounds remain absorbed in soot particles. Soot particles act as a disease carrier containing toxic trace metals like Be, Cr, Mn, Ni, U, Cd, Fe and V.

(ii) Inorganic particulate matter

Inorganic particulate includes aerosols and flyash.

- (a) **Aerosol:** Smaller solid particles in the liquid droplets are called as aerosol.

Sources:

- Sea spray and dust are natural sources of aerosol.
- Man made activities of aerosols are from Industrial combustion and condensed organic vapour.
- The size of aerosols ranges from 1 nm to $100 \mu\text{m}$ in diameter.
- Aerosols include dust, smoke, fly ash, pollen grains, cloud, mist, fog, smog.

▪ **Condensed aerosol:**

The gases which are present in the atmosphere undergo chemical reactions and forms liquid of solid particles and it is known as condensed aerosol. Cloud, mist, smog are the examples for condensed aerosol.

▪ **Sea Sprays:**

Waves are produced in the sea due to the vigorous action of wind and number of bubbles are formed and collapsed frequently. This is known as white cap or sea spray or sea salt aerosol.

▪ **Aerosol dust:**

Solid materials like clay minerals and finely divided organic matter undergo fragmentation by the action of wind and mixed with atmosphere and are known as aerosol dust.

▪ **Aerosol mist:**

They are formed by the oxidation of SO_2 to H_2SO_4 by dust and forms cloud.



Tetraethyl lead which is present in leaded gasoline combines with oxygen and alkyl halides and forms condensed aerosols.

Effects of Aerosol

- It produces respiratory problems
- It reduces the visibility
- It is responsible for acid rain, photochemical smog

(b) Fly ash

The smaller ash particle which enters into the atmosphere and pollutes the air is known as fly ash.

Source

- combustion of fossil fuel
- industrial combustion processes
- ash contains the oxides of inorganic particulate matter

Effects

- it is carcinogenic in nature
- they cause asthma and respiratory problems

CHLORO FLUORO CARBONS

Chlorofluorocarbons are a class of compounds initially developed in 1930 to replace the toxic and inflammable refrigerants such as methyl chloride, ethyl chloride, propane and sulphur dioxide.

Freon-12, the new compound, CCl_2F_2 is marketed under the popular brand names such as “Freon” and “Genetron”.

Common CFCs and their formula:

Base	Type of CFC / Halon	Formula
Methane based	CFC – 11	CFCl_3
	CFC – 12	CF_2Cl_2
	CFC – 22	CHClF_2
Ethane based	CFC – 113	$\text{CF}_2\text{Cl} - \text{CFCl}_2$
	CFC – 114	$\text{CF}_2\text{Cl} - \text{CF}_2\text{Cl}$
	CFC – 115	$\text{CF}_2\text{Cl} - \text{CF}_3$
Halon	H – 1211	CF_2BrCl
	H – 1301	CF_3Br
	H - 2402	$\text{CBrF}_2 - \text{CBrF}_2$

Properties of CFCs and Halons

Production and consumption of large quantity of CFCs and halons are particularly due to their exceptional properties such as:

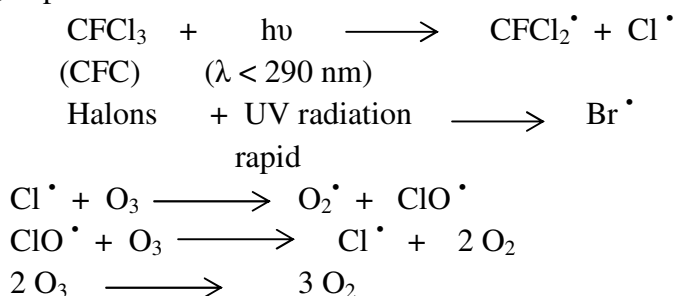
- Low toxicity
- Low chemical reactivity
- Non-inflammability
- Inexpensive
- Efficient
- Easy to handle

Sources, Uses of CFCs and Halons

- Propellant Refrigerant
- Air conditioner Cleaning agent
- Sterilizing agent Fast food packing
- Fire extinguisher Blowing agent Solvent

Harmful effects of CFC – Ozone depletion by CFC

CFC is unreactive biologically and chemically in the earth as well as in the troposphere. But in the stratosphere it undergoes photolytic decomposition by UV light and it causes the O₃ depletion.



Harmful effects of CFC

- Decreasing the ozone level all over the globe and not just in Antarctic or Arctic.
- CFC decreases number of O₃ in ozone layer and therefore it allows UV radiation to strike on earth.
- CFCs are the major component of green house gases and are responsible for global warming.
- Because of ozone depletion, the amount of H₂O₂ in the stratosphere increases and causing acid rain.
- Ozone depletion and increased ultraviolet solar radiation would also increase photochemical processes which in turn would create poisonous urban smog.

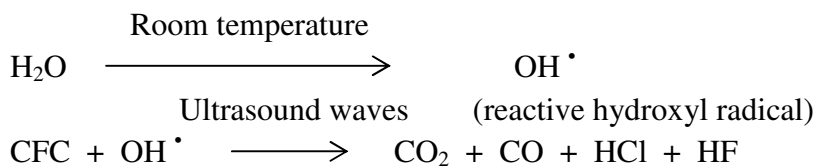
Control Measures

Alternative technology and substitute chemicals for CFC

1. HFC (Hydro fluoro carbon)
2. Cyclopentane
3. A mixture of propane and isobutene
4. LPG

Safe disposal of CFC

- Efforts are on towards disposal of ozone depleting CFCs because large amounts of ozone depleting substances are being collected from aerosols, insulation foams and cooling units for their final finish.
- Breaking down this non-inflammable and inert chemical requires expensive methods involving high temperatures and jets of hot ionized gases.
- A group of Japanese Scientists have discovered that CFCs can be broken down in water at room temperature using ultrasound waves at a frequency of 200 kilohertz.



The product formed can be easily disposed off. Researchers found that the technique destroyed 80% of the dissolved CFC-113, the commonest of CFCs, within half an hour, increasing the temperature of water from 22° C to 30° C only.

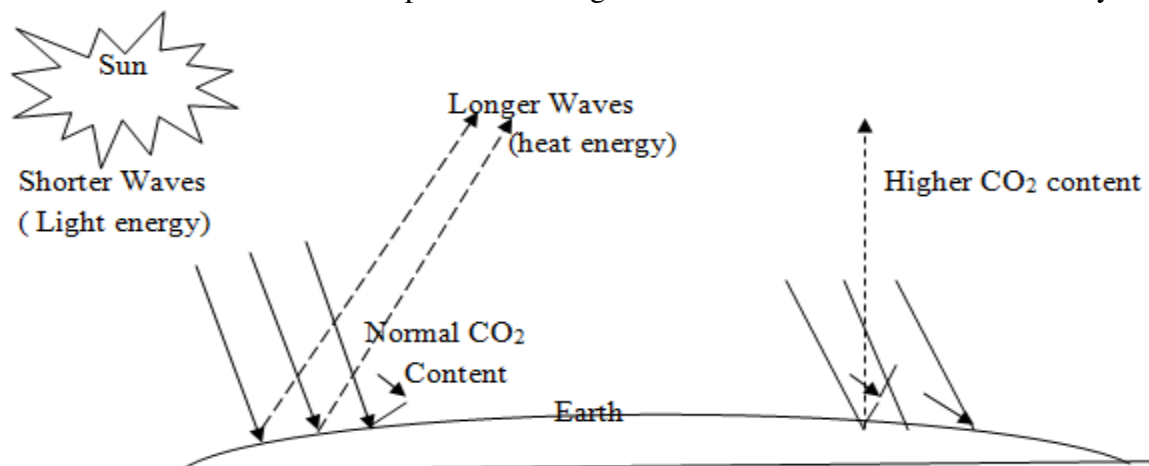
GREEN HOUSE EFFECT

Greenhouse is the body which allows the short wavelength incoming solar radiation to come in, but does not allow the long wave outgoing terrestrial infra red radiation to escape.

Green House Effect

The earth's atmosphere bottles up the energy of the sun, and is said to act like a greenhouse, where CO₂ acts like glass windows. CO₂ and water vapours in the atmosphere transmit short wavelength solar radiation but reflect the longer wavelength heat radiation from warmed surface of the earth.

These CO₂ molecules are transparent to sunlight but not to the heat radiation. So they trap



and re-emit the solar heat stimulating an effect which is popularly known as Greenhouse effect. **“The progressive warming up of the earth's surface due to blanketing effect of manmade CO₂ and other gases in the atmosphere”.**

Sources of Green House Gases

Green house gas	%	Concentration (ppm)	Sources
CO ₂	50	356	- Volcanic eruption - Power stations based on burning of fossil fuels - Burning of fire woods and deforestation - Factories and automobiles (diesel, petrol)
CH ₄	19	1.7	- Leakage from coal mines and natural gas - Incomplete combustion of fuel, Anaerobic bacteria from cow and sheep's mouth, Rice, paddies
CFC	17	CFCl ₃ (CFC-11) 2.3 x 10 ⁻⁴ ppm	Propellant, Refrigerant, Air conditioner, Cleaning agent, Solvent, Sterilizing agent, Fast food packaging, Fire extinguisher, Blowing agent
O ₃	8	1.3	- Reaction between hydrocarbons and NO_x - Reaction between NO₂ and photon (hν) - Deforestation,
N ₂ O	4	0.3	- Nitrification (ammonia to nitrate) - Denitrification (nitrate to dinitrogen) - Burning of fossil fuels, Deforestation - Microbial action on nitrogenous fertilizers in soil
Water vapour (Aerosols)	2	0.15	Incomplete combustion products of the industry deforestation

Greenhouse Gases

The four major greenhouse gases, which cause adverse effect are, Carbon dioxide (CO₂), Methane (CH₄), Nitrous oxide (N₂O), Chlorofluorocarbons (CFC). Among these CO₂ is the most common and important greenhouse gas.

Global Warming Potential (GWP)

The potential of a greenhouse gas to cause greenhouse warming is expressed by “Global Warming Potential” (GWP). It is a function of both the infrared sorption characteristics and the lifetime of the gas. The greenhouse gases can be arranged in GWP sequence as follows:

$$\begin{array}{ccccccc} & 10000x & & 150x & & 25x & \\ & \text{CFC} & > & \text{N}_2\text{O} & > & \text{CH}_4 & > & \text{CO}_2 \end{array}$$

Green house Effect on Global Climate

- Earth temperature rises, oceans get warm up and sea level would raise flooding low lying regions.
- Rainfall will be enormous but problems of desertification, drought and soil erosion further worse.
- In temperate regions, winter will be shorter and warmer and summer will be longer and hotter.
- The tropics may become wetter and the subtropics are expected to be drier.
- Plants and animals will also be affected resulting in disruption of the whole ecosystem.
- Most obvious effect of climate changes will be on agriculture.
- Because CO₂ is a natural fertilizer, plants will grow larger and faster with increasing CO₂ in the atmosphere. But with increase in the yield, soils may become impoverished or poor more rapidly.

Consequences of Green House Effect and Global Warming

- Temperature effect of CO₂ and water vapour combined has long range impact on global climate.
- With rise in level of CO₂, temperature on the earth's surface rises causing more evaporation of surface water, leading to further temperature increase.
- Thus it is expected that the combined effect will bring 2 to 3° C rise in surface temperature for doubling of CO₂ concentration around 2080 AD.
- It has been estimated that with this rise in surface temperature, results in receding many glaciers, melting of ice caps in polar regions.
- With increase in concentration of CO₂ and much warmer tropical oceans, there may be more cyclones and hurricanes and early snow melt in mountains will cause more floods during monsoon.
- If unchecked, it could alter earth's temperature, rainfall and rising levels of sea.
- The U.S. would produce less grain while Mexico, India, Africa would be warmer and wetter, enabling them to produce more grains.
- It is also reported that rice-growing season and area under rice cultivation could increase.
- A slight increase in global temperature can adversely affect the world food production.
- Biological productivity also decreases due to warming of surface layer.

Examples of Consequences of Global Warming

- In India, Gangotri glacier which feeds rivers like Ganges is retreating by 18 meters per year and already lost one third of its 15 mile length in last 40 years.

- At Antarctica, shrinking sea ice has reduced the population of Adelie penguin by 33% in 25 years
- Arctic ocean, the area covered by sea ice has declined by about 6% between 1978 and 1995.
- Global warming is responsible for melting more than 50 billion tones of water per year from Greenland ice sheets, leading to a 59 cm rise in global sea level over the last century.
- Australia's greenhouse gas emissions have increased by at least 17% during 1990-98 and a storm dumped nearly 30 cm of rain on Sydney during August 15, 1998. This is three times more than the total rainfall of that month.
- In Canadian Arctic, the Peary Cariban population has decreased due to heavy rainfall from 24,000 in 1961 to 1100 in 1997.
- In Alaska, the average number of snow less days in summer in Barrow has increased from fewer than 80 in 1950s to more than 100 in 1990s.
- In Spain more than half million hectares of forest went up in flames during a severe drought. In Indonesia, upto 0.8 million hectares of land burned in 1998.
- New York city had its warmest day in 1999 and driest July on record, with temperature climbing above 35° C for 11 days.
- In Colombia, mosquitoes carrying dengue in the Andes now appear even at 2195 meters and formerly limited to not higher than 1000 meters.
- In Kenya, hundreds of people died in 1997 due to malaria where previously not experienced.
- Global warming could worsen the rain fall, transport systems, allergies, vegetation, world economy and is threatening the Earth's thermostat.

Control and Remedial Measures of Greenhouse Effect

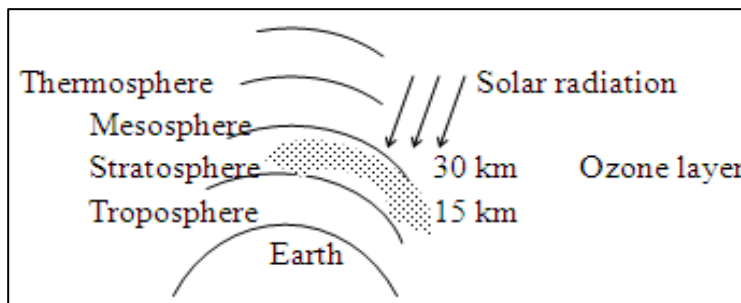
- Reduce the consumption of fossil fuels such as coal and petroleum.
- Non-conventional renewable sources of energy such as wind, solar, nuclear and bio-gas energies.
- Disposing off the greenhouse gases in the atmosphere.
- International co-operation for attempting the reduction of green house gases
- Learn to adapt and accept the changing climate.

The following measures may also be adopted to reduce the ever increasing greenhouse effect:

- CO₂ level can be decreased by drastic cut in the consumption of fossil fuels in the highly developed and industrialized countries.
- There should be restriction on emission of CO₂ and CFCs, from factories and automobiles.
- Advanced researches are necessary to assess and evaluate potentials of methanol, obtained from methane, which can be used in transport sector.
- By the use of unleaded petrol, catalytic converters and other antipollution devices in automobiles and industrial plants.
- In tropical and subtropical countries, where sunlight shines during most of the period in year, the solar energy may be developed as alternative to conventional fossil fuels.
- Biogas is another alternative source of conventional energy for domestic use.
- Enhancing forestation will certainly reduce the CO₂ level as trees are the big natural sink of CO₂, they can utilize CO₂ during photosynthesis.

OZONE LAYER

Ozone is naturally occurring gas found throughout the atmosphere, with a maximum ratio at altitudes ranging from 15 to 40 km above the earth. This region is frequently known as **ozone layer**. Ozone concentration differs by about 100 ppm in the stratosphere compared to 0.05 ppm in the troposphere.



This presence of ozone layer in stratosphere is of vital significance for all biota, because the harmful solar radiations, such as ultraviolet rays, which are lethal to life on the earth, are not allowed to enter the earth's atmosphere by ozone layer or ozone umbrella. In the absence of this layer, all the ultraviolet rays of the sun will reach the earth's surface and consequently the temperature of lower atmosphere will rise to such an extent that the biological furnace of the biosphere will turn into a blast furnace.

Thus ozone layer strongly absorbs or blocks the short wave ionizing ultraviolet rays between 200 nm and 315 nm and so protects the life on earth from severe radiation damage. The thickness of ozone layer is measured in Dobson Units (DU), where 1 **DU** = **0.01 mm of the compressed gas at 0°C and 760 mm Hg pressure.**

Formation of Ozone Hole or Ozone Depletion

A world average value of ozone layer is approximately 300 DU, ranging from 250 in tropics to 450 in more northern and southern regions of the planet. The levels above 200 DU are considered to be normal. When ozone level decreases below 200 DU, the condition generally referred as ozone hole. The term ozone hole is used to describe the thinning of the region of the ozone layer.

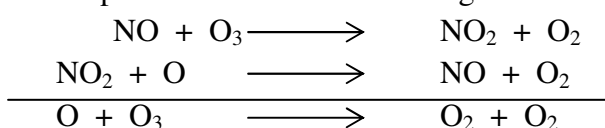
Mechanism of ozone depletion

Natural Process

A dynamic equilibrium existing between the production and decomposition of ozone molecules constitutes one of the most important mechanism.

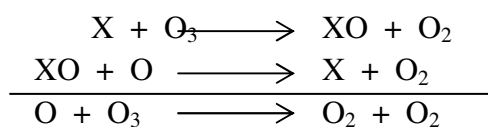
The natural process of ozone depletion involves the conversion of atmospheric nitrogen into NO_x due to solar activity.

The nitric oxide cycle contributes in a significant way in the natural destruction of stratospheric ozone in the lower region of the stratosphere.



Anthropogenic process of depletion of ozone

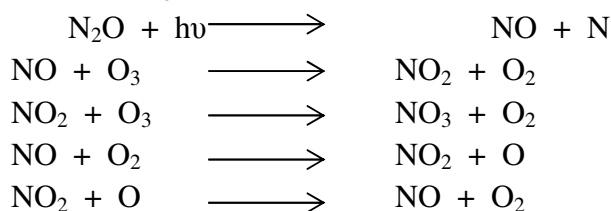
The depletion of ozone layer, have a general mechanism of the following type.



Here X is the reacting species. The most common of these species have been identified to be free radicals such as HO_X (H[•], OH[•], HOO[•]), NO_X (NO[•], NO₂[•]) and ClO_X (Cl[•], ClO[•]). Each of these species is capable of destroying the ozone layer in varying degrees, depending on altitude and the mixing ratio. Thus, near the stratopause the HO_X radicals are responsible for 70% of total ozone destruction. Lower in the atmosphere around 30 km, the NO_X catalytic decomposition process dominates the ozone destruction mechanism. The NO_X cycle in the region just above the tropopause has been found to be responsible for about 70% of the ozone depletion.

(a) Nitrogen oxide (NO_X) cycle

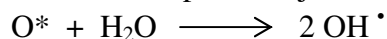
Supersonic aircrafts fly in stratosphere at altitudes of about 17 to 20 km because of low air resistance which is essential to maintain the speed. Their exhaust gases directly provide water vapours and NO_X into the ozonosphere. The supersonic transport (SST) fleet could add significant quantities of oxides of N₂ leading to about 40% reduction in ozone concentration. Nuclear explosions also produce large quantities of NO_X which directly enter into stratosphere. Studies indicate that nuclear tests conducted by USA and USSR also reduced O₃ concentration.



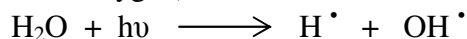
Species of molecules such as NO₂, *OH, NO₂ are highly reactive. The net result is that NO_X increases the rate of O₃ destruction with no change in the concentration of NO. This catalytic cycle could go on indefinitely, causing reduction of a large number of O₃ molecules.

(b) Hydroxyl radical (HO_X) cycle

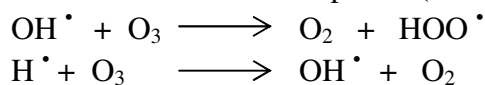
The HO_X radicals are generated by photochemical reactions and from water vapours released from supersonic jets exhaust.



(excited oxygen)



These reactive free radical species (OH and H) decompose ozone as,

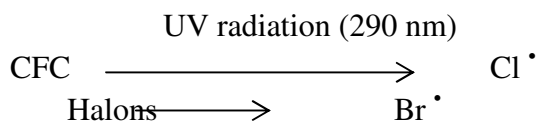


(c) Chlorine radical (ClO_X) cycle

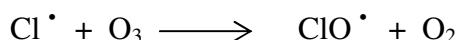
The CFCs and halons are entirely man made with wide application in air conditioning, refrigeration, aerosols, electric and metal cleaning, foam blowing and modern fire fighting. They act as a catalyst in the destruction of ozone layer in stratosphere. A wide range of

chlorofluorocarbons are known, but widely used are CFCl_3 and CF_2Cl_2 . Apart from CFC, volcanoes also inject HCl into the stratosphere to produce free chlorine atom.

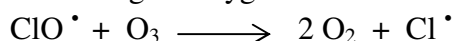
CFCs are almost completely inert both biologically as well as chemically in the troposphere and it takes about 20-40 years to escape into the stratosphere. Once these CFCs and halons reach at higher altitudes, they are capable of undergoing UV photolytic decomposition. The ultraviolet light breaks up the tight chemical bonds of CFCs and halons and a chlorine or bromine radical is set free.



These radicals act as catalyst in breaking down ozone into oxygen molecule and chlorine monoxide.



The chlorine monoxide formed is eventually split up by interaction with another ozone molecule to give oxygen molecule and a freed chlorine atom.



This chlorine atom again repeats the cycle of destruction of ozone in the stratosphere.

(d) Sulphate hypothesis

Sulphates aerosols emitted through volcanic eruptions and chimneys of several factories accumulate in atmosphere mainly between the altitudes of 15 km to 22 km. These sulphates aerosols catalyze the transformation of O_3 to oxygen ($\text{O}_2 + \text{O}$), thus depleting its concentration.

EFFECTS OF OZONE DEPLETION ON ENVIRONMENT

It is universally accepted that the ozone layer in stratosphere protect us from harmful UV radiations coming from sun. Ozone accounts for only three parts in ten millions of Earth's atmosphere, but plays several crucial roles in the radiation balance of the planet.

(a) Effect on Plants

- Plant proteins are susceptible to UV injury, as they absorb around 280 nm.
- 20-50% chlorophyll reduction and harmful mutation.
- Intense UV radiation causes greater evaporation of surface water through the stomata of leaves and decreases the soil moisture content.
- Increased UV-B radiation reaching the earth through depleted ozone layer &
- Reduce the effectiveness of plant photosynthesis as well as multiplication of phytoplankton.

(b) Effect on Human beings

- There is danger of increase in the flux of UV radiation over earth's biosphere.
- The range of wavelengths particularly affected by the changes in atmospheric ozone is from 290 nm to 320nm. All the known effects of these radiations are harmful for man's life.
- UV radiation-the narrow spectral band which is thought to cause most biological damage-appears to trigger two quite distinct immunological effects.

- One is confined to patches of skin that are actually irradiated, while the other develops in the immune system as a whole.
- Each 1% reduction in O₃ may cause 6% increase in skin cancer cases.
- Skin cancer are rapidly climbing in human diseases caused by UV rays.
- UV radiations break langerhans cells in the epidermis of human skin and break down the defenses in the skin. Thus sun-bathing on the beach may not be a pleasure anymore.
- UV radiations cause blood vessels near the skin's surface to carry more blood, making skin hot, swollen or red, causing sun burns and skin ageing.
- UV radiations are also absorbed by cornea and lens in the eye leading to photo keratitis and cataracts.

(c) Effect on Biotic Community

- Many micro-photo planktons would die.
- Thus reduction in photo plankton adversely affect zoo planktons and marine animals, fishes etc. will starve in the absence of sufficient supply of food.
- Loss of fish population would directly affect the inhabitants of coastal areas.
- On microorganism, irreversible and photo reversible types of injury are caused.

ANTARTIC AND ARCTIC OZONE HOLE FORMATION

- ❖ Ozone hole is actually an area of thinner than normal ozone. Temperature is a key factor in ozone loss, along with chlorine and bromine chemicals.
- ❖ Ozone depletion is more in Antarctic than to other regions. This is because in polar regions a special type of cloud called **polar stratospheric cloud** is formed and it leads to more ozone depletion.
- ❖ Ozone depletion is annual phenomenon over the South Pole during August and September when coldest temperatures are prevalent. This clouds in these cold conditions and chemical reactions on cloud particles help chlorine and bromine gases to rapidly destroy ozone. By only October, temperature typically starts to warm and the ozone layer starts to recover.
- ❖ Size and duration of the hole depend on natural year to year variability of atmospheric conditions.
- ❖ In Antarctic and Arctic, conditions exist where spring time ozone depleting events may occur, resulting in a major loss of ozone over a relatively short time.
- ❖ The conditions include a complex combination of climate factors as well as the accumulation of reservoir species which react vigorously to bring about a major loss of ozone, especially at the onset of the first day light of the polar spring. During winter the ozone level begins to recover to pre-hole levels.

SMOG

Smog is a general term referring to air pollution in which atmospheric visibility is partially obscured by a haze consisting of solid particulates and liquid aerosols. The name smog is derived from a combination of Smoke + Fog.

Two types of smog have been recognized.

- (a) Sulphurous Smog or London Smog
- (b) Photochemical Smog or Los Angeles Smog

(a) SULPHUROUS SMOG or LONDON SMOG

Sources

- Combustion of coal and burning of wood and agricultural wastes.
- It is also called as **Classical smog**, because it is associated with use of traditional fuel, coal.
- **Sulphurous smog** is so named as smoke formed mix with fog in presence of SO₂ only forms smog which is irritating in nature.
- Most severe air pollution ever occurred was due to sulphurous smog caused in London during December 1952 and last for several weeks led to the death of more than 4000 persons and made thousands ill. Thus the smog is called as **London smog**.
- Mixture of smoke, fog and SO₂ which affected London so badly for several years after introduction of coal as a fuel, is chemically a reducing mixture, so it is also called as **Reducing smog**.

Components of Smog

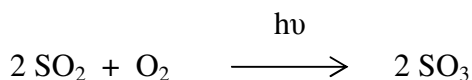
- SO_x
- Particulates, such as soot, ammonium sulphate
- Humidity or water from fog

Mechanism of Sulphurous smog

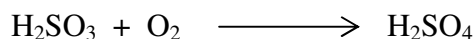
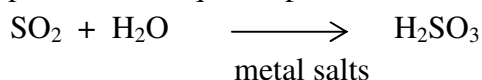
- Most widespread disaster due to SO_x occurs when it is accompanied by smoke.
- The combination occurs during temperature inversion.
- It is generally worst in early hours of morning and appears to be worsening shortly after sunrise, because of photo chemically induced oxidation of SO₂ into sulphur trioxide followed by combination with moisture to give acidic aerosol.



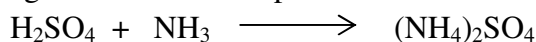
Formation of SO₃ takes place in presence of sunlight or particulates.



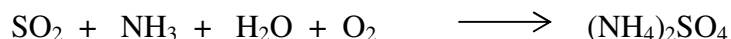
SO₂ may combine with moisture from fog to form sulphurous acid. Oxidation of this to sulphuric acid is quite rapid in metals salts.



Sulphuric acid so formed may reacts with particulate matter such as soot, ammonia to form smog of ammonium sulphate.



Some times SO₂ combines with NH₃ in moist air.



Effect of Sulphurous smog

- London smog produces much more dangerous effects than the sum total of SO₂ and particulates.
- It constitutes an example of synergistic effect.
- It causes damage to plant, poses human health hazard and creates numerous corrosion problems.
- Sulphates are found to lower the temperature by reflecting away incoming solar radiation

- Responsible for boosting the number of cloud droplets, thereby increasing cloud reflectivity.
- This decrease in temperature due to sulphate aerosols and is regional.
- It creates acid rain and depletes ozone layer.

(b) PHOTOCHEMICAL SMOG or LOS ANGELES SMOG

- Los Angeles in California experienced a new and serious form of air pollution during 1944 which was characterized by reduced visibility, eye irritation and plant damage
- It is not related to smoke or fog.
- It is worst in the sunshine and it peaks in the afternoon.
- This smog is called as **Los Angeles Smog**.
- It is also called as **Photochemical smog** because an important reaction, dissociation of NO_2 , is caused by sunlight

Sources

- emissions from petroleum combustion
- principally from motor vehicles
- Followed by a sequence of chemical and photochemical reactions taking place under specific conditions.
- The smog contains high levels of oxidants and carbon containing reaction products.
- Photochemical smog requires the presence of unburned gaseous hydrocarbons and nitrogen oxides,
- Both of which are emitted by internal combustion engines.
- Specific climate conditions are also needed like warm temperature.

Components

- Inorganic gases, such as ozone, nitrogen oxide (NO_x), hydrogen peroxide and carbon monoxide
- Organic peroxides, organic hydroperoxides, peroxy acetyl nitrate (PAN), peroxy benzoyl nitrate (PBN), peroxy propionyl nitrate (PPN)
- Other components are peracetic acid, acetyl peroxide, ethylhydro peroxide, n-butylhydro peroxide and tertiary butyl hydroperoxide

Mechanism of Photochemical smog formation

- The formation of oxidants, ozone, PAN and NO_x are indicative of smog formation.
- Photochemical reactions during smog are initiated by UV light.
- Photochemical reactions start with the combination of N_2 and O_2 in the air.
- They do not react with each other at ordinary temperature, but when the temperature rises to the levels in hydrocarbon combustion flames, they react to form nitric oxide. These are the primary photo chemical products leading to smog formation.
- Hydrocarbons, which are emitted in automobile exhausts, are reacting by free radical mechanism either with nascent oxygen or ozone to form secondary pollutants such as aldehydes, ketones and organic peroxides.

Step I - Formation of Hydroxyl radical

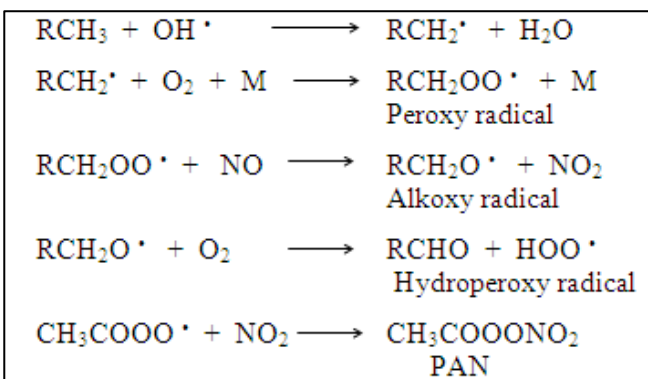
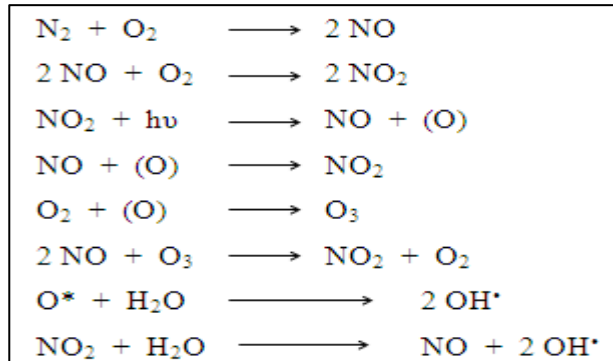
- The chemical reactions for smog formation generally centre on hydroxyl radical.

The hydroxyl formation is because of the reaction sequence which starts with the formation of nitric oxide.

- The excited oxygen or NO_2 react with water vapour to form hydroxyl radical.

Step II - Oxidation of Hydrocarbons

- The vehicle exhausts emit unburned volatile hydrocarbons and these are oxidized through reactions initiated by hydroxyl radical.
- Such a photochemical smog contains primary pollutants as NO_x and hydrocarbons as well as secondary pollutants such as O_3 , PAN, HNO_3 , particulate nitrate, aerosol particles and sulphate.
- Hot climate ($>24^\circ\text{C}$) with low humidity is favorable for the formation of photochemical smog.



Effects of Photochemical smog

(a) On Plants

- PAN, PBN, PPN which are produced photochemically is very damaging to plants, attacking younger leaves
- Ozone, which is the most toxic photochemical oxidant, cause injury to plants, bleaching on the upper surface of leaves, leaves tip burn etc.
- Smog contains ozone, PAN etc. produce early maturity in plants.
- NO_x and PAN cause death to forest trees.
- Ozone is also harmful for vegetables like radish, carrot, tobacco and carnation.

(b) On Human being

- NO_2 causes nose and eye irritation and pulmonary discomfort.
- ozone and peroxy acetyl nitrate (PAN), peroxy benzoyl nitrate (PBN) cause irritation of the eyes creating lachrymation
- affect severely the respiratory tract of human beings.
- Aromatic hydrocarbons vapours are much more irritating to mucous membranes and their inhalation causes systematic injury to trachea.
- They cause spread of diseases like asthma and bronchitis in epidemic form.
- People also suffered from emphysema and total surface area available for gaseous exchange is reduced causing chronic breathlessness resulting in death.

(c) On Materials

- Degradation of materials - "weathering"
- Smoke containing fog, dust, mist and soot in the smog reduces the visibility
- causes corrosion of metals, stones, buildings, textile, paper, rubber, leather and painted surfaces.

- Many organic polymers including rubber, natural and synthetic textiles materials are subjected to chemical alteration upon exposure to varying quantities of ozone.
- Cause carbon chain breaking and cross linking, making material less elastic and more brittle.

Control of Photochemical pollutants

The control of primary precursors, such as NO_x and hydrocarbons will ultimately control ozone and PAN which are secondary pollutants

(a) Chemical method

By spraying some chemical compound in the atmosphere which contains smog and they release free radical. These free radicals react with smog and collapse it. Example: Diethyl hydroxyl amine.

(b) Petrol reformulation

Petrol is mixed with additives such as methyl tertiary butyl ether which allows the petrol to burn completely and efficiently at low temperatures and decreasing the emission of NO_x and hydrocarbons.

(c) Catalytic converters

The use of catalytic converters in automobiles reduces the release of hydrocarbons and NO_x . Various techniques like incineration, absorption, adsorption and condensation methods are employed to control hydrocarbon emission from stationary sources.

ACID RAIN

Normal rain water is always slightly acidic because of the fact that CO_2 present in atmosphere gets dissolved in rain forming carbonic acid (H_2CO_3). The pH is approximately 5.6

Due to presence of SO_2 and NO_2 gases as pollutants in atmosphere, pH of the rain is further lowered, often to as low as 2.4 pH and this type of precipitation of lower pH is called Acid Rain. Literally it means the presence of excessive acids in rain water.

Source of acid rain

Natural source

- Volcanoes
- Swamps
- Plankton in the oceans
- Lightning

Man made activities

- combustion of fossil fuels automobile exhaust coal fired thermal power plants
- steel industry domestic fires smelters

Formation of acid rain

Actually acid rain is the one phase of acid deposition which can either be wet or dry.

• Precipitation by dry deposition:

Dust particles containing sulphates and nitrates, settled on the earth is called dry deposition.

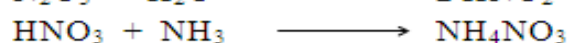
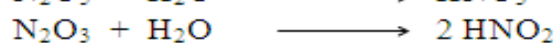
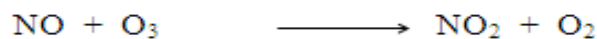
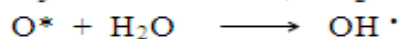
• Precipitation by Wet deposition:

Acid rain, snow, dew, fog, frost and mist represent the wet form of deposition. However, the wet rain is much more common compared with dry deposition.

THEORY OF ACID RAIN

(a) Formation of Nitric acid in atmosphere

- In day time, nitric oxide is oxidized by O_2 , O_3 and $ROO\cdot$.
- The NO_2 so formed, subsequently contributes to ozone and $OH\cdot$ radical formation which is partially responsible for smog. In this process, Nitric oxide is regenerated and available once again.
- The mechanism for the removal of nitrogen oxides from the atmosphere is by conversion to nitric acid, through oxidation of NO_2 by $OH\cdot$ radical.
- According to another sequence of photochemical reactions in the atmosphere, the nitrate radical can also react in the following way.
- Under humid conditions of air, N_2O_5 invariably reacts with water vapour to form droplets of HNO_3 . Some HNO_2 is also formed.
- Nitric acid is removed from the atmosphere by wet or dry deposition and is one of the main contributors to acid rain or acid precipitation. Nitric acid may also react with NH_3 , the principal source of which is urea in animal urine and other organic reduced nitrogen sources.
- NH_4NO_3 act as condensation nuclei for the formation of water droplet or may be deposited as solid aerosol.

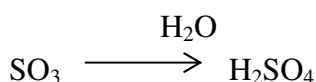
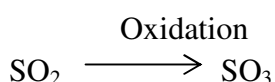


(b) Formation of Sulphuric acid in atmosphere

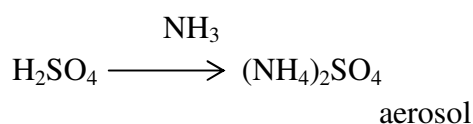
Formation of sulphuric acid in the atmosphere can take place with a wide range of reduced as well as partially oxidized sulphur compounds such as H_2S , COS , CS_2 , CH_3SH etc. These compounds are released from the oceans and from soils under reducing conditions because of various microbiological processes.

Volcanic eruptions release SO_2 directly into the atmosphere, increasing the density of the aerosol leading to significant depression in the global temperature by blocking solar radiation. Sulphur compounds in the stratosphere also take part in chemical processes causing diverse environmental consequences.

By means of the oxidation processes it is possible to oxidize any reduced sulphur compound. One of the main products is SO_2 . The foremost chemical reaction of SO_2 in air is oxidation to SO_3 which reacts with moisture to give H_2SO_4 . The H_2SO_4 or its sulphates occur as aerosols.



acid rain



Starting with SO₂, oxidation step appears to proceed in homogeneous or heterogeneous reactions that produce sulphuric acid.

Effects of Acid Rain

Acid rain exerts both direct and indirect effects on the organisms and materials it comes in contact with. The direct effects of acid rain are determined by concentration of pollutants in the air.

(a) On Aquatic Biota

- Significant reduction in fish population.
- Decrease in the variety of species in food chain.
- There are 15,000 fishless lakes in Sweden because of increased acidity.
- About 237 lakes in the Adirondack region of USA have a pH below 5.
- It can also leach aluminium from the soil. It is highly toxic to aquatic animals
- Northern pike and chain pickerel are quite tolerant to low pH levels.
- Bacteria and blue green algae are killed due to acidification, disrupting whole ecological balance.
- Destroyed trees in a wide wrap across Europe.

(b) On Terrestrial Ecosystem

- In 1958, pH of the rain at Europe was 5.0 and in 1962 the pH of rain at Netherlands was 4.5.
- This acid rain had damaged leaves of plants & trees and had retarded the growth of Swedish forests.
- In West Germany, about 10% of forests died and 18 million acres of forests were severely affected.
- Actually nutrients like calcium, potassium, iron and magnesium have been leached away from soil by acids which are most essential for plant growth.
- Reduced rate of photosynthesis and growth and increased sensitivity to drought and disease.
- North America and Europe, it destroyed crops and forests, reducing agricultural productivity.
- Retarded growth of crops such as pea, beans, radish, potato, spinach and carrots.
- The activity of symbiotic nitrogen fixing bacteria present in the nodules of leguminous family is inhibited, thereby destroying the fertility of the soil.
- Root systems are damaged by the uptake of aluminium released from the soil.
- Nitrates may be leached from the soil by acid run off waters.
- Acid deposition weakens the trees like pine, spruce, ashes and birch which can be easily attacked by pathogens and drought.
- Acidification of soil changes its biology and chemistry.
- Plants can easily absorb cadmium from the acidified soil and its high levels are injurious for animals and human beings.

(c) On Lake Ecosystem

- It causes complications in ponds, rivers and lakes.
- In summer, rapid snow melt gives jolt of acid water to lakes.
- This acid jolt damages to young fish, algae, and insects and to the food chain.
- Acid lakes have low levels of phytoplankton.
- Snails, clams, oysters etc having their shells of calcium carbonate are affected.
- Black flies, mosquitoes, deer flies, dragon fly larvae, water boatman and other aquatic worms occur abundantly where fishes are eliminated.
- The activity of the bacteria and other microscopic animals is reduced in acidic water.
- Dead materials and accumulated substances lying on bottom of lakes are not rapidly decomposed.
- Thus essential nutrients as nitrogen and phosphorous stay locked up in plant and animal remains.
- Thus Biomass production is reduced and fish population declines.
- Aquatic plants such as broad-leafed pond weeds do not grow in acid water. This affects the feeding and breeding habits of aquatic species.

(d) On Buildings

- It causes extensive damage to buildings and structural materials of marble, limestone and mortar.
 - Limestone is attacked rapidly and the attack on marble is termed as “**Stone leprosy**”.
- $$\text{CaCO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$$
- In Greece and Italy, invaluable stone statues have been partially dissolved by acid rain.
 - Taj Mahal in Agra is suffering due to SO_2 and H_2SO_4 acid fumes released from Mathura refinery.
 - Acid rain also corrodes houses, monuments, statues, bridges, fences and railings that cost the world 1500 million dollars a year.
 - British parliament building also suffered damage due to acid rains.
 - Due to acidity, levels of heavy metals such as Al, Mn, Pb, Cd, Cr and Cu in water increases beyond the safe limit which indirectly affects the buildings.
 - Much of the falling snow in Britain is now highly acidic and if it does not melt, it may turn into pollution time bomb.
 - Precious stonework at St. Paul’s Cathedral is being eaten away
 - Ornamental stone work on a church in Bristol is corroded by acid in the air.
 - H_2S tarnishes silver and blackens leaded house paints.
 - Traces of radioactive elements present in rain severely damage the buildings.

(e) On Human beings

- Acidic rain has been found to be very dangerous to the living organisms as it can destroy life.
- Acidification can play havoc with human nervous system, respiratory system and digestive system by making the person an easy prey to neurological diseases.
- Acids produce highly toxic compounds which contaminate the potable water and enter man’s body.

- Acidic rains containing air pollutants contribute to a variety of safety hazards, associated with reduced visibility due to smog etc.
- These contaminants can be nuisance in several aspects and cause adverse health effects.

Control of Acid Rain

- There is an urgent need for proper regular monitoring to provide timely warnings about acidification of our environment.
- Short term control of acid deposition problem can be achieved by using lime.
- New York has been liming the lakes and ponds since 1959.

QUESTIONS

PART-A

1. Define Air Pollution.
2. Sources of air Pollution.
3. 1° & 2° Pollutants with example.
4. Effects of Air pollution.
5. What are particulate air pollutants. What are 1° sources of particulate matter?
6. What are the causes & effects of air pollution? How will you control it?
7. Define suspended particulate matter.
8. Differentiate between mist & fog.
9. What are gaseous pollutants?
10. Define photochemical smog. How it is obtained?
11. How CFC's are accumulated in atmosphere? Give its effects.
12. Briefly describe about the impact of CO on human health.
13. Briefly explain about origin & ill effects of the following: CO, SO₂.
14. Differentiate kinds of air pollution from Industries & automobiles.
15. Write the sources of carbon monoxide.
16. Explain air pollutant emerging from domestic sources, Transport, Industries.
17. What are the two types of smog that occur in industrial areas?
18. Mention the products of photochemical smog.
19. Give the effects of SO₂ on human being.
20. What is the permissible concentration of NO₂ in industrial area?
21. List five air pollutants commonly released in air.
22. Discuss the chemistry of photochemical smog.
23. Discuss briefly how air pollution can be checked.
24. Explain indoor air pollution & its effects.
25. Explain air pollution episode, Bhopal gas tragedy.
26. Why CO₂ is not being considered air pollutant?
27. Difference between 1 & 2 air pollution / Smog & Photochemical Smog.
28. List the major pollutant in the automobile exhaust and discuss the ways & means to control the same.
29. Give any seven air pollutant gases? Mention its source & Effects.
30. What are aerosols? Give example.
31. Give the case study of London Smog.
32. What is PAN?

33. Give the reaction of Photo Chemical Smog.
34. $\text{HC} + \text{O}_3, \text{O}_2, \text{NO}_2, \text{NO} \longrightarrow ?$
35. How deforestation connected with air pollution?
36. How shall SO_2 be removed from polluted air?
37. What is the main cause of increased CO_2 concentration in the atmosphere?
38. What do you understand by fixed & mobile sources of air pollution?
39. What is pollutants matter? How it affects human health.

PART - B

1. What is Ozone? 2. What is the importance of Ozone layer to us?
3. What do you meant by ozone depletion?
4. Give the reaction involved in the formation of ozone hole. **(May 14)**
5. Give the causes and effects of ozone depletion on environment. **(Jan 15)**
6. How is the stratospheric ozone measured? What is Dobson unit?
7. Explain the formation & destruction of ozone in the stratosphere.
8. Give formation of Antarctic ozone hole & role of polar stratospheric clouds (PSCs).
9. Why is ozone hole formation less severe in arctic region compared to antartica?
10. Write the possible reaction responsible for the destruction of ozone by CFCs.
11. As an individual what can you do to alleviate the ozone hole problem.
12. Which are the agents responsible for ozone depletion?
13. Name three most important types of CFC which are responsible for ozone depletion. Ozone depletion potential – Define.
14. What is acid rain mention its impacts?
15. Explain the mechanism of acid rain formation and enumerate its effects.
(Jan 15, May 14, May 11, Jan 11)
16. How acid rain forms? Give reason. How acid rain affects buildings?
17. Explain cause, effects of Acid Rain. **(May 11)**
18. What are the two principal acids present in the acid rain?
19. Define dry & wet deposition? Is there any difference in damage due to these two types of deposition?
20. Write a short note on green house effect? **(Jan 11)**
21. List the major green house gases & their sources. **(May 11)**
22. Explain the factors responsible for the global warming? **(Jan 15)**
23. Define the term global warming? What are the consequences of warming? **(Jan 11)**
24. Explain the global warming & factors contribution. **(May 14)**
25. Give the measures taken at the global level to control the emission of green house gaseous and global warming. **(Jan 15)**
26. Discuss the effects of green house effects. **(Jan 15)**
27. What are the major air pollutants? Enumerate any two effects of oxides of sulphur and hydrocarbon. **(May 14)**
28. Discuss the method of control of air pollution from automobiles. **(May 14)**
29. Enumerate the biochemical effects of oxides of nitrogen and sulphur. **(May 13)**
30. What are the effects of photochemical smog? **(May 13)**
31. List major air pollutants and explain their effects on human beings. **(May 13)**
32. Mention the different types of control measures of gaseous ad particulate air pollutants. **(May 11)**

Water pollution - Definition

- The presence of any toxic substance in water which affects temporarily or permanently its quality.
- Presence of any foreign material which changes either physical or chemical properties of water.

Types of water pollution**(a) Physical pollution**

The physical pollution of water brings about changes in water with respect to its color, density, odour, taste, turbidity and thermal properties.

(b) Chemical pollution

- Due to the presence of inorganic and organic chemicals in water.
Ex: acids, alkalis, toxic inorganic compounds, dissolved inorganic compounds, suspended inorganic compounds, suspended organic compounds and dissolved organic compounds.
- Causes changes in acidity, alkalinity or pH, dissolved oxygen
Organic pollutants: The organic pollutants can be biodegradable or non-biodegradable.
Inorganic Pollutants: The wastes of industries like fertilizers, gases, liquids, coke ovens, alkali producing industries etc. Ex: free chlorine, chloramines, H_2S , NH_3 ,

(c) Biological pollution

- It is a secondary pollution caused by microorganisms.
- Due to the presence of pathogenic bacteria, certain fungi, pathogenic protozoa, viruses, parasitic worms and plant toxins.

(d) Physiological pollution

- It is also a secondary pollutant
- Caused by several chemical agents present in trace amount such as Cl_2 , SO_2 , H_2S , It causes unpleasant taste and odour to water.

SOURCES OF WATER POLLUTION

Water has the property to dissolve many substances in it. Therefore, it can easily get polluted. Pollution of water can be caused by

(a) Point Sources

- Industries
- Power Plants
- Underground coal mines
- Offshore oil wells etc.

(b) Non-Point Sources

- Surfaces runoff from agricultural fields
- Over flowing small drains
- Rain water sweeping roads and fields
- Atmosphere deposition

Water Pollutants	Sources	Harmful effects
Sewage and Domestic wastes	• Sewage waste contains mineral and organic matter. • Includes human excreta, soap, detergent, metals, glass, rubbish garden waste etc. • Drinking water contains sunscreen agents, antiseptics, pain killers etc. • Municipal waste contains Cu, Cr, Pb, etc. Sewage contains fatty acids, esters, amides, amino acids, etc.	• Stimulate the growth of pathogenic bacteria, viruses and protozoa. • Chronic diseases in man and animals. • Cause diseases like typhoid, cholera, malaria. • Unfit for drinking and domestic use. • Depletion of DO & affects the aquatic flora. • Reduces the photosynthetic activity. • Cause Eutrophication.
Industrial Effluents	• Contains toxic chemicals, hazardous compounds, corrosive alkalis, oils, greases, dyes, non-biodegradable matter, radioactive and thermal pollutants etc. • The industries are pharmaceuticals, tanneries, textiles, paper, plastic, chemical, fertilizer, coal, lime, cement, steel, paints and rubber, etc.	• Affects kidneys, liver, lungs, brain and reproductive system. • Cause cellular degeneration in brain which results in coma and numbness. • Hg poisoning causes Minamata in Japan. • Effluents lower the photosynthetic rate of aquatic communities. • Effluents containing acids and alkalies make water corrosive. • Disinfectants cause mortality of fish, planktons.

Water Pollutant	Sources	Harmful effects
Agriculture discharges	• Plant nutrients, pesticides, insecticides, herbicides, fertilizers, farm wastes, manure slurry, sediments. • Agricultural discharges are exposed to be three times more than domestic sewage.	(a) Effects on Man • Excessive use of fertilizers leads to the accumulation of nitrates in water. • It causes Blue Babies or methaemoglobinaemia (lack of oxygen). i.e., a bluish coloration of the skin. • Nitrate poisoning in grazing animals. (b) Effect on eco and aquatic system • Disruption of nutrient balance in water • Decrease essential nutrients in soil. • Resulting in malnutrition. • Causes Eutrophication & increases BOD.
Detergents	Detergents are sodium or potassium salts of sulphonates of higher fatty acids. Produce foam and pollute water. Ex: TPBS (tetra propylene benzene sulphonate) ABS.	• Detergents are non-biodegradable. • The persistent surfactants interfere with the waste treatment process. • Water contaminated is anesthetic. • ABS kills species of may fly nymphs. • Cause Eutrophication.
Siltation	• Consists of dirt and dust particles. • Create high turbidities in water.	• Prevents the free movement of aquatic organisms. • Growth of fishes and their productivity.

Water Pollutant	Sources	Harmful effects
Toxic metals	• Emissions from the mining, metallurgical, chemical, leather, sugar, distilleries, battery industries, street dust, sewage, fossil fuel burning and thermal power plants.	• Methyl mercury causes Minamata disease • Abnormal distribution of chromosomes. • Heavy metals deposited in kidneys, liver and bones. • It disrupt metabolic functions of body. Causes low intelligence level, abnormal neuro behavior and impaired growth. • Affect aquatic flora.
Thermal pollutants	• The waste heat from atomic and thermal power plants. • Various industries with cooling requirement & Municipal sewage.	• The reduction in DO and increases BOD. • Animals are killed outright by hottest water. • The less hot water affects the fauna. • Bacteria multiply rapidly, causes excessive eutrophication.
Radioactive materials	• Nuclear power plants, nuclear reactors, nuclear installations, research labs, hospitals etc. Toxic radioactive substances like Pu, Am, Cm, Np, Co, Sr, Th, Ra, Cs, Zr, Ru, etc are produced from neutron bombardment • Explosions of nuclear weapons.	• Skin cancer, carcinoma, melanoma, breast cancer, leukemia, DNA breakage and cataracts are the list of diseases in man. • It destroys biological immune system & increase in rate of mutation in plants. • Causes somatic and genetic disorders, in man and animals.

CONTROL OF WATER POLLUTION

- Use of agrochemicals reduce surface runoff and leaching
- Use of nitrogen fixing plants
- Adopting integrated pest management
- The nutrient rich water used as fertilizer
- Separate drainage for sewages and rainwater
- Planting trees would reduce pollution
- Recycling of waste water to generate electricity, fuel, fertilize etc.
- Solar energy to kill micro organisms by using catalysts
- Industrial water purified by reverse osmosis before mixing with river

CAUSES AND EFFECTS OF ORGANIC WATER POLLUTANTS

ORGANIC POLLUTANTS

The most common form of water pollution due to organic chemicals is because of the presence of fats, proteins, carbohydrates and other organic substances.

The organic pollution of water is mainly due to the presence of

- (a) Suspended organic substances
- (b) Dissolved organic substances

(a) Suspended organic substances

Industrial wastes:

- Paper mills, Tanneries
- Plant nutrients, Domestic sewage
- Detergents, Biocides
- Food additives, Pharmaceuticals
- Plasticizers, Paints
- Solvents, Fibers
- Waste from slaughter house
- Meat packing plants
- Food processing plants
- Run off from crop lands
- Decomposed products of organic compounds.

The vegetable wastes:

- Algae decayed leaves, fungi
- They impart acidity, green colour and taste to water.

Animal wastes:

- Decomposition of dead animals, insects etc.
- Responsible for growth of bacteria in water.

(b) Dissolved organic substances

- Industrial wastes, decay of plant and animal wastes.
- contains large amount albuminoidal, free ammonia and chlorides
- produce pathogenic bacteria in water

Organic pollutants	Sources	Harmful effects
Carbohydrates & Proteins	Carbohydrates` • Sewage and domestic wastes. • Wastes from sugar, glucose, breweries, dairies, starch, cellulose, sucrose, glycogen, dextrin etc. • Proteins, amino acids, albumin, gelatin, casein, keratin etc.	• Putrefaction by bacterial action & release S and P compounds. • Produce gases like H₂S and SO₂. • Cause putrid and musty smell in water.
Oils	• Petroleum drilling and Oil spills. • Off-shore exploration and production of oil. • Accidental fires in ships and leakage from oil pipe lines.	• (i) Water soluble oils e.g. Greasing oils • Form milky emulsions, stable and non-destroyable. • Degreasing oils have high pH value and affects aquatic life. • These oil films coat the gills of fish affecting their respiration. • (ii) Water insoluble oil e.g. Cutting oils • Cutting oils used as emulsifying agents • Enter into streams from engineering factories as lubricants and coolants. • It floats on the surface and reduces light transmission and photosynthesis in marine plants. • Decreases DO and affects the sea food which enters the human food chain.

Organic pollutants	Sources	Harmful effects
Polychlorinated biphenyls (PCB)	- Transformers, electrical capacitors, Dielectrics, lubricants, plasticizers etc. - Indiscriminate dumping in unregulated landfills.	- In animals, it affects the central nervous system and respiratory tract. - Increased the birth defects and low birth weight. - Cause several physiological disturbances in man - Retard the growth of fetus through nerve & development impairment. - They are resistant to biodegradation.
Phenolic compounds	They enter into the water system through trade wastes.	Low concentration produces bitter taste in water Higher levels result in pain, renal irritation, severe shock and death.
Aldehydes	Released from the thermal decomposition of fats, oils and glycerol. Ex: Acetaldehyde, furfural, formaldehyde, etc.	- Cause odour in water - Inhibit algal growth - Toxic to fish and other aquatic animals They irritate entire respiratory tract.
Aromatic hydrocarbons	Petroleum industry and textile industry.	Carcinogenic effects on man and animals. Benzene like compounds causes odour pollution in water.

PESTICIDE / INSECTICIDE POLLUTANTS

- Organic chemicals used to control unwanted and dangerous species of plants and animals.
- Increasing pressure to increase food production to meet current demand requires protection of crops from pests.
- Use of pesticides is inevitable and constitutes integral part of modern crop management practice.

Pesticides include chemicals to protect crops from

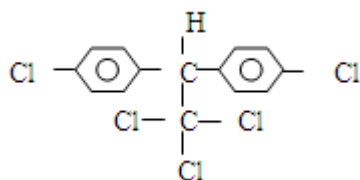
- Insects (insecticides), Weeds (herbicides), Disease causing microorganisms (fungicides and bactericides), Mites (acaricides), Nematodes (nematicides) and Rodents (rodenticides).
- Pests hazardous to animals and humans, vectors- such as mosquitoes, flies, fleas, ticks and lice that transmit diseases.

Classification of Pesticides

(a) Organochlorine compounds

Example: Aldrin, dichloro diphenyl trichloroethane (DDT) and benzene hexachloride (BHC), etc.

Structure of DDT:

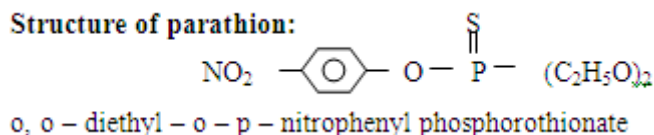


1,1,1-trichloro – 2,2-bis (p-chlorophenyl) ethane

(b) Organophosphorous compounds

Example: Ethion, parathion, malathion,

Structure of parathion:



o, o - diethyl - o - p - nitrophenyl phosphorothionate

(c) Organo Carbamates compounds

Example: Carbaryl, Baygon, dimetilan, propoxar, carbofural etc.

(d) Chlorophenoxy acids

Example: 2,4-dichlorophenoxy acetic acid and 2,4,5-trichlorophenoxy acetic acid.

Characteristics of an Ideal pesticide

- Should be biodegradable.
- Should not kill the friendly insects.
- Should selectively kill the harmful pests only.
- Should not be toxic to man.
- Should be cheap and easy to use.
- Be completely harmless to people, domestic animals, wildlife, and other aspects of environment.
- Its residues would only last for very short periods.
- Non-flammable, non-corrosive and it would have no undesirable odor.

Sources of Pesticide pollutants in water

(a) Rain water

Pesticides contaminated in vapour phase in air get adsorbed on to the dust particles which ultimately reach soil or water only with the rain water. The residues of pesticides such as BHC, DDT, DDE and dieldrin were detected in rain water.

(b) Spray drift

Roughly 30 to 40% of pesticide used in aerial spray reaches the target while the rest fled into air. Water contaminates heavily with pesticides during accidental spray.

Wind may also carry the drift to considerable distances while the major fall of pesticides takes place near the site of application and polluting water system.

(c) Runoff from agricultural fields

Generally water soluble pesticides are transported in water system while the insoluble ones get bound to the particulates and carried by water streams. Large quantities of DDT in water were found about 70 to 440 mg/L in mud.

(d) Industrial effluents

A huge amount of pesticides are escaped by pesticide producing industries. It was also found that, a wood factory released 12 to 70 mg/L of dieldrin in water.

(e) Domestic sewage

Pesticides during their usage may be carried by sewage drains also.

(f) Other sources

The other factors that influence the extent of pesticide pollution in water bodies are the amount of suspended particulates, chemical air crafts, accidental spillage, evaporation from soils and plants, distances between the area treated with pesticide and the receiving water, forest cover, time lapse between treatment and rain fall and residues in horticultural products.

Effects of Pesticides

(a) Effects of Pesticides on human

Greater than maximum reducible limit (MRL), cause some disorder or disease in the body. Thus pesticides are classified into four categories depending upon their mode of poisoning action.

(i) Physical poison:

- Kill the living organism by physical action.
- For example, endrin penetrates specifically through the epidermis of the skin and produces lethal effects.

(ii) Nerve poison:

Ex: DDT, MIC, malathion, parathion, diazinon, systox, etc.

- DDT initiates extreme nervous excitation, causes release of excessive neuro active substances,
- Disrupting nerve activity.
- Causes cancer, leukemia, sterility, tumor and hypertension etc.

(iii) Protoplasmic poisons:

- Cause precipitation of protein in the body resulting in liver damage and ultimately death.

Ex: Insecticides and chemicals such as endrin, ziram, lead arsenate and sodium arsenite etc.

(iv) Respiratory poisons:

Some pesticides inactivate the respiratory enzymes like oxidases, peroxidases and reductases.

For example, Fumigants such as hydrocyanic acid gas, methyl bromide, ethylene dichloride and dibromide cause acute suffocation and block the respiratory tract.

(b) Effects of Pesticides on eco system and aquatic system

- Foods like wheat, rice, groundnut, fish, meat, butter, ghee and cheese contain a pesticide residue.
- Contaminate the soil and affect its fertility
- Contaminate nearby water bodies, the accumulated pesticides disturb the ecosystem severely.
- Contaminate groundwater and when this water is used for irrigation, it affects crops.
- Accumulated by lower organism and are consumed by the higher animals such as birds, fishes and mammals and cause serious effects.

DETERGENTS

A **detergent** is a surfactant or a mixture of surfactants with "cleaning properties in dilute solutions." These substances are usually alkylbenzenesulfonates, that are similar to soap but are more soluble in hard water.

Detergent ingredients

- Surfactant, Builder, Zeolite, Filler, Material anti deposition agent and Additives

Detergents are classified into three broad groupings, depending on the electrical charge of the surfactants. Anionic detergents, Cationic detergents and Non-ionic and zwitterionic detergents

Sources / Applications of Detergents

- Laundry detergents
- Fuel additives
- Biological Reagent
- Soapless soap
- Green cleaning
- Dishwashing liquid
- Hard-surface cleaner
- Cleavable detergent
- Dispersant

Effects of detergents on Human beings

- Detergent can cause a negative impact both on **health and the environment**.
- Ingredients of the detergent have direct and indirect effects on humans and the environment.

- Cationic surfactants are toxic if ingested compared with anionic surfactants and non-ionic.
- Chlorobenzene is a chemical compound that is toxic and harmful to health.
- ABS has high risk to environment and difficult to destroy by microorganisms (non biodegradable).
- Detergent has the ability to dissolve the ingredients are carcinogens, such as 3.4 Benzonpyrene
- Detergent decomposition process will result in residual benzene when reacted with chlorine to form chlorobenzene compounds are very dangerous.

Effects on Aquatic biota

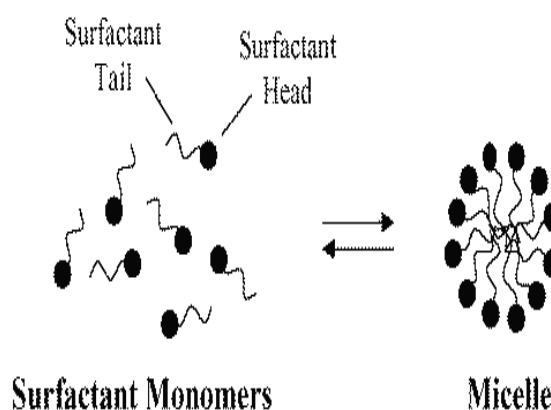
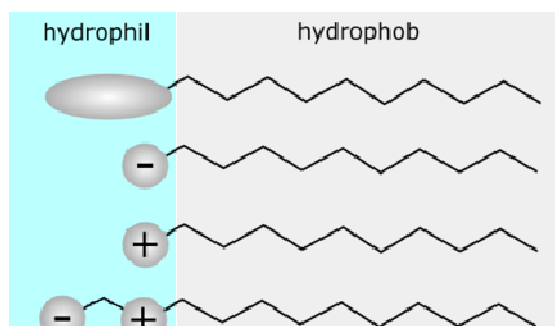
- Composed of complex phosphonates posing eutrophication problems
- Detergent has a toxic effect in the water.
- Destroy the external mucus layers & cause damage to the gills.
- Most fish will die if the concentration of 15 parts per million Detergent.
- Detergent with low concentrations kill fish eggs.
- Detergent also negative effects on water biota.
- Aquatic invertebrates are also severely affected by the detergents.

SURFACTANTS

Surfactants (also called surface active agents or wetting agents) are organic chemicals that reduce surface tension in water and other liquids.

Uses of surfactants Surfactants play an important role as cleaning, wetting, dispersing, emulsifying, foaming and anti foaming agents in many practical applications and products, including

- Detergents, Fabric softeners, Hair conditioners, Toothpastes
- Emulsions, Paints, Adhesives, Inks, Anti-fogs, Ski waxes, snowboard wax
- Agrochemical formulations
- Herbicides, Insecticides
- Cosmetics, Shampoos
- Firefighting, Pipelines, Ferrofluids, Leak Detectors



Sources of Surfactants

- Discharges from Industrial sources laundries and households
- Discharges from sewage treatment works
- Agricultural applications
- Surfactants are compounds composed of both hydrophilic and hydrophobic or ipophobic groups.

Effects of surfactants

- Surfactants can cause rough skin surface, the loss of natural moisture on the skin surface
- Increase the permeability of the outer surface
- Cationic surfactants are toxic compared with anionic surfactants
- Chlorbenzene toxic and harmful to health
- ABS has a high risk to the environment, this material now replaced with LAS
- Toxic to animals, ecosystems, humans, marine organisms and algae
- Bioaccumulation factors for aquatic organisms
- Linear alkylbenzene sulfonates (LAS) and the alkyl phenol ethoxylates (APE) break down in the aerobic conditions
- Degraded by soil micro-organisms, interfere with the life-cycles of some aquatic organisms.

CAUSES AND EFFECTS OF INORGANIC WATER POLLUTANTS

Inorganic Water pollutants

- Inorganic salts, Metal compounds
- Minerals acids
- Finely divided metals
- Trace elements
- Detergents.

Sources of Inorganic pollutants

(a) Acids and Alkalis:

- Acid manufacturing Industries
- Manufacturing of SO_2 , NO_x , NH_3
- Bases of Na, K, Ca etc
- Effluents from paper, tanning and textile industries.

(b) Soluble salts:

- Salts containing anions of carbonate, bicarbonate, acetate, nitrate, halides, sulphates and phosphate
- Sulphides of Cu, Cd, Hg, Zn, Ni, Mn, Co, Pb, Sn, Fe and As
- Certain alloys of Fe, Cd, Na, Li and Si are extremely toxic.

(c) Insoluble salts: Insoluble salts of CaSO_4 , CaCO_3 , dust and particulates etc

(d) Polyphosphates: Large amount of polyphosphates through detergents

(e) Acid mine drainage:

- coal producing areas, Coal mines discharge considerable quantities of sulphuric acid and ferric
- Iron pyrites when exposed to air in mining operations, oxidation reactions occur & produces acid.

(f) Inorganic pesticides:

- Inorganic pesticides having Cu, Cd, Mn, Fe, Zn, Co, S, As etc

(g) Other Chemicals:

- Organo-metallic compounds.
- Various industrial products
- Discarded plastics, synthetic fibers, packaging materials, textiles and toys of PVC

Effects of Inorganic pollutants

A. Effects of acid water

(a) On eco system

- Leaching of nutrients from the soil
- Biological uptake of Al from soil

(b) On aquatic system

- destroy most invertebrates and micro organisms at pH below 4
- Acidic water increases the solubility of Fe, Al and Mg salts and poses deleterious effect on aquatic life.
- Aquatic organisms are adversely affected by change in pH conditions of river water.

(c) On Man

- Acid water along with toxic metals affects the human nervous system, respiratory system, digestive system and reduces the visibility.

(d) On Buildings

- Causes damage to concrete structures and water pipes
- Acid water affects the marble construction,
$$\begin{array}{ccc} \text{CaCO}_3 + \text{H}_2\text{SO}_4 & \longrightarrow & \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2 \\ \text{Lime stone} & & \text{Stone leprosy} \end{array}$$
- Affects Sand stone containing iron oxide
$$\text{Fe}_2\text{O}_3 + 3 \text{H}_2\text{SO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3 + 3 \text{H}_2\text{O}$$

B. Effects of Alkaline water

(a) On Boilers

- Reacts with iron material of boilers and leads to caustic embrittlement.
- Excess of inorganic pollutants make the water hard and unsuitable for boilers.
- Brings excessive corrosion of plumbing system, piers, boats and metals.

(b) On Aquatic life

- Raise the pH of water upto 12 & destroy bacteria and microorganisms in water.
- Strong alkalis produce asphyxiation by the coagulation of gill secretions of fish.

(c) On Agriculture and others

- High alkaline water damage agricultural fields.
- The soluble salts cause serious diseases in man.

C. Effect of soluble and insoluble salts and pesticides

- Affect the aquatic life and in turn cause serious diseases in man.
- Nitrate causes methaemoglobinemia (blue baby) in children
- Iron and manganese salts make water turbid and unsuitable for drinking and boilers.
- Aluminium salts and alums affect the fish life.
- Excessive salts is discharged into rivers, becomes brackish damage fish life.
- Many of the fresh water fish varieties die in salt water due to osmotic changes.
- Inorganic pesticides cause thinning of shells of bird eggs.

	Mercury	Lead
Natural sources	• 0.00003% by weight of the earth's crust. • Extracted from its sulfide ore, cinnabar (HgS). • Dispersed in the atmosphere during eruptions.	• Much abundantly in Solar atmosphere and the atmospheres <u>hot sub dwarfs</u> • In <u>ore</u> with <u>zinc</u>, <u>silver</u> & <u>copper</u> • Lead <u>mineral</u> is <u>galena</u> (PbS).
Man made sources	• Combustion of fossil fuels release about 3000 tones • Extensive use in chlor-alkali industry, disperse million pounds. • Electrical and electronic industries. • Plastic industry, paper and pulp industry, certain fungicides	• Preparation of pipes, containers for corrosive liquids, lead chambers, Steel, storage batteries, cable coatings, varnishes and paint industries. • Lead arsenate is used as pesticide and Lead borate in plastic industries. • The burning of gasoline containing the anti-knock additive lead tetra ethyl, $Pb(C_2H_5)_4$, exhaust fumes as volatile lead halides, settles 40% on the ground & 60% in the atmosphere. • Mining and smelting of lead ores, emission from automobile exhausts and lead pigments.

	Mercury	Lead
Effects on plants	• Enters in the food chain • Affect the growth of many plant species	• Plants retain lead in their roots, reduces calcium and phosphorus in the soil. • Coats the surface of the leaf and reduces the amount of light reaching it. • Reduction in the rate of photosynthesis.
Effects on animals	• Birds and mammals eat fish are more exposed predators that eat fish-eating animals highly exposed. • At high levels of exposure, harmful effects on these animals include death, reduced reproduction, slower growth and development, and abnormal behavior	• CNS of animals inhibits ability to synthesize red blood cells. • Extended period of time, cause death in animals. • Affects Grazing animals • Move into the micro-organism and grazing food chains.
Effects on human beings	• Enter the blood stream. • Affects the digestive system & the lungs. • Cause brain damage, often fatal. • Minnamata disease • More than 50 people died in Japan due to mercury poisoning.	• Enters the bone & affects the metabolic activities. • Causes anemia, kidney damage. • Interference with heme synthesis, leads to hematological damage. • High concentration produce congenital malformations. • High mortality rate in infants due to brain damage.

	Chromium	Copper
Natural sources	• In air, rocks, water, soil, biological materials and volcanic emissions. • Found in variety of plants and Animal protein. • Plants like bryophytes, grasses, shrubs have high Cr levels • Found in human tissues, Lung, skin, ovaries and heart are the largest stores of Cr. • The form of chromium with medicinal value	• Earliest known metal. • Present in land and many organisms. • Blood proteins of snails. • Fruits, vegetables, eggs and dairy products have less than 3 ppm. • Necessary for blood formation in addition to iron. • Enzymes such as tyrosinase, cytochrome oxides, uricase and ascorbic acid oxidase.
Man made sources	• In atmosphere by electroplating, oil burning of fossil fuels. • Used in explosives, preservation of wood. • Refinery, textile mill and dying Manufacture of stainless steel, chrome-plating, copper stripping, photography • Corrosion inhibitor, textile manufacture, leather tanning paints and ink. • Porcelain, chromic potassium sulphate • Oxalate used in colouring glass, fertilizer.	• Chalcophile found in sulfide deposits with Pb, Cd and Zn. • Manufacture of alloys, paints, ceramics and pesticides. • Industrial smelters used for producing Zn. • Iron and steel industry, fertilizer industry and burning of wood • discharge of mine tailings • Disposal of fly ash and municipal and industrial waste.

	Chromium	Copper
Effects on plants	Reduction in growth & photosynthesis in algae.	• Toxic to aquatic biota.
Effects on animals	• Affect hematological characteristics & biochemical process of fresh water fish. • Cr (IV) in drinking water cause toxic effects in rats and rabbits report by WHO.	• An allergic contact dermatitis by direct exposure of Cu salt or dust. • Toxic for fresh water fish.
Effects on human beings	• Carcinogenic, skin ulcers, kidney inflammation, affects nervous system, aquatic life is strongly influenced by Cr in water Chromium deficiency • Plays a vital role in insulin metabolism as the glucose tolerance factor (GTF). • Impaired growth and longevity • disturbs glucose, lipid and protein metabolism.	• Workers engaged in copper mining affected by Cu poisoning • Cause greenish black discoloration of hairs. • Attack of Wilson's disease in excess concentrations in tissues. • Continuous ingestion induce chronic copper poisoning Cu deficiency • Causes an increase in the embryonic deaths in rats and chicks. • Anemia, diarrhea and demineralization of bones.

ANALYSIS OF WATER POLLUTANTS

To control water pollution necessarily requires quantitative measurements of water pollutants. So different types of examinations such as physical, chemical, biological and microbiological examinations of water are necessary. Water analysis is usually expressed in milligrams per litre and parts per million.

DISSOLVED OXYGEN

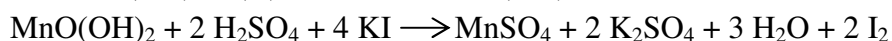
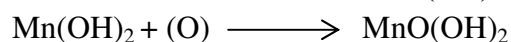
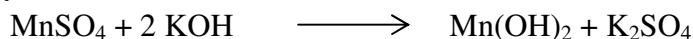
The dissolved oxygen levels in natural as well as waste water depend on physical, chemical and biological activities of the water body. The analysis of dissolved oxygen is very important in water pollution control as well as waste water control. Methods used for determination of DO are Winkler's or Iodometric method and Electrometric using membrane electrode.

DETERMINATION OF DO BY WINKLER'S METHOD (IODOMETRIC METHOD)

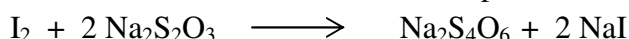
- Oxygen dissolved in water in varying concentrations.
- The solubility of oxygen depends on temperature, pressure and salinity of water.
- It is essential for the life of fishes and other aquatic animals.
- DO content is more in pure water than in polluted water
- It is more in winter than in summer season.
- It is usually ranges from 6-8 ppm.

Principle

This method is based on the fact that atomic oxygen decomposes potassium iodide and liberates free iodine. This can be titrated with standard sodium thiosulphate solution using starch as indicator. But oxygen present in water is in molecular form, its conversion into atomic state is necessary. This is done by reacting the water sample with manganous sulphates in strongly alkaline medium.



This iodine then reacts with standard thiosulphate solution.



Interference due to oxidizing agents such as NO_2^- , SO_3^{2-} present in waste water, may be eliminated by addition of NaN_3 to alkaline KI solution. On acidification NO_2 is decomposed.

Procedure

250 ml reagent bottle is filled with water sample leaving 10ml space. 2ml of MnSO_4 solution is added followed by 2ml of alkaline KI solution. Precipitate of manganous hydroxide will be formed. The bottle is shaken well to facilitate the reaction of Mn(OH)_2 precipitate with the dissolved oxygen present in the water sample and gives brown precipitate of basic manganic hydroxide. Concentrated sulphuric acid is added drop by drop to dissolve the precipitate till yellow solution of iodine is obtained.

100ml of this iodine solution is taken into a clean conical flask and titrated against standard thiosulphate solution using starch as an indicator till the blue colour just disappears.

Amount of Dissolved Oxygen present in water sample

= Normality of Dissolved oxygen X Eq. Wt of Oxygen X 1000 ppm

BIO CHEMICAL OXYGEN DEMAND (BOD)

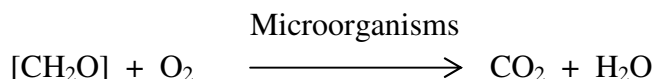
Excessive nutrients, such as nitrates and phosphates, commonly originate in domestic sewage, runoff from domestic fertilizers, waste material from animal feed lots, packing plants etc. These nutrients are responsible for water pollution primarily because they stimulate growth of microorganisms which often increase the BOD. Thus there is a close relationship between organic matter and dissolved oxygen in water. Thus water pollution is measured by BOD.

Definition

BOD is a standardized measurement of the amount of oxygen required by microorganisms to cause the break down or decomposition of organic matter in water sample over a period of 5 days at 20° C.

Principle

BOD is calculated by measuring the DO consumed by bacterial and chemical action in a closed sample of water maintained at 20° C for five days.



The selection of micro organisms (seed) is very important and the results are obviously not reproducible. The purpose of seeding is to introduce into the sample a biological population capable of oxidizing the organic matter in the waste water.

BOD values may range from 3 to 10 ppm depending on the quality of water. Clean water sample have low BOD compared to the polluted water which will have high BOD values.

BOD level (ppm)	Water Quality
1-2	Very Good (Contains no organic waste)
3-5	Fair: Moderately Clean
6-9	Poor: Somewhat polluted
100 or greater	Very Poor: Very polluted (Contains organic waste)

Procedure

Two bottles of 300 ml capacity are taken and filled with water. The DO of first bottle is determined immediately by Winkler's method.

The second water bottle is seeded with a little diluted domestic waste water. This bottle is sealed and incubated in the dark for 5 days at 20° C. During incubation the bottle should be protected from entry of air. After 5 days, the DO of the water in the bottle is measured.

$$\text{BOD} = \text{DO before incubation} - \text{DO after incubation}$$

BOD of 75 mg/L means that biodegradation of organic matter in a litre of sample consumes 75 mg of O₂.

Presence of acid or base has marked effect on bacterial activity. So before finding out DO, the pH of sample water should be brought to the range of 6.5 to 8.3. Also sample water should be dechlorinated.

BOD values are useful in process design and loading calculations, measurement of treatment efficiency and operation, stream pollution control and in determining self purifying capacity of a stream.

CHEMICAL OXYGEN DEMAND (COD)

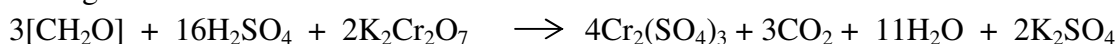
Although BOD test indicates the amount of total organic matter, there are some drawbacks. To compensate this, chemical oxygen demand test is carried out. It is the index of total organic content of oxygen demanding substances in water. COD implies the oxygen requirement of a sample for oxidation of organic and inorganic matter. The oxidisable inorganic matter is usually negligible in comparison with the quantity of organic matter.

Definition

COD is generally defined as the amount of oxygen required to chemically oxidize all the oxidisable impurities completely using an oxidizing agent like acidified potassium dichromate.

Principle

The method is based on the chemical oxidation of the sample with a strong oxidizing agent, such as potassium dichromate in 50% sulphuric acid, which is capable of oxidizing all the organics.



A known amount of potassium dichromate is added to a measured amount of the sample and the mixture is boiled with concentrated sulphuric acid. After boiling, the amount of unreacted dichromate is determined by titration against a standard ferrous ammonium sulphates (Mohr's salt) solution using ferroin as the indicator.

Sometimes Ag_2SO_4 is added in the oxidation process, which catalysis the oxidation of straight chain aliphatic compounds, aromatic compounds and pyridine. Interference of chloride can be removed by adding mercury salt. HgSO_4 is added to oxidize nitrites to nitrates and it also removes chloride.

Procedure

Two round bottom flasks are taken. In one RBF, 20 ml of distilled water, 10 ml of 0.25 N $\text{K}_2\text{Cr}_2\text{O}_7$ and 15 ml of concentrated H_2SO_4 are taken. In another RBF, 20 ml of sample water, 10 ml of 0.25N dichromate and 15 ml of concentrated sulphuric acid are taken in presence of HgSO_4 and Ag_2SO_4 . Both the RBF are refluxed for about 3 to 6 hours and cooled for 20 minutes.

The excess of dichromate in both the flasks are treated against FAS using 3 drops of ferroin indicator and formation of wine red colour is the end point.

$$\text{COD} = \frac{(A - B) \times N \times 8 \times 1000}{\text{Volume of sample}} \text{ ppm}$$

where A is the volume of FAS consumed by the water sample

B is the volume of FAS consumed by the distilled water

N is the normality of FAS

Volume of sample is 20 ml and equivalent wt. of oxygen is 8.

COD has high value compared to BOD as it takes into account both biologically oxidisable and biologically inert organic matter. COD converts all the organic pollutants in water to CO_2 and H_2O .

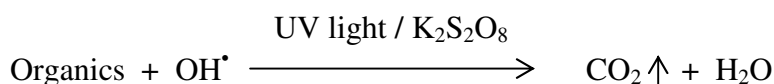
Chemical oxygen demand has been found to be more scientific than the biological oxygen demand. COD is important in management and design of treatment plant because of its rapidity in determination. It is used in calculating the efficiency of treatment plants and proposing standard for discharging domestic effluents in various types of water streams.

TOTAL ORGANIC MATTER (TOC)

Total organic matter is the best method for the measurement of total organic matter in water samples. As it is an instrumental method, it is superior to the COD method and is quite popular.

Principle

In this method all the organic carbon are converted to CO_2 by the oxidizing agent $\text{K}_2\text{S}_2\text{O}_8$ and UV light. The oxidation is promoted by the free radical $\text{OH}^\bullet$ generated from UV light.



By measuring the amount of CO_2 by gas chromatography or conductivity or NDIR, the amount of total carbon in the water sample can be calculated.

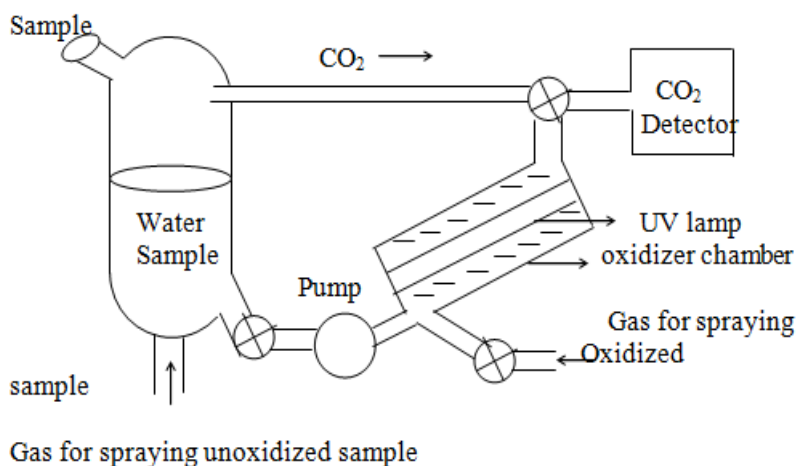
Procedure

Potassium thiosulphate and phosphoric acid are added to the water sample. Phosphoric acid is sprayed with air or N_2 to expel CO_2 formed from carbonate and bicarbonate in solution. After spraying, the sample is pumped to a chamber containing an ultraviolet lamp emitting radiation at 184.9 nm.

The $\text{OH}^\bullet$ radical formed by UV irradiation is the active species which brings about the rapid oxidation of dissolved organic compounds. At the end of oxidation, CO_2 is liberated from the system and measured with a gas chromatography or conductivity meter or non-dispersive IR spectrophotometer.

By measuring amount of CO_2 , the amount of total carbon in the water sample can be calculated.

TOC is an important water quality parameter, which gives an indication of the overall amount of organic pollutants.



LAND (OR) SOIL POLLUTION

Definition

The addition of unwanted substance to the soil, which adversely affects the physical, chemical and biological properties of soil and reduces its productivity.

Sources of Land pollution

(a) Chemical Pollutants

- Chemicals such as CaCO_3 , bicarbonates, calcium sulphate etc,
- Tanneries, synthetic drug factories and distilleries discharge lot of suspended and dissolved salts.

(b) Metallic Pollution

- Cu, Steel, Cd, Zn from factories .

- CO, Ni, Pb, Ba, Hg,

(c) Industrial Effluents

- Sugar, Textile, Steel, Paper, Chemical, Pesticides, Petroleum, Cement, Glass industries

(d) Agricultural Waste

- Excessive use of chemical fertilizer.
- Fertilizers increases the salt content of the soil.
- Excess Fertilizers reduce productivity of the soil.

(e) Pesticides

- Insecticides, Fungicides, Algaecides, Rodenticides and Weedicides

(f) Radioactive Waste

- Mining and nuclear power plants.

Effects of Land pollution

- Pollute the soil ultimately affect human health.
- Affect soil fertility.
- Affects non target organisms, soil flora and fauna.
- Reduce soil productivity
- Pathogenic bacteria, viruses cause various types of diseases.
- Radioactive pollutants cause abnormalities.

Eg: Strontium-90 instead of Ca gets deposited in bones and become brittle and prone to fracture.

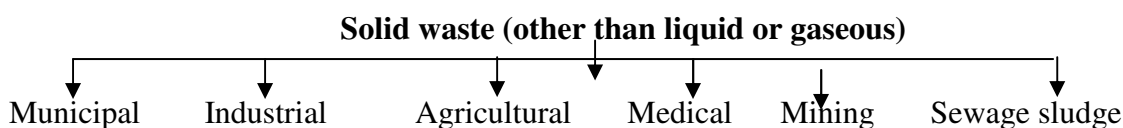
- Nitrogen and phosphorous from fertilizers in the soil reach nearby water bodies with agricultural runoff and causes eutrophication.

Control of Land pollution

- Effluents should be properly treated before discharging them on the soil.
- Solid wastes should be properly collected and disposed of by appropriate method.
- From the waste, recovery of useful products should be done.
- Bio degradable organic waste should be used for generation of bio gas.
- Cattle dung should be used for methane generation.
- Night – soil can also be used in the bio gas plant to produce inflammable methane gas.
- By Microbial degradation of biodegradable substance.

Solid waste and its classification

Higher standards of living of ever increasing population have resulted in an increase in the quantity and variety of waste generated. It is now realized that if waste generation continues indiscriminately then very soon it would be beyond rectification. Management of solid waste has therefore, become very important to minimize the adverse effects of solid wastes.



SOURCES OF SOLID WASTES

URBAN WASTES

- **Waste from homes: (Domestic waste)**

Contain variety of discarded materials like polyethylene bags, empty metal, aluminum cans, scrap metals, glass bottles, waste paper, diapers, cloth / rags, food waste etc.

- **Waste from shops:**

Waste paper, packaging materials, cans, bottles, bags, peanut shells, egg shells, tea leaves etc.

- **Bio medical waste:**

Anatomical wastes, pathological waste, infectious wastes etc.

- **Construction / demolition waste:**

Debris & Rubbles, wood, concrete etc.

- **Horticulture waste:**

Vegetable parts, residues, remain of slaughtered animals etc.

Urban waste materials that can be degraded by micro organisms are called biodegradable wastes. Wastes that cannot be degraded by micro organisms are called non-biodegradable wastes.

INDUSTRIAL WASTES

Industrial waste consists of large number of materials including factory rubbish, packing materials, organic wastes, acids, alkalis, metals etc.

- Chemical industries metal, internal processing industries.
- Thermal power plants (fly ash).
- Nuclear power plants (Radioactive waste).
- Solid waste from industries include scrap metal, rubber, plastic, paper, glass, wood, oil paints, tars, dyes, scrub leather, ceramics abrasive slag, asbestos, batteries etc.

EFFECT OF SOLID WASTE

Uncompleted dumping of wastes not only gives an ugly look but also pollutes the environment. The main impacts of waste accumulation are,

- **Spoilage of landscape**

Municipal wastes heap up on roads due to improper disposal system. People clean their own houses and litter their immediate surroundings which affect the community including themselves. Every year several tons of solid waste is dumped along highways and other places, there by spoiling land space.

Pollution

Dumping of waste on the land may pollute ground water and also the water bodies present in the vicinity. Toxic chemicals present in the waste may percolate in the ground and contaminate ground water.

- **Health hazards**

Heaps of domestic and industrial waste is dumped on vacant and un used land in residential areas which causes unhygienic conditions and ultimately results into out break of diseases like cholera, gastroenteritis, malaria, dengue etc.

Decomposition of organic waste produces foul smell and allows breeding of various types of insects and infection organisms. Rats thrive on these dumps and multiply rapidly causing menace to the people by destroy their property and spreading several diseases

- **Effect on soil**

Much waste may spread on the soil and changes physico-chemical and biological characteristics of the soil. It adversely affects the fertility of the soil.

- **Effect on terrestrial and aquatic life**

Urban and industrial waste often contains a variety of toxic chemicals. Such chemical may enter into the food chain and effect both terrestrial and aquatic life.

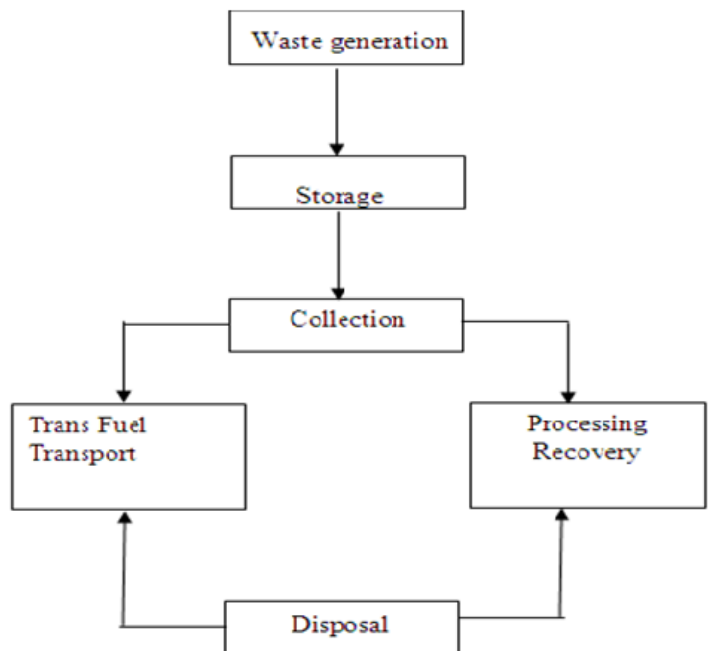
CONTROL OF SOLID WASTE MANAGEMENT

Need for solid waste management

The main objective of solid waste management is to minimize the above adverse effects before it becomes too difficult to rectify in future.

Steps involved in the solid waste management:

The waste materials get collected in municipal dust bins, other garbage bin, trucks and transported to a central location where the processing and recovery of waste material takes place. Finally the disposal is done by the following method



METHODS OF SOIL WASTE MANAGEMENT

Sanitary landfills

In sanitary landfill, garbage is spread out in thin layers, compacted and covered with clay or plastic foam. In modern landfills, bottom is covered with impermeable lines, usually several layers of clay, thick plastic & sand. The lines protects ground water from being contaminated due to percolation of leachate. Several wells are drilled near landfill site to monitor if any leakage is contaminating ground water.

Advantage

- Simple, Economic
- Segregation not required
- CH₄ produced by anaerobic decomposition used to produce heat & electricity.

Disadvantage

- Deep pit should be away from residential area, otherwise bad odors, dust, noise.
- Fire hazard due to formation CH₄ in wet weather.
- Large no. of mosquitoes, flies, rodents, snakes are produced.

Incineration

First the solid waste is segregated into combustible and non-combustible material. Then the combustible solid waste is burnt in a furnace called incinerator at 850° C in the presence of O₂ and convert them into ash.

Advantages

- The heat produced in incinerator is reused in the form of steam power for generation of electricity throughout turbines.
- The residue is only 20 -25% of original weight.

Disadvantages

- Capital & Operating cost high.
- Formation of smoke, dust, ash need further disposal.

Composting

It involves degradation of organic waste by micro-organisms in the presence of O₂ and provide a number of alternative features. The organic waste is converted into compost & CO₂. The urban solid waste can also be disposal off through biotechnology, produce valuable organic manure and CH₄.

Method

The organic waste materials grinded using mechanical processes and the ground material is shredded prior to be placed over 30 cm thick refuse layer. The alternate layers are filled with cow dung & finally covered with 5 to 10 cm earth layer at the top. The completion of bacterial action requires 34 months to produce compost manure.

WHO has setup a compost plant in new Delhi in 1982 with a capacity to handle 90 to 100 tonnes of waste every day. And the prepared compost was supplied to nurseries, kitchen garden, horticulture etc.

Advantages

- Compost manure increases water retention & ion – exchange capacity of soil.
- Recycling occurs.

Disadvantages

- The non-organic have to be disposed separately.

Vermiculture

This technique is popularly known as earth warm farming. It is an important biotechnology for converting solid wastes into compost with the help of earthworms.

Method

Made a pit where the bottom of the pit is filled by broken and coarse sand. Above this moist loamy soil is placed. The layer forms the vermibed.

Advantages

- It is excellent organic manure with rich nutrients.
- Healthy food grains without toxins can be produced.

Pyrolysis

The combustible organic waste is decomposed thermally in a pyrolysis reactor in the absence of O₂ at 600 to 1000 °C. This process yields compounds like tar and oils, gaseous phases containing fuel gases like H₂, CH₄, CO. The solid residue is carbon.

Advantages

- The product like tar is used to lay roads.
- The product produced is used as fuel.

Recycling

Recycling is an integral part of waste management and is truly an eco friendly technique. In recycling a product at the end of its service life (i.e.) waste is converted into another useful product. It involves separating, collecting, processing, marketing and using the material that could have been thrown away.

A sheet of paper can be recycled into other paper products, bottles & pouches can be recycled for product from waste is important because it does not produce unwanted by products.

Advantages

- Reduces our relevance on landfills, incinerators.
- Protects our environment by effective handling of waste.
- Conserves natural resources because it reduces the need for fresh raw material.

3R – APPROACH: Reduce, Reuse, Recycle

For waste management we stress on three ‘R’ – Reduce, Reuse and Recycle before destruction and safe storage of wastes.

REDUCTION IN USE OF RAW MATERIALS

Reduction in the use of raw materials will correspondingly decrease the production of waste. Reduced demand for any metallic product will decrease the mining of their metal and cause less production of waste.

Advantages

- Decrease the production of waste.
- Cattle dung is used in gobar gas plants & manure.
- Aquatic weeds (water hyacinth) used as fertilizer, biogas, paper etc.
- Sugar cane waste used to produce electricity, paper boards etc.
- Fly ash used as cement substitute.

REUSE OF WASTE MATERIALS

Reuse of waste materials will decrease the production of waste. Because of financial constraints poor people reuse their materials to the maximum.

Examples

- The refillable containers which are discarded after use can be reused.
- Villagers make casseroles and silos from waste paper and often waste materials.
- Making rubber rings from discarded cycle tubes which are used by newspaper vendors, instead of rubber bands, which reduces the waste generation during manufacturing of rubber bands.

RECYCLING OF MATERIALS

Recycling is the reprocessing of discarded materials into new useful products.

Examples

- Formatting of old type of products (eg) old aluminum cans and glass bottles are melted into new useful products.
- Cellulose insulation from paper.
- Fuel pellets from kitchen waste.
- Preparation of automobiles & construction materials from cans.

Advantages of 3R approach

- Saves money, energy, raw materials, land space
- Reduces pollution
- Reduces deforestation (recycling of paper)
- Reduces mining (reuse of metals)

LEGAL PROVISIONS FOR HANDLING & MANAGEMENT OF WASTE

There are several legal provisions in one country which safe guard one environment. Indian constitution in the world which contain specific provision for the protection of environment. Some of the legal provisions related to waste management are,

Factories act 1948, amended in 1987

This act safeguards one environment. The basic objectives of this act are. Cleanliness of the industrial establishment & factor.

- Paper & safe disposal of waste & effluents.
- Safe use & handling hazardous substances, Penalizes any disobedience of law.

The environment (Protection) act 1986

This is meant for the protection of air, water, soil quality and control of the environmental pollutants including wastes.

THERMAL POLLUTION

Definition

Thermal pollution can be defined as presence of waste heat in the water which can cause undesirable change in the natural environment.

Sources of Thermal pollution

Radioactive Substance

The drainage from

- Nuclear Power Plants and Nuclear reactors
- Nuclear installation & Processing
- Hospitals and Research institutions discharge.

Coal fired power Plants

The condenser coils of the power plants are cooled with water from nearby water system and discharge the hot water (15° C temp. increases) back to the water system.

Industrial Effluents

Heat producing industries like refineries, steel, textile, sugar, paper etc release huge amount of hot water (8-10 °C temp increases).

Domestic Sewage : It has high temperature than the water in the water bodies.

Effects of Thermal pollution

Reduction in DO

The dissolved oxygen content of water is decreased as the solubility of oxygen in water is decreased at high temperature.

E.g.: 10 °C – DO Content 11ppm and 30 °C - DO Content 7 ppm

Change in water properties

- The vapor pressure, settings rate of suspended particles increases.
- Viscosity, density, solubility of gases decreases.

Interference with Biological Activities

- The composition of flora and fauna changes because the species sensitive to increased temperature due to thermal shock will be replaced by temperature tolerance species.

Undesirable change in Algae population : Excess nutrients from the washout water from agricultural lands combined with thermal pollution cause excessive algal growth or eutrophication.

Longevity : The rise in temperature of water will induce the living organisms to increase in their activity which will exhaust the organisms and shorten their life.

Effect on Water Quality

- Decreases in DO affect the microorganism in turn affecting water quality.

Interference with Reproduction

- In fishes, several activities like nest building, hatching, migration and reproduction are seriously affected by high temperature.

Increased rate to Disease

- The activities of pathogenic micro organisms are accelerated which cause various diseases.

Control measures of Thermal pollution

- **Cooling Ponds**

Water from condensers is stored in ponds where natural evaporation cools the water which can then be re-circulated or discharged in nearby water body.

- **Spray Ponds**

The water condenser is received in spray ponds. Here the water is sprayed through nozzles where fine droplets are formed. Heat from these fine droplets is dissipated to the atmosphere.

- **Cooling Towers**

Wet Cooling Tower: Hot water is sprayed over baffles cool air entering from sides takes away the heat and cool the water.

Dry Cooling Tower: The heated water flows in a system of pipes. Air is passed over these hot pipes with fans. This is very costly.

RADIOACTIVE POLLUTION

Definition

Radioactive pollution is a special form of physical pollution of air, water and soil with radioactive materials. Its nature of contamination and effects are different from those of the other kinds of pollution.

Sources of Radioactive pollution

Natural sources

- Cosmic rays emission from space.
- Terrestrial radiations from radio nuclides present in the earth crust.
E.g.: Radium 224, Uranium 238, Thorium 232, Potassium 40 and Carbon 14.
- Soil, rocks, air, water, food which contain one or more radioactive substances.

Man – made sources

- Mining and refining of plutonium and thorium.
- Explosion of nuclear weapons.
- Nuclear power plants.
- Fuels power plants.
- Fuels and preparation of radio active isotope.
- Use of radio isotope in medical, industries and research application.

Name	Sources	Effects
Np, Ru	Neutron bombardment of atomic fuel	Affects photosynthesis in plants and metabolic pathway in human.
Pu	Neutron bombardment of atomic fuel	Lung cancer.
Cs-137	Continuous removal of surface soil, deep plugging, leaching of minerals, use of fertilizers.	Radiation sickness (Vomiting, diarrhea, weakness)
Th-232	Nuclear bombs fission fusion products.	Birth abnormalities, blood changes, genetic disorder.
Sr -90	Nuclear power plants & Nuclear Installation	Affects bones & teeth.
Am	Testing and use of nuclear weapons	Affects (central nervous system) CNS
Co-60, Ir - 132	Radio in research	It cause chronic health effects.
P-32	Radio isotopes in agricultural use.	It affects photo synthesis.
Ca - 45, S-35	Radio isotope in industry	Damage teeth & bones
Zn-65, Fe -65	Nuclear Reactors	Accumulate in animals, fishes and cause serious metabolic problems.

Effects of Radioactive pollution

Ionization radiation can affect living organisms by causing harmful changes in the body cells and also changes at genetic level.

- **Genetic Damage:** Genetic damage is caused by radiation, which induce mutation in the DNA there by affecting genes and chromosomes. The change may be transmitted to several generations.
- **Somatic Damage:** burns, miscarriages, eye cataract and cancer of bone, thyroid, breast.

Control Measures of Radioactive pollution

- As there is no cure of radiation damage, all effects should be made to prevent radioactive pollution.
- Leakage of radioactive elements from reactors and laboratories, processing or using them should be totally checked. All safety measures should be strictly enforced.
- Disposal of radioactive waste must be safe. They should be changed into harmless form and stored in safe place, where they may regularly decay in a harmless manner.
- Regular monitoring through frequent sampling and quantitative analysis has to be ensured in sink areas.
- Safety measures against nuclear accidents should be strictly enforced.
- Nuclear explosion and use of nuclear weapons should be completely banned.
- Appropriate steps should be taken to prevent occupational Hazards.

Safe Disposal

First radioactive waste are stored in tanks and buried under soil for 10-13 years. After 13 years it is taken out and reactive nature of radioactive waste is checked. If it is inactive then it is disposed in air or ocean.

QUESTIONS

PART – A

1. Name any four major water pollutants.
2. What are non-point sources of water pollution?
3. Define Eutrophication.
4. What is meant by 5R principle?
5. Give the significance of DO in rivers.
6. What is meant by thermal pollution?
7. How arsenic contaminates the water.
8. Mention the effects of solid waste.
9. Mention any two ill effects of chromium and mercury.
10. How can be nuclear hazards disposed safely?
11. Name the sources and effects of marine pollution.
12. Classify the water pollutants.
13. Define radioactive pollution. Mention the sources of radioactive pollution.
14. Discuss on effects of pesticides on environment.
15. Give examples for radioactive waste.
16. What is Bio magnification?
17. Write short note on recycling of MSW.
18. Difference between recycling and reuse.
19. Classify solid waste. What are the sources of urban and industrial solid waste?
20. List five ways to reduce Surface and ground water pollution?

21. What is the purpose of a land fill?
22. What are the benefits of waste minimization?
23. What are the causes and effects of nuclear pollution?

PART - B

1. What are the different types of water pollutants? Discuss briefly the sources, effects and the various methods of disposal of it. **(May 14, May 13)**
2. Explain the experimental procedure for estimating COD. **(Jan 11)**
3. Write experimental procedure for estimating BOD of water. **(Jan 15)**
4. Distinguish between BOD & COD. **(May 14)**
5. Explain the procedure for the determination of TOC in water.
6. Describe Winkler's method of estimating Dissolved Oxygen. **(May14, May 11)**
7. Explain various sources and effects of heavy metal pollution.
8. Explain effects of Organic pollutants and pesticides on receiving water bodies.
9. How Inorganic pollutants and radioactive pollutants affect the water bodies?
10. List and explain the various types of water pollutants.
11. Write an explanatory note on the effects of toxic metals and pesticides.
12. Explain causes, effects of Organic and Radioactive pollutants.
13. Discuss on Eutrophication and Biomagnification.
14. Enumerate the toxic metals and their effect on human beings.
15. What are the effects of fluorides and nitrates in water?
16. Distinguish between point and non point sources of water pollution?
17. Write note on (i) Minamatta disease,(ii) Blue baby syndrome, (iii) Itai-Itai disease (iv) Love canal tragedy.
18. How do nutrients from agricultural fields effect water shed?
19. Enumerate with examples major sources of surface and ground water pollution.
20. Discuss adverse effects and control of water pollution.
21. What would happen if soil pollution is not prevented?
22. What do you meant by thermal pollution? What are the countries cause major thermal pollution?
23. What do you understand by the term thermal pollution? Discuss various effects and control measures of thermal pollution.**(Jan 15, May 14, Jan 11, April 10)**
24. What is Chernobyl disaster?
25. Why nuclear power plants are considered a thread to ecological security?
26. Explain about any one nuclear holocaust. **(Jan 15)**
27. Discuss about the significance of hazardous waste management. **(May 14)**
28. Mention preventive measures for nuclear accidents.
29. Oceans are the ultimate sink for most of the waste we produce.
30. Classify solid waste? How can the solid waste be managed?**(Jan 15, May 13)**
31. Enumerate the various methods of solid waste disposal. **(Jan 14)**
32. Write short notes on land filling and incineration methods of solid waste disposal. **(May 14)**
33. Give a general composition of solid waste in urban Indian conditions.
34. What are the advantages of vermi composting over conventional composting?
35. Write a detailed note on the 3R-approach.
36. Write short notes on the sources and effects of detergents and surfactants?

Principles of Air pollutants monitoring Methods

When a pollutant is emitted as a result of carelessness and especially when it is extremely necessary to measure its concentration at site of emission, various laboratory methods fail to give reliable information.

There are some laboratory methods that can be used on the outdoor location and it is also possible to use basic principles underlying laboratory methods for testing the pollutants, so that a suitable and feasible method can be employed for continuous measurement. This type of continuous measurement is generally known as monitoring.

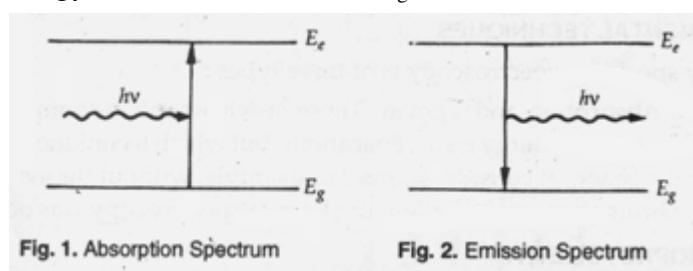
While measuring air pollution on a regular, constant and continuous basis, some physical properties of the pollutants are necessary to be identified for the measurements. These properties include light absorbance or transmittance, electrical conductivity, change in potential, formation of some material by passing electric current or radioactivity.

SPECTROSCOPY – INTRODUCTION

- Spectroscopy is analysis of electromagnetic radiation scattered, absorbed or emitted by molecules.
- The interaction of light with matter results in physical changes (such as reflection, absorption, diffraction or emission) or chemical changes (such as dissociation or isomerisation).
- During the interaction, the energy is absorbed or emitted by the matter.
- Absorption and emission form the basis of the spectral analytical techniques and the subject specialization is called **spectroscopy**.

Absorption and Emission Spectrum

Absorption spectrum results when an atom or molecule undergoes a transition from lower energy level to higher one with the absorption of a photon of energy $h\nu$. Provided $h\nu$ is exactly equal to the energy difference, $(\Delta E = E_e - E_g)$ between the two levels.



Emission
the atom

spectrum results when
or molecule falls from the

excited state to the ground state with the emission of a photon of energy $h\nu$.

Atomic spectrum is given by electronic transitions, whereas a **molecular spectrum** is given by vibrational, rotational and electronic transitions.

Absorption law

Beer-Lambert's law

When a monochromatic light of intensity I is passed through a solution of concentration, C molar and thickness, dx , then intensity of transmitted light changes by dI . Then, probability of absorption of radiation is given by:

Mathematically, the law is stated as,

$$-(dI/I) = kCdx$$

On integrating the above expression, we get :

$$\log(I_0/I) = k/2.303 Cx$$

$$A = \epsilon Cx$$

$(I_0/I) = A$ (absorbance or optical density)
 $\epsilon Cx = k/2.303$ (molar absorptivity coefficient)

This equation is known as Beer's law.

Various types of Molecular spectra

Spectra	Transition	Region of spectrum	Criteria	Source	Sample form
Rotational (or Microwave)	Between rotational energy levels	Microwave (1 – 100 cm ⁻¹)	Molecule must possess permanent dipole moment (e.g. HCl, H ₂ O)	Klystron valve	Gas or vapour
Vibrational & Vibrational rotational (or Infrared)	Between vibrational energy levels	Infrared (500 – 4000 cm ⁻¹)	Dipole moment of molecule must change during vibrations	Electrically heated rod of rare-earth oxides	Gas or liquid or dilute solution or (solid+KBr) in the form of disc
Electronic (or UV-Visible)	Between electronic energy levels	UV – (25,000 – 70,000 cm ⁻¹). Visible – (12,500 – 25,000 cm ⁻¹)	Presence of chromophore in a molecule	UV – hydrogen discharge lamp. Visible – Tungsten filament lamp	UV – Gas or dilute solution. Visible – Gas or dilute solution

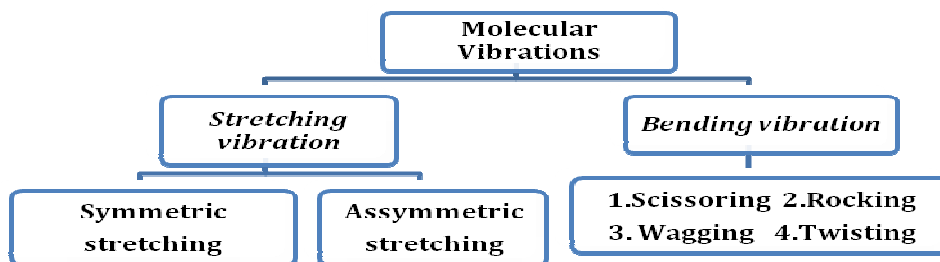
INFRARED SPECTROSCOPY (IR)

Principle

- Absorption of energy by a molecule in infrared region brings about changes in vibrational and rotational energy levels of the molecule.
- IR spectroscopy is otherwise known as vibrational spectroscopy. A single vibrational energy change is accompanied by a large number of rotational energy changes. Thus, vibrational spectra appear as vibrational-rotational bands.
- The ordinary infrared region extends from 4000 - 667 cm⁻¹.
- The region from 12500 - 4000 cm⁻¹ is called near infrared.
- The region from 667 - 50 cm⁻¹ is called far infrared.

Molecular Vibrations

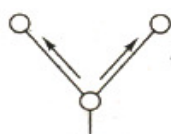
The absorption of infrared radiations causes vibration in a molecule. The fundamental vibrations in a molecule are stretching and bending.



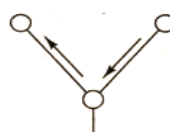
(a) **Stretching vibration:** Distance between two atoms increases or decreases but the bond axis remains same.

Symmetric stretching: Movement of atoms w.r.to a particular atom in a molecule is in the same direction.

Assymmetric stretching: One atom approaches the central atom while the other departs from it.



Symmetric



Asymmetric

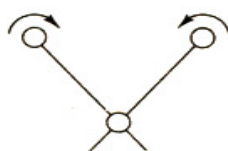
(b) **Bending vibration:** Bond angle changes but the bond length remains unaltered.

Scissoring: Two atoms approach each other.

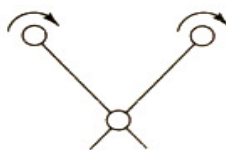
Rocking: Movement of both atoms take place in same direction.

Wagging: Two atoms move up and below the plane w.r.to the central atom.

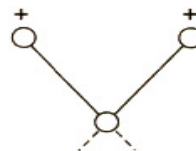
Twisting: One of the atoms move up the plane while the other moves down the plane w.r.to central atom.



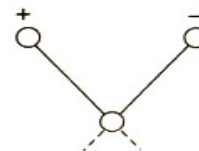
Scissoring



Rocking



Wagging



Twisting

Number

of Fundamental Vibrations

Polyatomic molecules may exhibit more than one fundamental vibrational absorption bands.

The number of these fundamental bands is related to the degrees of freedom in a molecule. The total number of degrees of freedom of a molecule will be equal to $3n$ where n is the number of atoms in a molecule.

$3n$ degrees of freedom = Translational + Rotational + Vibrational.

In the case of **linear molecule** there are only two degrees of rotation.

Total degrees of freedom = $3n$

Translational degrees of freedom = 3

Rotational degrees of freedom = 2

Vibrational degrees of freedom = $3n - 3 - 2 = 3n - 5$

In the case of **non-linear molecule**, there are three degrees of rotation as the rotation about all the three axes will result in a change in the position of the atoms. So, for non-linear molecule

Total degrees of freedom = $3n$

Translational degrees of freedom = 3

Rotational degrees of freedom = 3

Vibrational degrees of freedom = $3n - 3 - 3 = 3n - 6$

Selection Rules

Infra-red light absorbed only when a change in dipole character of the molecule takes place. If a molecule has a **centre of symmetry**, then vibrations are centro symmetric and inactive in the infra-red but are active in the Raman. The vibrations which are **not centrosymmetric** are active in infra-red but inactive in Raman.

Instrumentation

Radiation source

- (i) Nernst glower, consists of a rod of sintered mixture of oxides of Zr, Y and Er. This rod is electrically heated to 1500° C to produce IR radiation.
- (ii) Nichrome wire.

Monochromator

It allows the light of a particular wavelength to pass through it.

Example: Optical prism made of NaCl, KBr and grating.

Sample cells

These must be transparent to IR radiation.

Detectors & Recorder

These convert thermal radiant energy into electrical energy.

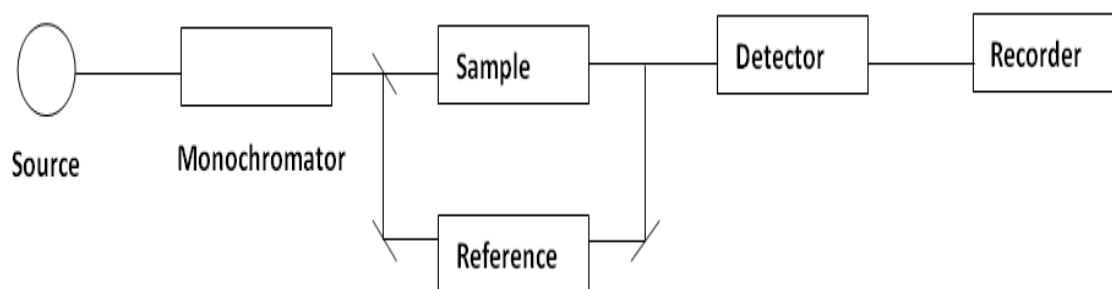
Example: Photoconductivity cell and Photoelectric detectors.

Working

Light from the source is split into two beams. One of the beams which passes through sample is called sample beam and the other beam is called reference beam.

When the beam passes through the sample, it becomes less intense due to the absorption of certain frequencies. Now, there will be a difference in the intensity of the two beams. When the two beams recombine, they produce an oscillating signal, which is converted and measured as electrical signal with the help of the detector. The signal from the detector is passed to the recording unit and recorded.

Block diagram of IR spectrometer



Applications

- In establishing identity of compounds
- In detecting impurities in a sample
- To determine hydrogen bonding in a molecule
- It provides valuable information of molecular symmetry, dipole moments, bond length, etc.
- To detect functional groups
- Distinguishing positional isomers of a compound

VISIBLE AND ULTRAVIOLET SPECTROSCOPY

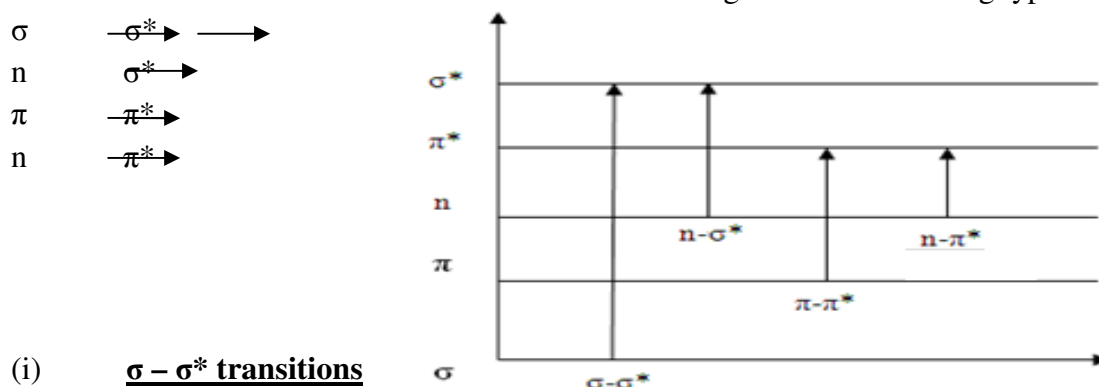
Principle

Molecules containing π -electrons or non-bonding electrons (n-electrons) can absorb the energy in the form of ultraviolet or visible light to excite these electrons to higher anti-bonding molecular orbitals. The more easily excited the electrons, longer the wavelength of light it can absorb.

Types of Electronic Transitions

When a molecule absorbs energy in the UV or visible region, its electrons are promoted from a bonding molecular orbital to an anti-bonding molecular orbital.

The electronic transitions involved in UV and visible region are of following types:



(i) $\sigma - \sigma^*$ transitions

These transitions occur by the promotion of sigma electrons from the bonding to antibonding (σ^*) orbitals. It requires high energy and thus absorbs radiation in the far region (126nm-136nm).

(ii) $n - \sigma^*$ transitions

These transitions are seen in saturated compounds, containing heteroatoms in addition to $\sigma - \sigma^*$ transitions. The energy required is less than that for $\sigma - \sigma^*$ transitions.

(iii) $n - \pi^*$ transitions

Unsaturated compounds containing heteroatoms such as O, N and S show this type of transition.

(iv) $\pi - \pi^*$ transitions.

The electron transfer occurs from bonding π - orbital to the antibonding π^* orbital. It is only associated with compounds containing double or triple bonds and aromatic compounds.

The energy required for various transitions obeys the following order:

$$\sigma - \sigma^* > n - \sigma^* > \pi - \pi^* > n - \pi^*$$

Instrumentation

Radiation source

Hydrogen or deuterium lamp.

Monochromator

Selects appropriate wavelength of light (Prism, Grating).

Cells (sample and reference)

It is made out of quartz or fused silica.

Detectors

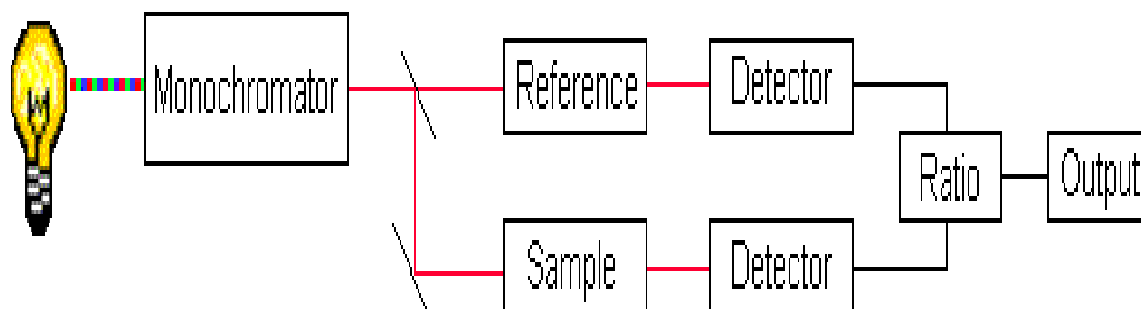
Photomultiplier is commonly used detector, converts light radiations into electrical signals.

Recorder

Working

- Radiation from a hydrogen or deuterium lamp is allowed to pass through monochromator unit, which allows a narrow range of wavelength to pass through it.
- The beam of light coming out of monochromator is split into two halves. One-half of the beam (sample beam) is directed through a transparent cell containing a solution of sample under analysis.
- The other half (reference beam) is directed through an identical cell containing the solvent.
- The instrument is so designed that it compares intensities of the sample beam and reference beam.
- If the sample absorbs light at a particular wavelength, then the intensity of sample beam I will be less than the intensity of reference beam I_0 .
- The instrument gives the output graph-a plot of wavelength Vs absorbance A , and is known as the absorption spectrum.

Block Diagram of UV-visible spectrometer



Applications

- determining the structure of several vitamins
- detecting steric hindrance
- studying rates of reactions
- determination of dissociation constants of acids and bases.

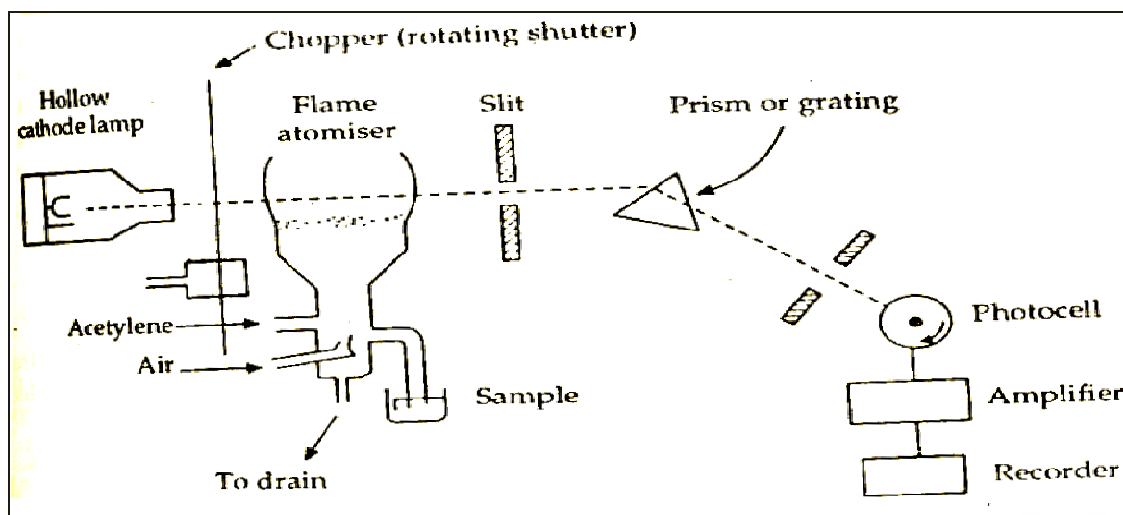
ATOMIC ABSORPTION SPECTROSCOPY (AAS)

Principle

It is based on the atomization of the sample followed by the absorption characteristic radiation by the ground state atoms. Atomic absorption spectrometric (AAS) technique involves the following three basic processes:

- (i) Generation of characteristic radiation of the element.
- (ii) Creation of atomic vapour of the sample.
- (iii) Detection and measurement of the absorption signal.

Block diagram of Atomic Absorption Spectroscopy



Instrumentation

- **Radiation source**

It is a hollow cathode lamp consisting of a glass tube containing noble gases like argon which acts as an anode and a hollow cathode made of the metal to be analyzed.

- **Chopper**

It is a rotating shutter placed between source and flame. It breaks the steady light from the lamp into a pulsating light (because recorder will record only the pulsating current).

- **Burner**

It is used to convert the liquid sample into gaseous state. Burners commonly used are total combustion burners and premixed burners.

- **Monochromator**

It selects a particular wavelength of light emitted from source and allow to pass through Prism/grating.

- **Detectors**

It converts light energy into electrical energy. Most commonly used detectors are photomultiplier tubes.

Working

- Radiation obtained from the hollow cathode lamp passes through the flame.
- Metallic compounds are decomposed in flame & reduced to elemental state forming cloud of atoms.
- The atoms absorb a fraction of radiation in the flame.
- Absorption radiant energy increases with increase in concentration of metal in the sample.
- The unabsorbed radiation is allowed to pass through a monochromator, and then to a detector, amplifier and recorder.

Applications

- It can be used to determine about sixty trace metals in single sample.
- It is widely used in metallurgical and general inorganic analysis.
- It is an ideal method for analysis of many ores and minerals.
- Determination of trace leads in ferrous alloy.
- Determination of vanadium in lubricating oil.

GAS CHROMATOGRAPHY

Introduction

Gas chromatography has proved to be a powerful tool for qualitative identification and quantitative determination of volatile organic compounds (VOCs) in air pollution monitoring. It is very handy for the analysis of low levels of pesticides in the environment and analysis of volatile liquids, vapour and gases.

Principle

Gas chromatography is based on the principle that when the vapours of sample in a gas stream are allowed to pass through a column containing a stationary liquid or solid phase; components of the mixture migrate at different rates due to differences in boiling point, solubility or adsorption leading to separation of mixtures in microgram quantities.

Gas chromatography consists of mobile phase as a gas. It is of two types, gas-liquid chromatography (GLC) and Gas-solid chromatography (GSC) depending on the stationary phase.

Components of GC

Carrier Gas

Chemically inert carrier gas is employed like He, N₂, Ar or H₂. Temperature of the separating column must be equal to average boiling point of least volatile component of the sample.

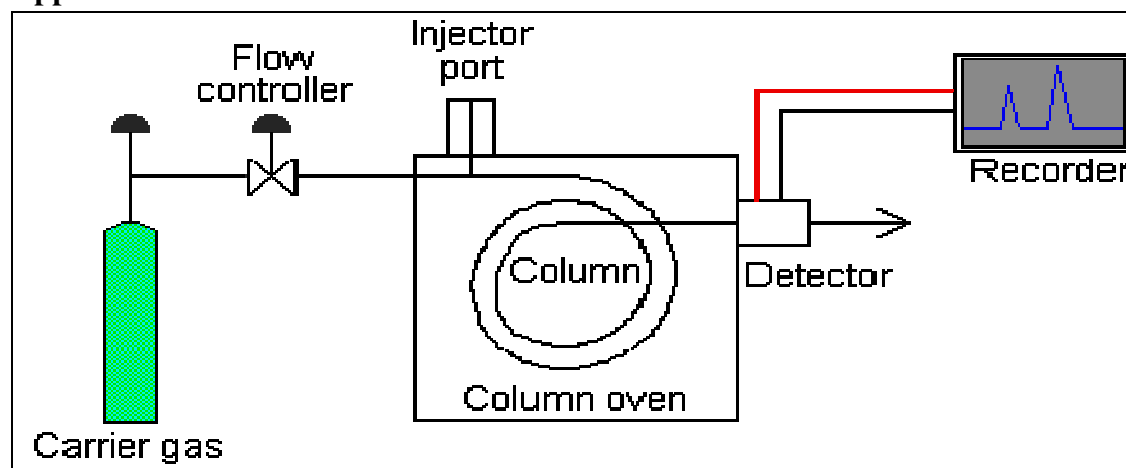
Stationary phase

A good stationary phase is thermally stable & chemically inert. It has low volatility and good degree of compatibility with analytes.

Detectors

Generally, thermal conductivity detector, flame ionization detector or electron capture detectors are used as they have good precision and accuracy, high signal to noise ratio.

Apparatus & Instrumentation



Working

Volatile samples which are thermally stable at the operating temperature are introduced into the gas stream via an injection port. Different constituents migrate at different rates due to their different distribution ratios between the mobile phase & stationary phase.

A continuous flow of gas elutes the components from the column in order of increasing distribution ratio from where they pass through a detector connected to a recording system.

Retention time

It is the time taken by the solute between the sample injection point to reach the detector.

$$t_R = \frac{\text{Length of the column}}{\text{Linear flow rate of solute migration}}$$

Application

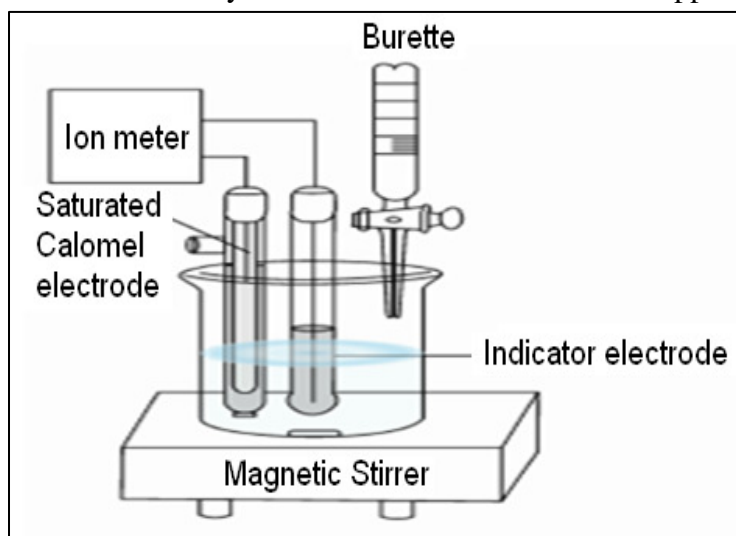
- Gas chromatography is a unique & versatile technique.
- Used for qualitative & quantitative identification of components in a sample.
- Used to investigate various parameters of a system, such as determination of partition coefficients, thermodynamic functions and adsorption isotherms.

Disadvantages

- Sample must be volatile & thermally stable below 400 °C
- Retention data published are not always reliable
- Most commonly used detectors are non-selective.

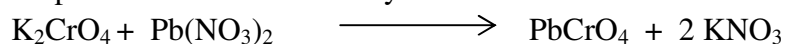
CONDUCTOMETRY

Conductometry is measurement of electrolytic conductivity to monitor progress of chemical reaction. It has notable application in analytical chemistry. In usual analytical chemistry practice, the term conductometry is used as a synonym of conductometric titration, while the term conductometry is used to describe non-titrative applications.



Principle

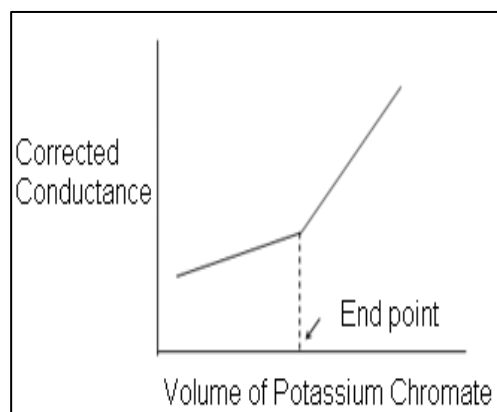
Solution of electrolytes conducts electricity due to the presence of ions. The specific conductance of a solution is proportional to the concentration of ions in it. When lead nitrate solution is titrated with potassium chromate the conductance increases slightly. It is due to the replacement of nitrate ion by chromate ion.



After end point the conductance increases rapidly due to the further addition of potassium chromate. A graph is drawn between volume of potassium chromate added and the conductance of the solution. The end point is the intersection of the two lines.

Procedure

The given lead nitrate solution is made up to the mark in a 100 ml volumetric flask. 20 ml of the lead solution is taken in a 100 ml beaker and it is diluted by adding 20 ml of conductivity water and conductivity cell is immersed in the solution. Conductance is measured periodically after each addition of potassium chromate solution. Conductance values are recorded and the observed conductance is corrected for dilution effect. The value of corrected conductance is plotted against the volume of titrant added. From the graph the end point is determined and hence the amount of lead present in the given solution is calculated.



Applications of Conductometry

- Solubility of sparingly soluble salts.
- Ionic product of water and basicity of organic acids.
- Salinity of sea water and can check water pollution in rivers and lakes.
- Alkalinity of fresh water.
- Estimate ash content in sugar juices
- Determination of atmospheric SO₂.

Air monitoring instruments and techniques

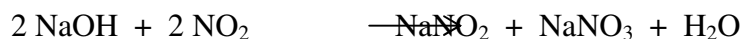
Pollutant measured	Instruments
NO _x	Colorimetric technique, chemiluminescence technique, ion selective electrodes, electrochemical transducer and non-dispersive infrared/ ultraviolet spectroscopy.
CO _x	Non-dispersive infrared (NDIR), spectrometry, IR fluorescence method, atomic absorption spectroscopy (AAS), electro catalytic method, electrochemical transducer, flame ionization detector (FID) method.
SO _x	Electrochemical transducer, chemiluminescence technique, gas chromatography and non-dispersive infrared (NDIR) spectroscopy and instruments based on measurement of colour, conductivity.
C _x H _y	Gas chromatography, chemiluminescence, NDIR, UV absorption, iodometric and thermister catalyst method.
Particulates	Gravimetry, piezoelectric balance method, thimble sampling method, nephelometry, acoustical particle counter, electrostatic precipitator.
Radioactive pollutants	Neutron activation analyser
Organic pollutants	Nuclear magnetic resonance (NMR), Electron spin resonance (ESR)

ANALYSIS OF NO_x

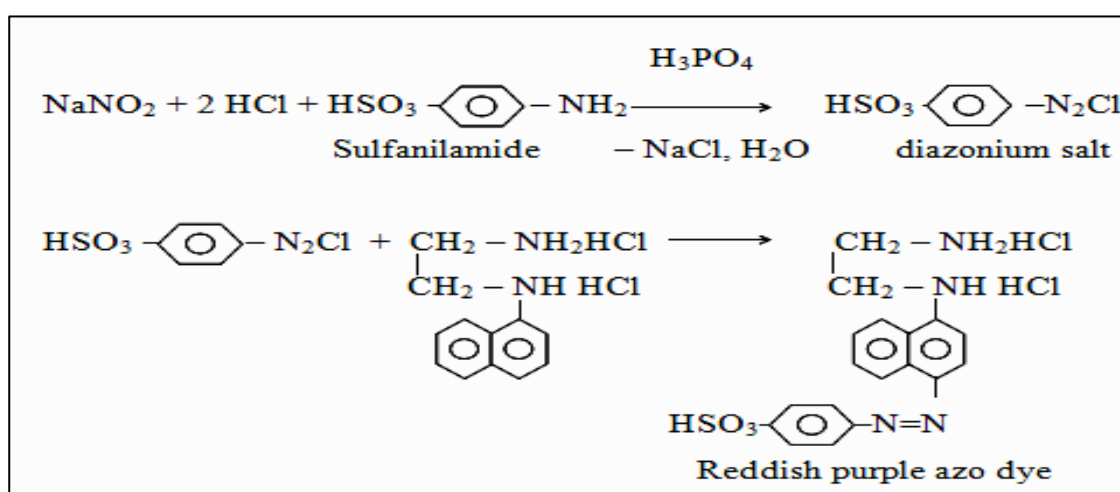
(a) Determination of NO_x by Spectrophotometer method

Principle

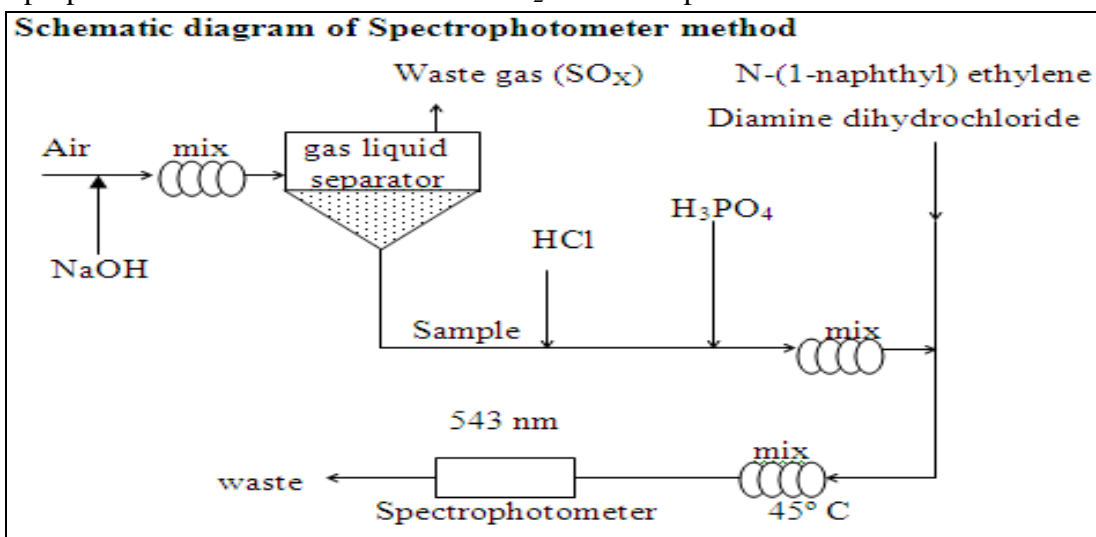
In this method, NO₂ which is present in impure gas is converted into reddish purple azo dye by using following chemicals. NO₂ is collected over NaOH solution and NO₂ is converted into NaNO₂ and NaNO₃.



The interference of SO₂ is removed by reaction with H₂O₂ to produce H₂SO₄ before analysis. The NaNO₂ is allowed to react with H₃PO₄, HCl and sulfanilic acid and the product is then treated with N(1-naphthyl) ethylenediamine dihydrochloride. The end product is a reddish purple azo dye.



The resulting azo dye is measured at 543 nm in spectrophotometer and this concentration is proportional to the concentration of NO₂ in the sample air.



Procedure

The sample air is bubbled through 50 ml of absorbing reagent, NaOH at 200 ml/min for 24 hours. It is then treated with 1 ml H₂O₂ to remove any SO₂. The product is treated with

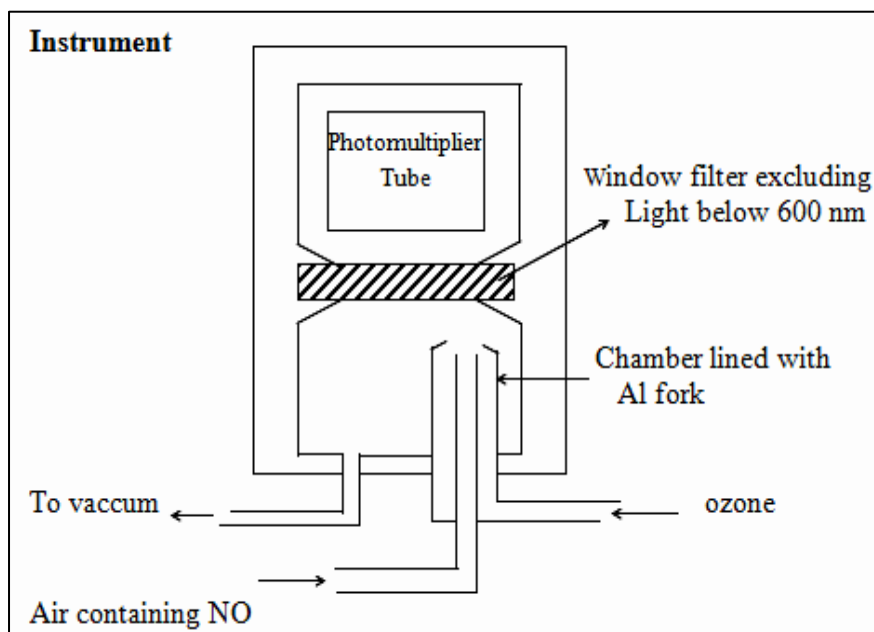
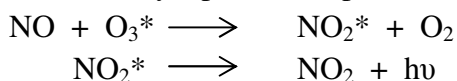
sulphanilamide, HCL, H₃PO₄ and N(1-naphthyl) ethylene diamine dihydrochloride with through mixing.

The product is allowed to stand for 1 hour for development of reddish purple azo dye. The concentration of azo dye is measured at 543 nm in spectrophotometer against a blank. This concentration is directly proportional to the concentration of NO₂ in the impure air. This method is applicable to 0.01 to 1.5 µg NO₂ per ml.

(b) Determination of NO_x by Chemiluminescence method

Principle

The emission of light from electronically excited species arising out of a chemical reaction is measured by a photomultiplier. NO present in the impure air reacts with O₃ to produce electronically excited NO₂ molecule, which in the next step loses energy and return to the ground state through emission of light of wavelength 600 to 3000 nm. The emitted light is measured by a photomultiplier, the intensity being proportional to the concentration of NO₂.

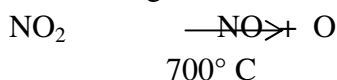


Procedure

In the determination of NO, ozone from an electro generator and the atmospheric sample are drawn continuously into a reaction vessel. The luminescence radiation is monitored by a photo multiplier tube.

A linear response is reported for NO concentration of 1 ppb to 10,000 ppm.

NO₂ is invariably associated with NO in air sample. It is therefore, necessary to convert NO₂ to NO before analysis. This can be effected by passing the air sample over a thermal converter which brings about the required chemical reaction.



Particulate interfere in this determination of NO₂ and this can be eliminated by using a membrane filter at the air inlet. The method is also applicable to analysis of O₃ by interaction with NO.

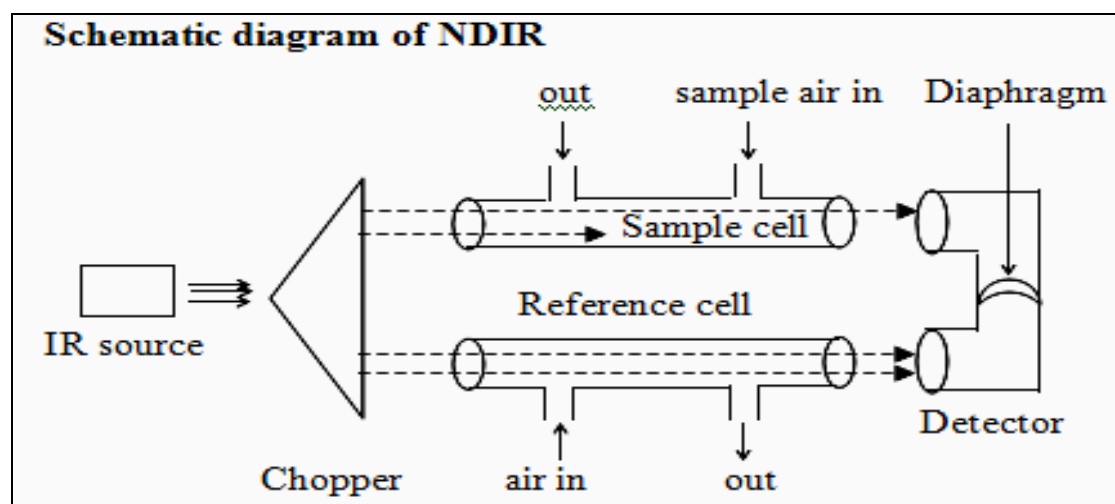
ANALYSIS OF CO_x

(a) Determination of CO by Non-Dispersive Infrared Spectrometer

It is called as non-dispersive IR spectrometer as the IR radiation from the sources is not dispersed according to wavelength by a prism or grating as in a standard IR spectrometer. It is made very specific for a given compound or type of compound by using the material under investigation as part of the detector.

Principle

NDIR is based on the principle that CO strongly absorbs infrared radiation at certain wavelength. When IR radiation is passed through the CO contaminated gas, CO absorbs certain amount of energy. The gas gets heated and in turn there is change in pressure when volume remains constant. The change in pressure is directly proportional to the intensity of radiation and thus concentration of CO can be calculated.



Procedure

The reference cell is filled with clean air and the sample cell with air containing CO and both cells are 1 metre long. Radiation from an IR source is chopped by a rotating device so that it alternately passes through a sample cell and a reference cell. The detector consists of two chambers filled with CO gas. The chambers are divided by a thin flexible metal diaphragm.

When IR radiation is passed through sample cell containing a trace of CO, part of the energy is absorbed depending on the amount of CO in sample. Thus there is a difference in absorption between the reference and sample air. This determines the absorption of IR radiation by CO gas in the two chambers of the detector.

Due to the difference in absorption, pressure difference will arise in the two compartments and the diaphragm bulges towards one chamber. This movement of the diaphragm is detected and recorded. The output is a series of pulses which is proportional to the concentration of CO in sample air.

(b) Determination of CO by Gas Chromatography

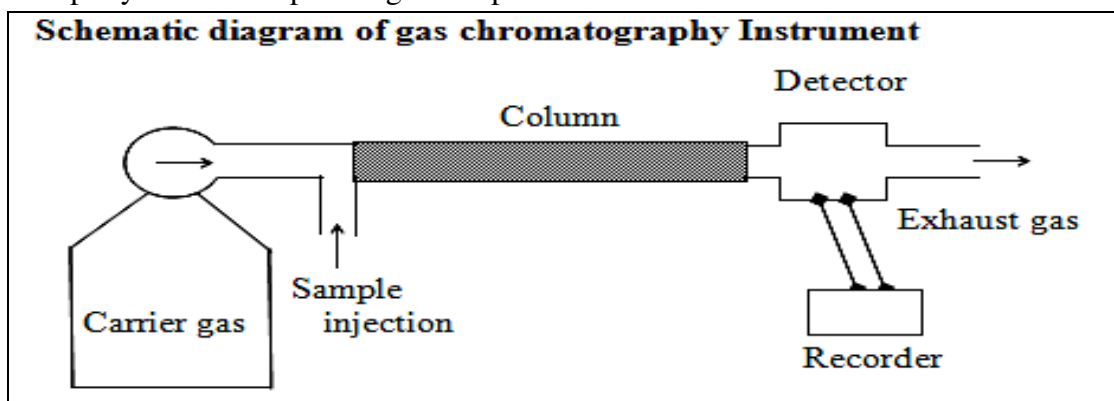
Principle

In Gas Chromatography, a mixture of volatile material transported by a carrier gas is led through a column containing an absorbent solid phase or an absorbing liquid phase coated on a solid material. Each volatile component will be partitioned between the carrier gas and the solid

or liquid. Depending on the retention times in the column, the volatile component will emerge from the column at different times and finally be detected by a suitable detector.

Instrument description

The carrier gas or moving phase may be Ar, He, N₂ and the flow rate of the carrier gas is controlled at 60 to 80 ml/min. The sample is injected as a single compact plug into the carrier gas stream. In case of liquid samples, the injection chamber must be heated to vaporize the liquid rapidly and avoid spreading of the peaks.

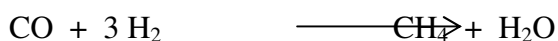


In gas-solid chromatography (GSC), the fixed phase consists of a solid material such as granular SiO₂, Al₂O₃ or carbon. The more common type is gas-liquid chromatography (GLC) in which the fixed phase is a non-volatile liquid such as silicon oil, polyethylene glycol, etc. which is held as a thin layer on a solid support such as crushed firebrick. The packed column is usually 4 mm tubing of stainless steel or glass bent in a V-shaped or coiled. Lengths of such tubes are from 120 cm to 150 cm. The selectivity as well as sensitivity of GC is determined by the detector. The most common detector is thermal conductivity cell which responds to changes in the thermal conductivity of gases passing over it.

Procedure

The sample is fed into the sample valve and the carrier gas is used to transport the sample into the column. The volatile gases are separated out between the carrier gas and the solid or liquid absorbent phase of the column. Depending upon the retention time in the column the volatile gases will come from the column at different times and finally detected by the detector and recorded in a recorder. Thus this method permits both qualitative and quantitative analysis.

Low levels of CO, even less than 10 ppm can be conveniently measured after catalytic reduction by H₂ over a Ni catalyst at 360° C by gas chromatography with a flame ionization detector.



Catalytic reduction is conducted in a small tubular reactor, made of stainless steel heated to 360° C. Ni is deposited on fire brick dust in this tube. With temperature programming to 250°C, CO₂ can be determined.

ANALYSIS OF SO_x

Determination of SO₂ by West – Gaeke spectrophotometer method

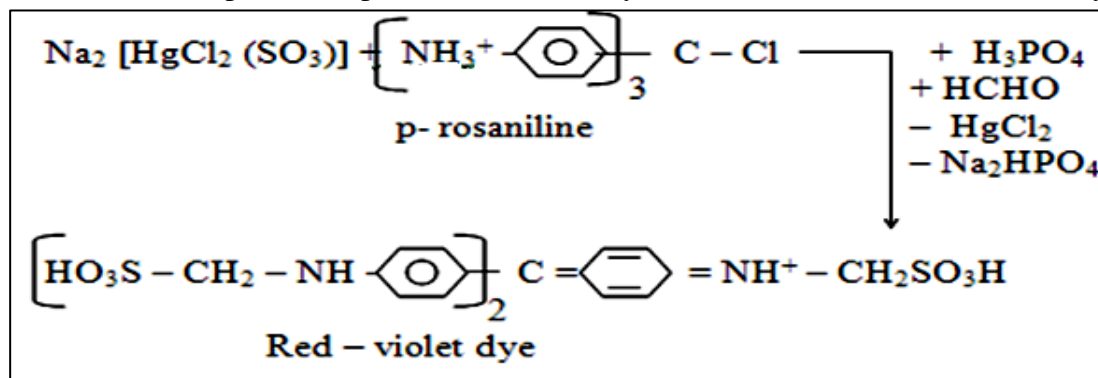
Principle

The sample gas is passed through a dilute aqueous solution of sodium tetra chloromercurate (II), Na₂[HgCl₄], containing sulfamic acid. Sulfamic acid destroys the major interference from

NO₂ by converting to N₂. SO₂ present in air converts the added mercurate into a stable non-volatile complex, sodium dichloro sulfitomercurate (II).



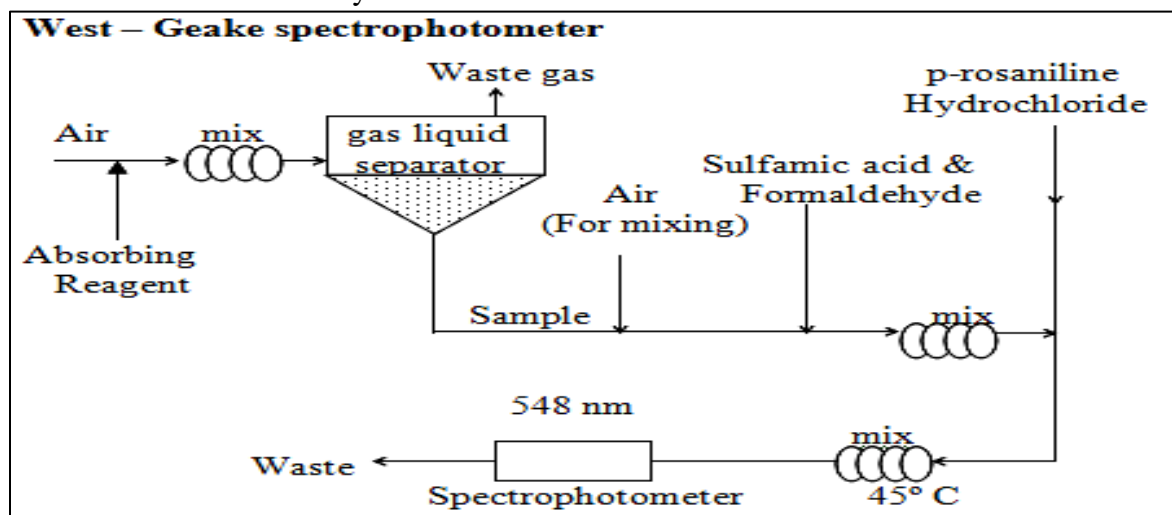
The solution is allowed to react with formaldehyde, phosphoric acid and then with para roaniline hydrochloride. The product is para rosaniline methyl sulfonic acid which is red-violet dye.



absorbance of the product red-violet dye is measured at 548 nm. Its concentration is determined using Beer-Lambert law. The concentration of para rosaniline methyl sulfamic acid (red-violet dye) is proportional to the concentration of SO₂ in the sample air.

Procedure

30 to 60 litres of air is pumped through 10 ml of the scrubbling solution (HgCl₂ and HCl) in a small impinger at 1 litres / min. 1 ml of dilute para rosaniline reagent solution is added along with sulfamic acid, phosphoric acid and formaldehyde solution. After 20 to 30 minutes, the absorbance of red-violet dye is measured at 548 nm.



The concentration of red-violet dye is proportional to the concentration of SO₂ in the sample air.

By using this method, SO₂ in very minute concentration in the range of 0.0005 to 5 ppm can be determined.

ANALYSIS OF H₂S

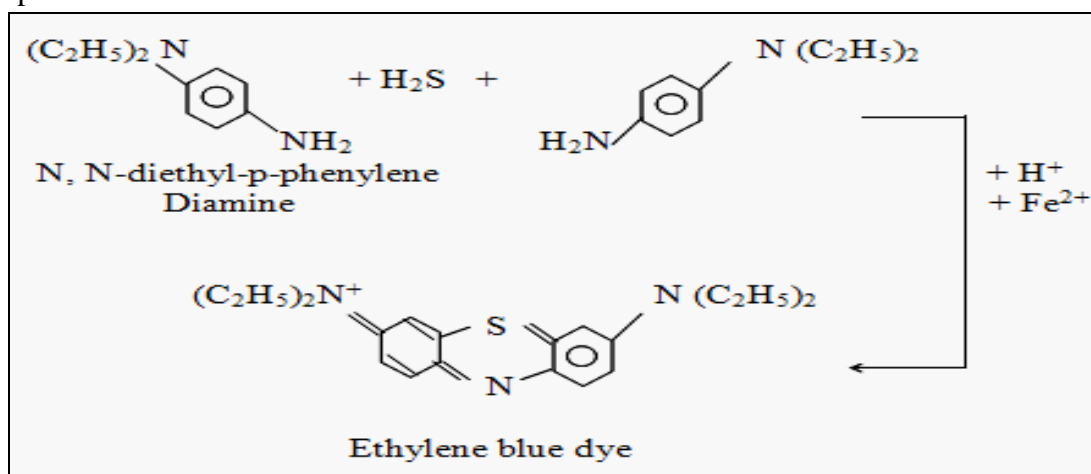
H₂S can be determined by reduction method in the absence of SO₂. If SO₂ is present, it can be removed by passing the gas on glass fibre filter impregnated with K₂CO₃. H₂S also form coloured precipitates with II group metals. In iodometric method, H₂S solution reacts with KI and liberates iodine that can be titrated against a standard solution of thiosulphate.

UV molecular absorption method is also quite sensitive. H_2S is first collected in zinc acetate and made acidic. The liberated gas is passed into N_2 in a UV molecular absorption spectrometer at about 202 nm and SO_2 is detected and recorded.

Determination of H_2S by Spectrophotometer

Principle

H_2S and SO_2 gases are coexisting in the atmosphere. H_2S is produced by the reduction of sulphate and organo-sulphur compounds by the action of the bacteria namely *Desulphovibrio desulphuricans*.



The ethylene blue dye formed is then measured at 670 nm in spectrophotometer.

Procedure

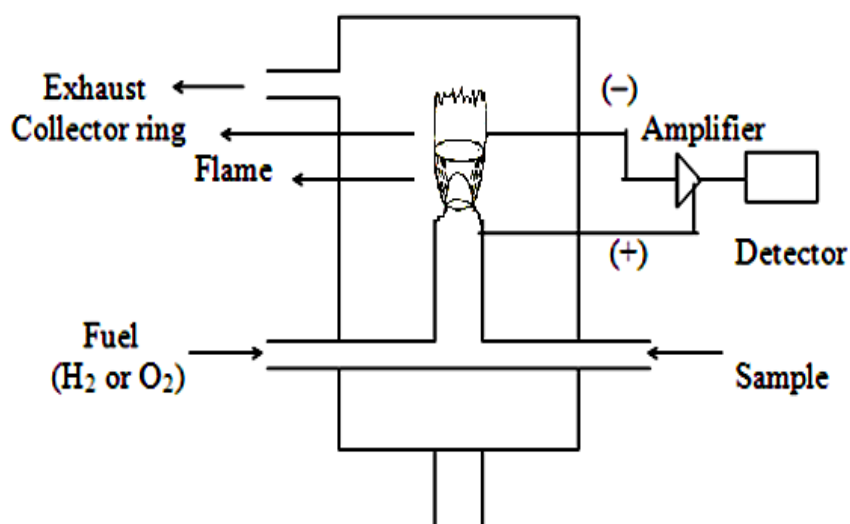
50 ml of cadmium sulphate and 50 ml of NaOH solution is taken in a standard gas wash bottle with fitted glass dispenser. 100 to 500 litres of air at 10 L/min is passed through the wash bottle and 2 ml of sodium chlorate is added and shaken well. 1 ml of N, N'-diethyl-p-phenylenediamine is added in presence of H_2SO_4 and FAS. After 10 minutes, solution is transferred into a separating funnel and extracted with chloroform. The absorbance is measured at 670 nm in spectrophotometer and absorbance is proportional to the amount of H_2S in sample air.

ANALYSIS OF HYDROCARBONS

Flame ionization Detector (FID)

Principle

In a flame, organic compounds form highly conducting fragments, such as C^+ . On applying a potential gradient across the flame, a small current is generated which is readily measured. The response of a flame ionization detector is proportional to the number of carbon atoms in the compound.



Procedure

Sample gas is injected into the flame. The flame is produced by H_2 in air or oxygen. When hydrocarbons are present, flame produces a complex ionization in which a large number of ions are present.

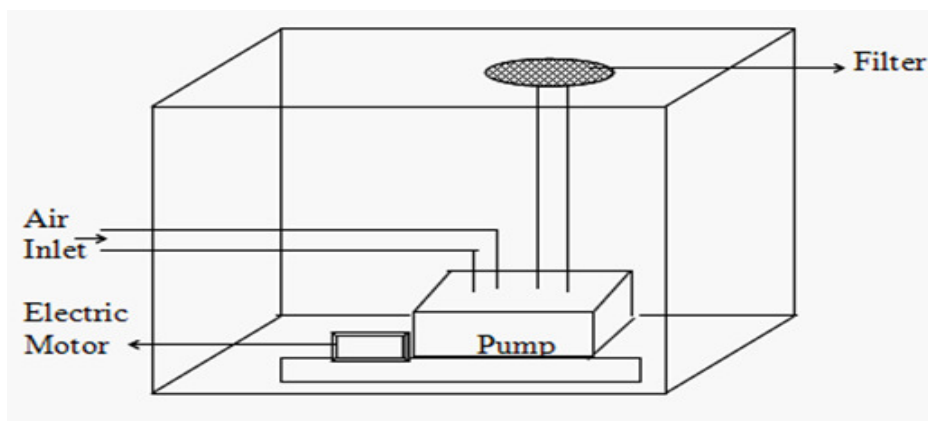
An electric field is set up in the vicinity of the flame by making the burner jet as +ve with respect to wire loop. The electric current induces ion migration between the electrodes and it produces small ionization current. This current is directly proportional to the hydrocarbon ions in the flame. The current is amplified and displayed in the meter. As little as $0.1 \mu g$ to $10^{-3} \mu g$ of hydrocarbons of air can be analyzed using FID.

ANALYSIS OF PARTICULATE MATTER

High volume sampler method

Principle and Instrument Description

High volume sampler consists of an electric motor and a pump. A gas meter attached to this unit would indicate the volume of air pumped through the sampler. This is fitted with a filter made up of glass fibres and the outlet of the pump is connected with the filter.



The sampler is capable of handling large volume of about 2×10^6 litres of air. When air passes through the filter, particles of diameter in the range of 0.1 to 100 micron are collected on the filter.

The difference in the weight of the filter before and after the passage of air will give the amount of particulate matter in the air. Particulate levels as low as $1 \mu g/m^3$ is measured by this method.

Procedure

Before sampling, the filter of high volume sampler is maintained at 15 to $35^\circ C$ at 50% relative humidity for 24 hours and then it is weighed. The filter is then kept in the high volume sampler and air is drawn through a portion of the filter at a rate of at least $1.7 m^3/min$, as measured by a rotameter and passes about 2×10^6 litres of air through it. After sampling, the filter is removed and equilibrated for 24 hours under the same conditions used prior to its fitting on the sampler. The filter is weighed again and the difference in the weight of the filter gives the amount of particulates per litre of air.

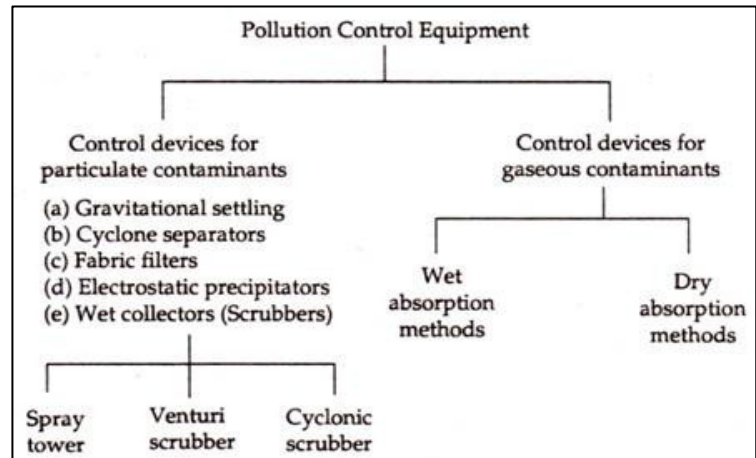
Particulate matter from filter may be extracted with a suitable solvent or incinerated or digested by oxidizing acid mixture before elemental analysis. The filter is used for subsequent chemical examinations.

AIR POLLUTION CONTROL

Control of gaseous pollutants

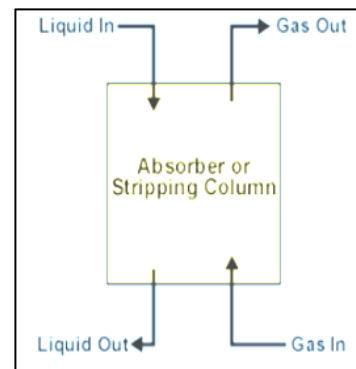
There are four commonly used control technologies for gaseous pollutants:

- Absorption,
- Adsorption,
- Condensation, and
- Incineration (combustion)



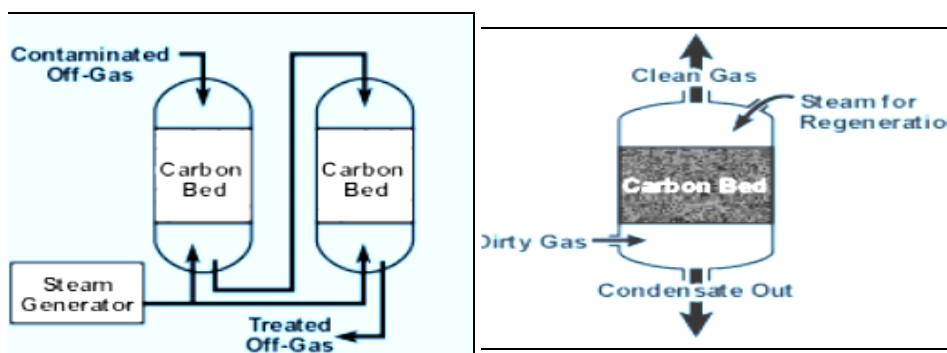
Absorption

- The removal of one or more selected components from a gas mixture by absorption
- Absorption is a process in which a gaseous pollutant is dissolved in a liquid.
- As the gas stream passes through the liquid, the liquid absorbs the gas,
- Absorbers are often referred to as scrubbers.
- types of gas absorption equipment include spray towers, packed columns, spray chambers, and venture scrubbers.



Adsorption

- When a gas or vapor is brought into contact with a solid, part of it is taken up by the solid. The molecules that disappear from the gas either enter the inside of the solid, or remain on the outside attached to the surface. The former phenomenon is termed absorption (or dissolution) and the latter adsorption.
- The industrial adsorbents are activated carbon, silica gel, and alumina
- Activated carbon is the universal standard for purification and removal of trace organic contaminants. Carbon adsorption systems are either regenerative or non-regenerative.
 - **Regenerative system** usually contains more than one carbon bed. As one bed actively removes pollutants, another bed is being regenerated for future use.
 - **Non-regenerative systems** have thinner beds of activated carbon. In a non-regenerative adsorber, the spent carbon is disposed of when it becomes saturated with the pollutant.



Condensation

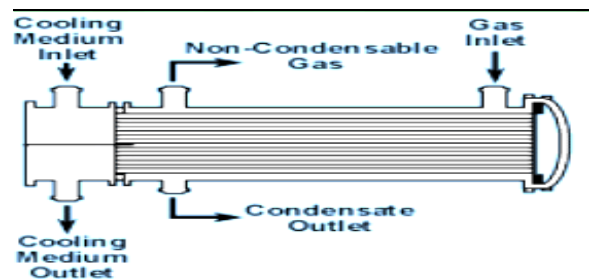
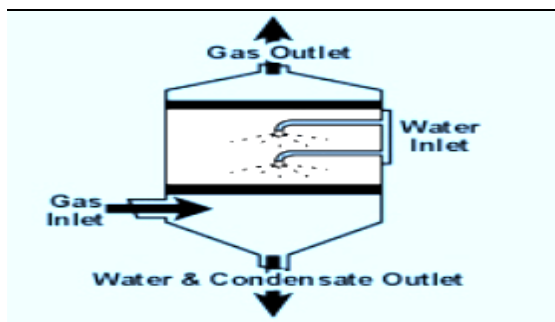
Condensation is the process of converting a gas or vapor to liquid. Any gas can be reduced to a liquid by lowering its temperature and/or increasing its pressure.

Condensers are typically used as pretreatment devices. They can be used ahead of absorbers, absorbers, and incinerators to reduce the total gas volume to be treated by more expensive control equipment. Condensers used for pollution control are contact condensers and surface condensers.

In a **contact condenser**, the gas comes into contact with cold liquid.

In a **surface condenser**, the gas contacts a cooled surface in which cooled liquid or gas is circulated, such as the outside of the tube.

Removal efficiencies of condensers typically range from 50 percent to more than 95 percent, depending on design and applications.



Incineration

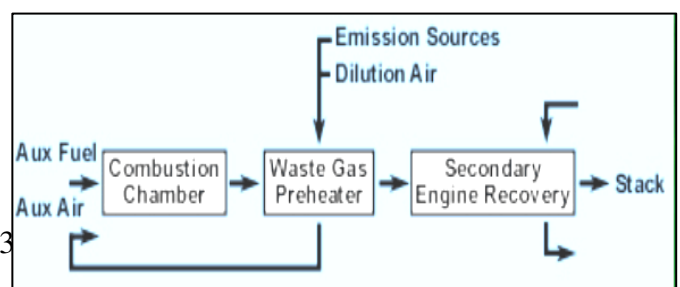
Incineration, also known as combustion, is most used to control the emissions of organic compounds from process industries.

- This control technique refers to the rapid oxidation of a substance through the combination of oxygen with a combustible material in the presence of heat. When combustion is complete, the gaseous stream is converted to carbon dioxide and water vapor. Equipment used to control waste gases by combustion can be divided in three categories:
 - Direct combustion or flaring, - Thermal incineration and - Catalytic incineration.

Direct combustor

- Direct combustor is a device in which air and all the combustible waste gases react at the burner. Complete combustion must occur instantaneously since there is no residence chamber.
- A flare can be used to control almost any emission stream containing volatile organic compounds.
- In **thermal incinerators** the combustible waste gases pass over or around a burner flame into a residence chamber where oxidation of the waste gases is completed. Thermal incinerators can destroy gaseous pollutants at efficiencies of greater than 99 percent when operated correctly.

Catalytic incinerators are very similar to thermal incinerators. The main difference is



that after passing through the flame area, the gases pass over a catalyst bed. A catalyst promotes oxidation at lower temperatures, thereby reducing fuel costs. Destruction efficiencies greater than 95 percent are possible using a catalytic incinerator.

Control of Particulate Emission

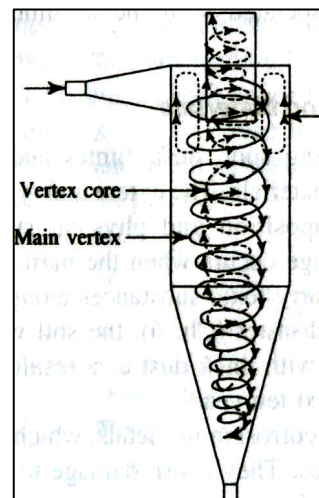
Methods of controlling particulate emissions:

The air pollution caused by particulate matter like dust, soot, ash, etc, can be controlled by using fabric filters, wet scrubbers, electrostatic precipitators and certain mechanical devices.

Mechanical Devices: It works on the basis of following:

- **Gravity:** In this process, the particulate settle down by the action of gravitational force and get removed.
- **Sudden change in the direction of air flow:** It brings about separation of particles due to greater momentum.
- **Cyclone collector**

A gas flowing in a tight circular spiral produces a centrifugal force on suspended particles, forcing them to move outward through the gas stream to a wall where they are collected. Thus it is possible to remove 95% particles in the diameter range 5-20 μ .

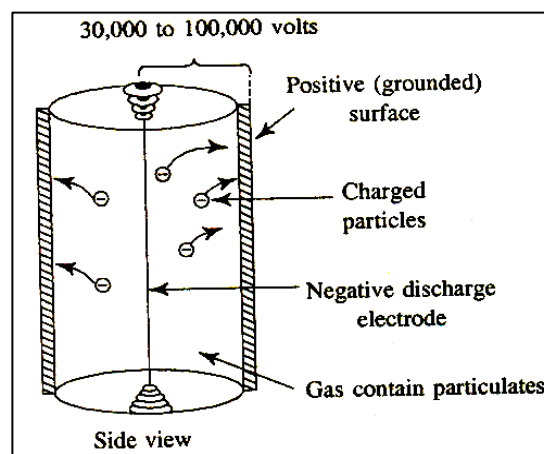


Electrostatic precipitators

When the polluted air containing particulate pollutants is passed through an electrostatic precipitator, it induces electric charge on the particles and then the aerosol particles get precipitated on the electrodes.

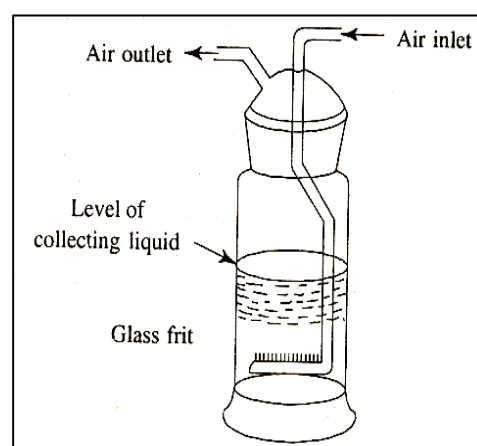
The removal of particulate matter from gas streams is an essential step for air pollution control. There are four types of equipment used for this purpose.

These are based on the principle that aerosol particles acquire charges when subjected to an electrical field. The particles acquire a charge when a gas stream is led through a high-voltage de-corona. The charged particles are attracted to a grounded surface from which they are removed. Ozone is a possible product of the corona discharge.



Wet scrubbers

They are used to trap SO₂, NH₃ and metal fumes by passing the fumes through water. These utilize a liquid (usually water) to help remove solid, liquid or gaseous contaminants. The extent of contact and interaction are increased by the use of spray as fine spray.



Fabric Filters: The particulate matter is passed through a porous medium made of woven or filled fabrics.

- The particulate present in the polluted air are filtered and gets collected in the fabric filters, while the gases are discharged.
- The process of controlling air pollution by using fabric filters is called 'bag filtration'.

Gravity settling chamber

Effluent gases are led into a chamber which is large enough to permit gas velocities to decrease and dust or droplets to settle. The chamber is generally in the shape of a horizontal rectangular tank articles with a diameter $> 50\mu$ are ordinarily removed in this manner. The method is however not suitable for fine particles which require longer setting time.

Some other methods of controlling Air Pollution

- (i) Tall chimneys should be installed in factories.
- (ii) Better designed equipment and smokeless fuels should be used in homes and industries.
- (iii) Renewable and non- polluting sources of energy like solar energy, wind energy should be used.
- (iv) Automobiles should be properly maintained and adhere to emission control standards.
- (v) More trees should be planted along roadsides and houses.

QUESTIONS

PART-A

- 1. State Beer-Lambert;s Law.
- 2. What are the functions of monochromators in UV-VIS spectrophotometer?
- 3. List out any two effects of NO as a pollutant.
- 4. What is meant by development of chromatogram?
- 5. How carbon monoxide and oxides of Nitrogen are estimated?
- 6. Mention the purpose of IR spectrometer.
- 7. What are disadvantages of AAS ?
- 8. What are the basic principles involved in AAS and IR spectroscopy?
- 9. Write explanatory note on Gas-Chromatography.
- 10. What is the basic principle of GC chromatographic technique?
- 11. Write note on gas chromatography.

PART - B

- 1. Explain the estimation of CO. (May 14)
- 2. With a neat sketch describe the instrumentation of atomic absorption spectroscopy. (May 14, Jan 15)
- 3. Explain elaborately the principle and instrumentation of UV visible spectroscopy. (May 14)
- 4. Write explanatory note on instrumentation of gas chromatography. (May 14)
- 5. Give a neat sketch describe the functions of the components of the FTIR spectrophotometer. (Jan 15)
- 6. Explain the procedure for estimating particulate matter in a sample of air.
- 7. Explain with reactions, estimation of SO₂ in a given sample.
- 8. Explain the procedure for the determination of CO₂ in air.
- 9. Explain the analytical procedure for SO_x and H₂S. (Jan 15)
- 10. How will you determine the concentration of HC in air? (Jan 15)
- 11. Explain the high volume sampler used for monitoring the air pollutants.

B. Tech DEGREE EXAMINATION, MAY 2015 (5781009)

Second Semester – Common for all branches

ENVIRONMENTAL SCIENCE

Time: 3 Hrs

Maximum Marks: 75

SECTION A – (10 x 2 = 20 marks)

1. Define sustainable development?
2. What is the economic importance of a forest?
3. Define food web with suitable examples.
4. What is meant by Hot spots of biodiversity? Give examples.
5. What is acid rain? Write its harmful effects.
6. How is photochemical smog formed? Write its problems.
7. Define BOD and write its importance in environmental studies.
8. What is meant by Eutrophication of water? Write its effects.
9. State Beer-Lambert law.
10. Write the principle of conductometry.

SECTION – B

UNIT I

11. (a) Discuss detail benefits and problems associated with construction of dams. (6)
(b) Name the various unconventional renewable energy resources and explain in detail the principle, merits and demerits of solar energy system. (5)

Or

12. (a) Discuss in detail the causes and effects of deforestation. (6)
(b) Narrate with a case study the environmental impact of modern agriculture. (5)

UNIT II

13. (a) Write short notes on structure and function of aquatic ecosystem. (6)
(b) Define biodiversity with suitable example and discuss the threats to it. (5)

Or

14. (a) Describe in-situ and ex-situ conservation of bio-diversity. (6)
(b) Explain the structure and functions of forest ecosystem. (5)

UNIT III

15. (a) Briefly describe the sources and effects of particulate air pollutants and explain any of two control methods. (6)
(b) Discuss the causes, consequences and mechanism of Ozone layer depletion. (5)

Or

16. (a) Discuss any two factors contributing for global warming and write its consequences.
(b) Explain sources, pollution effects and control measures of oxides of Nitrogen. (5)

UNIT IV

17. (a) Write an elaborate note on solid waste management. (6)
(b) Discuss the sources and ill effects of Hg and Pb. (5)

Or

18. (a) Explain the sources, effects and control methods of thermal pollution. (6)
(b) Give two organic water pollutants and discuss their sources and harmful effects. (5)

UNIT V

19. (a) Write principle of Gas Chromatography and explain its instrumentation with a diagram. (6)
(b) Explain the analytical techniques used for detecting CO_x and hydrocarbons. (5)

Or

20. (a) Describe the basic concepts and instrumentation of IR spectrometry. (6)
(b) Explain the analysis of SO_x and particulate pollutants. (5)

B. Tech DEGREE EXAMINATION, JANUARY 2015 (5781009)

Second Semester

ENVIRONMENTAL SCIENCE

SECTION A – (10 x 2 = 20 marks)

1. What are the functions of lithosphere?
2. State the environmental effects of extracting and using mineral resources.
3. Differentiate natural and artificial ecosystem.
4. Define endangered species.
5. What is acid rain? How does it form?
6. What is photochemical smog?
7. Define Eutrophication.
8. What is meant by 5 R approach?
9. State Beer-lambert's law?
10. What are the functions of monochromators in UV-Visible spectrophotometer?

PART-B (5 x 11= 55 marks)

Answer ALL questions, ONE from each unit.

UNIT- I

11. (a) What is deforestation? Give the ill effects of deforestation. (6)
- (b) Explain the adverse environmental impacts of modern agriculture. (5)

Or

12. (a) Discuss the effects of dams on forests and tribal people. (6)
- (b) Explain how the alternate energy sources play an important role in environmental impact? (5)

UNIT - II

13. (a) State the components of ecosystems and explain. (5)
- (b) Explain the different types of ecological pyramids giving suitable example. (6)

Or

14. (a) Name and discuss the values that can be assigned to biodiversity. (6)
- (b) Explain the threats to biodiversity. How can they be prevented? (5)

UNIT III

15. (a) Discuss the sources and effects of air pollution. (6)
- (b) Explain the factors responsible for the global warming? Discuss the safety measures recommended to control the effect. (5)

Or

16. (a) Explain the mechanism of acid rain formation and enumerate its effect. (6)
- (b) What do you understand by ozone layer depletion? Give causes and effect of this phenomenon. (5)

UNIT-IV

17. Write a note on (a) Sources and effects of thermal pollution. (4)
- (b) BOD (4)
- (c) Significance of dissolved oxygen in rivers. (3)

Or

18. (a) Classify solid waste. How can the waste be managed? (6)
- (b) Explain about any one nuclear holocaust. (5)

UNIT V

19. (a) Give a neat sketch of the FTIR spectrophotometer and describe the functions of the components. (6)
- (b) Explain the working principle of AAS and gas chromatography. (5)

Or

20. Describe briefly how the following pollutants in the atmosphere are determined?
(a) SO₂
(b) H₂S (c) Hydrocarbons. (11)

B.Tech DEGREE EXAMINATION, JANUARY 2015 (4781009)

Second Semester

ENVIRONMENTAL SCIENCE

Time: Three hours

Maximum: 75 marks

SECTION A – (10 x 2 = 20 marks)

1. What are the objectives of alternate energy resources?
2. State the environmental effects of extracting mineral resources.
3. What is bio-diversity and what is its significance?
4. What are hot spots of bio-diversity? Name two hotspots of Bio-diversity.
5. Define photochemical smog?
6. List out green house gases.
7. Define thermal pollution.
8. What is meant by non-point pollution?
9. What are the causes of population explosion?
10. What is meant by Rain Water harvesting?

PART-B (5 x 11= 55 marks)

UNIT- I

11. (a) Discuss in detail the adverse effect of modern agriculture. (11).

Or

12. (a) What is deforestation? Explain causes and ill effects of deforestation. (11)

UNIT - II

13. (a) Describe the structure and functions of forest ecosystem. (7)

- (b) Write a note on ecological succession. (4)

Or

- 14 (a) Discuss the threats faced by Indian Biodiversity. (8)

- (b) Explain the different types biodiversity. (3)

UNIT - III

15. What are the major air pollutants? Enumerate the effects of CO and CO₂.
How can it be controlled? (11)

Or

16. (a) Explain the mechanism of ozone layer depletion. (6)

- (b) How is acid rain formed? Explain any two environmental effects. (5)

UNIT-IV

17. (a) Explain causes and effects measure of marine pollution. (6)

- (b) Discuss effects and control measures of noise pollution. (4)

Or

18. What are the different types of solid wastes? Discuss briefly the various methods of disposal of solid waste. (11)

UNIT V

19. Give the salient features of

- (a) Environmental Protection Act (b) Wild life protection act (c) Water Act. (11)

Or

20. (a) Describe the role of Information Technology in Environment. (6)

B.Tech DEGREE EXAMINATION, APRIL/MAY 2014 (5781009)
Second Semester – Common for all branches
ENVIRONMENTAL SCIENCE

Time: 3 Hrs

Maximum Marks: 75

SECTION A – (10 x 2 = 20 marks)

1. What are the environmental effects of mining?
2. State the problems caused by the construction of dams.
3. Distinguish between food chain and food web.
4. Name two hotspots of biodiversity in India.
5. List out green house gases.
6. How is photochemical smog id obtained?
7. Give the significance of dissolved oxygen in rivers.
8. Mention any to ill effects of chromium and mercury.
9. List out any two effects of NO as a pollutant.
10. What is meant by development of chromatogram?

SECTION – B

UNIT I

11. (a) Explain any two types of renewable energy resources. (6)
- (b) What are the effects of over utilization of ground water? (4)

Or

12. (a) Explain the environmental effects of deforestation. (6)
- (b) Write short notes-on the following (5)
- (i) Eutrophication (ii) Blue baby Syndrome

UNIT II

13. (a) Discuss the major threats to Indian biodiversity. (6)
- (b) List out different types of ecological pyramids and explain any one in detail. (5)

Or

14. (a) Show the structure and function of forest ecosystem. (6)
- (b) Explain any one of the strategy adopted to conserve biodiversity. (5)

UNIT III

15. (a) Discuss about ozone hole and ozone depleting substances and write any two effects of ozone layer depletion. (6)
- (b) Discuss the method of control of air pollution from automobiles. (5)

Or

16. (a) What are the major air pollutants? Enumerate any two effects of oxides of sulphur and hydrocarbon. (6)
- (b) How is acid rain formed? Explain its environmental effects. (5)

UNIT IV

17. (a) Write short notes on land filling and incineration methods of solid waste disposal. (6)
- (b) What is meant by thermal pollution and explain its environmental effects? (5)

Or

18. (a) What are the different types of water pollutants? Discuss briefly the sources effects and the various methods of disposal of it. (8)
- (b) Distinguish between BOD and COD. (3)

UNIT V

19. (a) With neat sketch describe instrumentation of atomic absorption spectroscopy. (6)
- (b) Explain the estimation of CO. (5)

Or

20. (a) Explain elaborately the principle and instrumentation of UV visible spectroscopy. (6)
- (b) Write explanatory note on the instrumentation of gas chromatography. (5)

B.Tech DEGREE EXAMINATION, JANUARY 2013 (4781009)

Second Semester (Common to All Branches)

ENVIRONMENTAL SCIENCE

Time: Three hours

Maximum: 75 marks

SECTION A – (10 x 2 = 20 marks)

1. Define Overgrazing.
2. What are the environmental effects of extracting mineral sources?
3. Distinguish between ecosystem and biome.
4. Name two hotspots of Biodiversity.
5. Classify air pollutants.
6. List out greenhouse gases.
7. What is meant by thermal pollution?
8. What is meant by non-point pollution?
9. What are the effects of population explosion?
10. What is sustainable development?

SECTION B (5 x 11=55)

UNIT I

11. (a) What is deforestation? Explain the environmental effects of deforestation. (6)
(b) Define land degradation with its effects. (5)

Or

12. (a) Explain the various types of renewable energy resources. (6)
(b) What are the effects of over utilization of fertilizers? (5)

UNIT II

13. (a) Explain the in-situ conservation of bio-diversity. (7)
(b) "India is a mega diversity nation" – Explain. (4)

Or

14. (a) Describe structure and functions of various components of an ecosystem. (7)
(b) Explain about ecological pyramids. (4)

UNIT III

15. (a) What are major air pollutants? Explain the control of gaseous pollutants? (7)
(b) How is acid rain formed? Explain any two environmental effects. (4)

Or

16. (a) Explain the Mechanism of ozone layer depletion. (6)
(b) Write a note on global warming. (5)

UNIT IV

17. (a) Explain causes and effects measure of marine pollution. (6)
(b) Discuss effects and control measure of noise pollution. (5)

Or

18. (a) Explain any two methods of disposal of solid waste. (7)
(b) Write a note on radioactive pollution. (4)

UNIT V

19. Give the salient features of (11)
 - (a) Environmental Protection Act
 - (b) Wild life protection act
 - (c) Water Act.
20. (a) Discuss the role of IT in Environment. (7)
(b) Write a short note on environmental ethics. (4)

B.Tech DEGREE EXAMINATION, APRIL/MAY 2014 (4781009)

Second Semester

ENVIRONMENTAL SCIENCE

Time: Three hours

Maximum: 75 marks

SECTION A – (10 x 2 = 20 marks)

1. Define biosphere.
2. What is desertification? Give two reasons for it.
3. Name the four ecosystems.
4. What is bio diversity and what is its significance?
5. Write the sources of carbon monoxide.
6. What are the sources of photochemical smog?
7. Name any four major water pollutants.
8. Name the sources and effects of marine pollution.
9. What are the advantages of rain water harvesting?
10. Mention the issues involved in enforcement of environmental legislation.

SECTION – B (5 x 11 = 55)

UNIT I

11. What are the effects of mining? (11)

Or

12. Give a brief account of non-renewable energy resources and their significance. (11)

UNIT II

13. Discuss the components of eco system. (11)

Or

14. How is conservation of biodiversity achieved? Give details. (11)

UNIT III

15. Write a short note on Global warming. (11)

Or

16. Explain acid rain. (11)

UNIT IV

17. Discuss the causes and effects of soil pollution. (11)

Or

18. Discuss about the significance of hazardous waste management. (11)

UNIT V

19. Write short notes on environmental ethics. (11)

Or

20. Enumerate the effects of population explosion. (11)

B.Tech DEGREE EXAMINATION, JANUARY 2014 (4781009)

Second Semester (Common to All Branches)

ENVIRONMENTAL SCIENCE

SECTION A – (10 x 2 = 20 marks)

1. Mention the different atmospheric layers.
2. Define the term “soil leaching”.
3. What are the criteria for recognizing hot spots?
4. What are the characteristics of estuaries?
5. Define Photochemical smog.
6. How CFC's are accumulated in atmosphere?
7. Distinguish between BOD & COD.
8. When does a sound cause noise pollution?
9. What is the purpose of environmental impact assessment?
10. Define population explosion.

SECTION B (5 x 11 = 55 marks)

UNIT I

11. Explain various types of renewable and non-renewable energy resources.(11)

Or

12. What is deforestation? Explain the environmental effects of deforestation and the effects of over utilization of ground water. (11)

UNIT II

13. Discuss the threats faced by Indian Biodiversity. (11)

Or

14. (a) Explain the energy flow of ecosystem with the help thermodynamics law.(4)
(b) Describe about the structures and functions of forest ecosystem. (7)

UNIT III

15. Discuss causes and effects of (a) Green house effect (b) Acid rain. (11)

Or

16. (a) Explain the mechanism of ozone layer depletion. (6)
(b) Discuss the effects of carbon monoxide and sulphur dioxide. (5)

UNIT IV

17. Define water pollution. Enumerate causes and effects of water pollution.(11)

Or

18. Enumerate the various methods of solid waste disposal. (11)

UNIT V

19. (a) What is rain water harvesting? Discuss the various types of rain water harvesting method. (7)
(b) List out the urban problems related to energy. (4)

Or

20. (a) Discuss the role of IT in Environment and Human health. (11)

B.TechDEGREE EXAMINATION, APRIL/MAY 2012 (ETC 09)

Second Semester

ENVIRONMENTAL SCIENCE

SECTION A – (10 x 2 = 20 marks)

1. What is biosphere?
2. Define the term 'overgrazing'.
3. What are tropic levels?
4. What are biodiversity hot spots?
5. What is photochemical smog?
6. List out the effects of SO₂ on environment.
7. Define the term 'thermal pollution'.
8. Mention any two effects of solid waste.
9. Define the term sustainable development.
10. What is population explosion?

SECTION B – (5 x 11=55 marks)

UNIT I

11. (a) Discuss any three renewable energy resources in detail. (6)
(b) What is meant by land degradation and mention causes for land degradation.(5)

Or

12. (a) Explain the impacts of modern agriculture. (6)
(b) List out the benefits of dams. (5)

UNIT II

13. (a) Explain the energy flow in an ecosystem. (6)
(b) Discuss the various threats to biodiversity. (5)

Or

14. (a) Discuss characteristics features, structures and functions of grass land ecosystem. (6)
(b) Write a note on food web. (5)

UNIT III

15. (a) Describe the important sources of air pollution. (6)
(b) What is the significance of ozone layer on the earth's surface and what are the causes for depletion of ozone layer? (5)

Or

16. (a) Explain how the acid rain affects the environment and mention the different control measures of acid rain. (6)
(b) Discuss the effects of greenhouse gases on global warming. (5)

UNIT IV

17. (a) Describe the causes, effects and control measures of urban solid waste. (6)
(b) Explain the measures to be taken to control soil pollution. (5)

Or

18. (a) Write a note on the sources and effects of noise pollution. (6)
(b) List out the sources of radioactive pollution and suggest some control measures to avoid radioactive pollution. (5)

UNIT V

19. (a) What is meant by rain water harvesting? Why is it necessary nowadays? (6)
(b) Discuss the role of information technology on human health. (5)

Or

20. (a) Explain the role of an individual in prevention of pollution. (6)
(b) Write short notes on forest conservation act and water act. (5)

B. Tech DEGREE EXAMINATION, APRIL/MAY 2011 (ETC 09)
Second Semester (Common to All Branches)
ENVIRONMENTAL SCIENCE

Time: Three hours

Maximum: 75 marks

SECTION A – (10 x 2 = 20 marks)

1. Define the term Lithosphere.
2. What are the advantages of rain water harvesting?
3. Enumerate some characteristics of an eco system.
4. What are the types of grassland ecosystem?
5. What are the sources of air pollution?
6. Give any 4 steps to control air pollution.
7. How arsenic contaminates the water?
8. Mention the effects of solid waste.
9. What are the effects of Global warming?
10. What are the objectives of watershed management?

SECTION B (5 x 11 = 55)

UNIT I

11. What is solar energy? Describe the methods of harvesting solar energy. (11)

Or

12. Explain about world food problems and changes caused by overgrazing and agriculture. (11)

UNIT II

13. Describe about energy flow in the ecosystem. (11)

Or

14. Explain the various threats of Indian biodiversity. (11)

UNIT III

15. Explain in detail about various air pollutants sources and their effects. (11)

Or

16. Describe the formation, effect and control measures of acid rain. (11)

UNIT IV

17. How does marine water get contaminated? Give any one case study. (11)

Or

18. Explain the steps involved in solid waste management. (11)

UNIT V

19. Discuss role of information technology in environment and human health. (11)

Or

20. Explain about reason for population explosion and environmental, social impact of growing population. (11)

UNIT V

19. Describe the need, methods of rain water harvesting and its advantages. (11)

Or

20. Explain the following :
(a) Wildlife Protective Act. (6)
(b) Forest Preservative Act. (5)

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B.Tech. DEGREE EXAMINATION, JANUARY 2016.

Second Semester

ENVIRONMENTAL SCIENCE
(For 2009-2010 to 2012-2013 batches)
(Common to all branches)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Explain Biosphere.
2. What is the role of an individual in conservation of natural resources?
3. Define Ecosystem.
4. What are the methods by which nitrogen fixation takes place in the nature?
5. Define air pollution.

6. What is the effect of air pollution on plant life?
7. What are the effects of radioactive waste in soil pollution?
8. Mention the activities involved in solid waste management.
9. Name some of the acts enacted by the Indian Government to protect the environment
10. What are the factors affecting watershed?

SECTION B — (5 × 11 = 55 marks)

Answer ALL Units,
choosing ONE full question from each Unit.

UNIT I

11. (a) Write about the types of fresh water resources. (6)
(b) Explain the effects of over utilization of water. (5)

Or

12. Explain the environmental effects of extracting and using mineral resources. (11)

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UNIT II

13. Explain about desert and aquatic eco systems. (11)
- Or
14. Briefly explain the conservation of biodiversity. (11)

UNIT III

15. Explain how the green house causes and effects on global climate? Explain their consequences. (11)
- Or
16. Describe the sources of air pollutants and its control measures. (11)

UNIT IV

17. What is thermal pollution? Explain its sources and effects on ecosystem. (11)
- Or
18. (a) Write the sources of urban and industrial wastes. (5)
(b) Explain the process of solid waste management with diagram. (6)

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UNIT V

19. (a) With a neat sketch explain about atomic absorption spectroscopy. (7)
- (b) Explain any two methods available for the measurement of CO. (4)

Or

20. (a) What are the components of Gas Chromatography? Explain in detail with a neat diagram. (7)
- (b) How is NO_x estimated? (4)

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8. Distinguish between BOD and COD.
9. List out the detectors of UV-Visible Spectroscopy.
10. List out the various types of H₂S analyzers.

SECTION B — (5 × 11 = 55 marks)

Answer ALL units, choosing ONE question from each unit.

UNIT I

11. (a) Discuss in detail the adverse effect of using pesticides and fertilizers. (8)
- (b) List out any three environmental effects of mining. (3)

Or

12. (a) Explain any three various types of renewable energy resources. (6)
- (b) Describe the effects of over utilization of ground water. (5)

UNIT II

13. (a) Describe the structure and functions of forest ecosystem. (7)
- (b) Write a note on any three values of biodiversity. (4)

Or

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B.Tech. DEGREE EXAMINATION, MAY 2016.

Second Semester

ENVIRONMENTAL SCIENCE

(Common to All Branches)

(From 2013-14 batch onwards)

Time : Three hours

Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

1. List out the ill effects of deforestation.
2. What is meant by Eutrophication?
3. What are the criteria for recognizing hotspots?
4. Define food chain. Mention its types.
5. Mention the effects of chlorofluoro carbons.
6. How is acid rain formed?
7. How can be nuclear hazards disposed safely?

14. (a) Discuss the threats faced by Indian Biodiversity. (8)
- (b) "India is a mega diversity Nation". Justify your answer. (3)

UNIT III

15. (a) Enumerate the biochemical effects of oxides of nitrogen and sulphur. (6)
- (b) Explain the Mechanism of ozone layer depletion. (5)

Or

16. (a) What is meant by global warming and green house effect? List out their effects. (7)
- (b) What are the effects of photochemical smog? (4)

UNIT IV

17. (a) What are the different types of solid wastes? Discuss briefly the various methods of disposal of solid waste. (8)
- (b) List out the cause and effects of urban waste. (3)

Or

18. (a) Discuss the effects and control measure of thermal pollution. (6)
- (b) With a neat diagram, explain trickling filter process. (5)

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UNIT IV

17. Write notes on
- Biomagnification and Eutrophication (4)
 - BOD and COD. (4)
 - Significance of TOC. (3)

Or

18. (a) State the effects of industrial waste water and methods to control thermal pollution? (6)
- (b) Explain the impacts of pesticides pollution. (5)

UNIT V

19. (a) With a neat sketch describe the principle and instrumentation of the UV-Visible spectroscopy. (6)
- (b) Write explanatory note on the instrumentation of gas chromatography. (5)

Or

20. Describe briefly how the following pollutants in the atmosphere are determined?

- NO_x
- CO_x
- Hydrocarbons
- H₂S. (11)

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- What is smog? Give the types and effects of smog.
- Mention any two ill effects of Mercury and lead.
- What are the 3 R's in solid waste management?
- State Beer-Lambert's Law. Give its importance.
- What is chromatography?

PART B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE question from each Unit.

All questions carry equal marks.

UNIT I

11. (a) What are the ecological benefits of forest? (6)
- (b) Environmental damage caused by mining last longer after the mine has closed — explain. (5)

Or

12. (a) Explain how the alternate energy sources play an important role in environment impact. (6)
- (b) Write short notes on the following
- Hydrosphere
 - Lithosphere.

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B.Tech. DEGREE EXAMINATION, MAY 2017.

Second Semester

Common to All Branches

ENVIRONMENTAL SCIENCE

(From 2013-2014 batch onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

- Give the classification of environmental pollution based on pollutant?
- Distinguish between renewable and non renewable sources of energy.
- What is red data book?
- What are the advantages of In-situ and Ex-situ conservation of biodiversity?
- Name any four air pollutants and their sources and impacts.

UNIT II

13. What are the major threats to biodiversity? Write a note on efforts taken for biodiversity conservation. (11)

Or

14. What is ecological succession? Show pyramids of energy and biomass through proper illustrations. (11)

UNIT III

15. (a) Explain the effects oxides of carbon and oxides of nitrogen as air pollutants. (6)
- (b) Elaborate the causes and impacts of ozone layer depletion. (5)

Or

16. (a) Discuss the reactions of hydrocarbons and particulates in the atmosphere. (6)
- (b) Write short notes on the following
- Photochemical smog
 - Environmental sequences of acid rain. (5)

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18. (a) What are industrial wastes? How will you control it? (6)
 (b) What are the health effects of Pb, Cr and Cu? What are the sources for these pollutants? (5)

UNIT V

19. (a) What are thermal and radioactive pollutions? How will they affect human? (6)
 (b) What is the principle in which the AAS works? What is the alternate for this instrument? (5)

Or

20. (a) What is the role of Gas chromatography in environmental monitoring? (6)
 (b) How will we collect samples to study the air pollution in a city? Discuss the methods and instruments available right now for air pollution studies? (5)

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7. Name the disease which is caused by Hg.
 8. What is BOD? And what do we infer out of it?
 9. What is the source of energy in an ecosystem?
 10. What is acid rain?

PART B — (5 × 11 = 55 marks)

Answer ALL the Units choosing One full questions from each Unit.

UNIT I

11. (a) What are the environmental impacts of modern agriculture? (6)
 (b) What are the environmental effects of copper and coal mines? (5)

Or

12. (a) What are the problems related to water resources in India and How will you manage water resources? (6)
 (b) What do that causes deforestation? Explain with examples. (5)

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B. Tech. DEGREE EXAMINATION, APRIL/MAY 2018.

Second Semester

ENVIRONMENTAL SCIENCE

(Common to All branches)

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. In which atmospheric layer ozone is present?
2. State any two alternate energy sources.
3. What is alpha diversity?
4. What is the role of human in biodiversity conservation?
5. Distinguish between pollution and pollutant.
6. What is global warming potential?

UNIT II

13. (a) Is it possible to replace the non-renewable energy resources with renewable energy resources at present? If not why? (6)
 (b) What are the characteristic features, structure and functions of desert ecosystem? (5)

Or

14. (a) What are the values of biodiversity? (6)
 (b) What are the major threats to biodiversity? (5)

UNIT III

15. (a) Compare the functions of aquatic and terrestrial ecosystems. (6)
 (b) Discuss in details Air pollution. (5)

Or

16. (a) What is global warming? What is the principle behind it? (6)
 (b) How will you minimize the pollutants such as NO, CO, SO, CFC? (5)

UNIT IV

17. (a) How do insecticide, pesticide, detergent and surfactant cause water pollution? (6)
 (b) How will you manage solid waste? What are the roles of people in solid waste management? (5)

Or

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UNIT IV

17. (a) What are the different types of solid wastes? Discuss briefly the various methods of disposal of solid waste. (8)
(b) Discuss the effects and control measure of thermal pollution. (3)

Or

18. (a) Define the terms BOD and COD. Distinguish between them. (3)
(b) List out the various sources of water pollution. Explain their effects and control measures. (8)

UNIT V

19. (a) With a neat sketch explain about atomic absorption spectroscopy. (7)
(b) Explain the method available for the measurement of SO_x . (4)

Or

20. (a) What are the components of Gas chromatography? Explain in detail with a neat diagram. (7)
(b) How is NO_x estimated? (4)

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6. List out the sources and any two effects of chlorofluoro carbons.
7. How can we dispose nuclear waste safely?
8. Name any two toxic metals that affect the environment. Give its ill effects.
9. What is meant by finger print region?
10. List out the various types of H_2S analyzers.

PART B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Discuss in detail the adverse effect of using pesticides and fertilizers. (6)
(b) List out the effects of over utilization of ground water. (5)

Or

12. (a) Define the term sustainable development and discuss the approaches to build up from unsustainable to sustainable development. (6)
(b) List out any three environmental effects of deforestation. (5)

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Second Semester

Common to all branches

ENVIRONMENTAL SCIENCE

(From 2013-14 onwards)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL the questions.

All questions carry equal marks.

- List out the different atmospheric layers.
- Wood is a renewable resources where as coal is not. Justify.
- What is the primary criterion to recognize "Hot spots of biodiversity"? Name any two Hot spots of biodiversity.
- Define food chain. Mention its types.
- How is acid rain formed?

UNIT II

13. (a) Choose any one of the ecosystem. Give its food chain, food web, ecological pyramid and energy flow chart. (8)
(b) Write a note on any three values of biodiversity. (3)

Or

14. (a) Discuss the threats faced by Indian Biodiversity. (8)
(b) Explain the classification of biodiversity. (3)

UNIT III

15. (a) Enumerate the biochemical effects of oxides of carbon and hydrocarbon. (6)
(b) Explain the mechanism of ozone layer depletion. (5)

Or

16. (a) What is meant by global warming and green house effect? List out their effects. (7)
(b) What are the effects of photochemical smog? (4)

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